

1. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI), has evolved drastically over the years, touching various aspects of our lives. It is a technology that has not only fascinated us but also significantly impacted how we live, work, and interact with the world around us. Within the vast landscape of AI, there exist several distinct Domains of Artificial Intelligence, each with its unique characteristics and applications. According to Statista, the global AI market, with a value of billion 113.60 GBP in 2023, is on a continuous growth trajectory, primarily fueled by substantial investments.

Artificial intelligence (AI) refers to the ability of a machine to learn patterns and make predictions.



In its simplest form, Artificial Intelligence is a field that combines computer science and robust datasets to enable problem-solving. AI does not replace human decisions; instead, AI adds value to human judgment. Think of AI as a smart helper that can understand things, learn from examples, and do tasks on its own without needing to be told exactly what to do each time. For example, AI can:

- **Understand Language:** AI can understand and respond to what you say, like virtual assistants such as Siri or Alexa.
- **Recognize Images:** AI can look at pictures and recognize what is in them, like identifying animals in photos.
- **Make Predictions:** AI can analyze data to make predictions, like predicting the weather or suggesting what movie you might like to watch next.
- **Play Games:** AI can play games and learn to get better at them, like playing chess or video games.
- **Drive Cars:** AI can help cars drive themselves by sensing the road and making decisions to stay safe.

What is not AI?

When we talk about machines, not all of them are considered Artificial Intelligence (AI). Here are some examples:

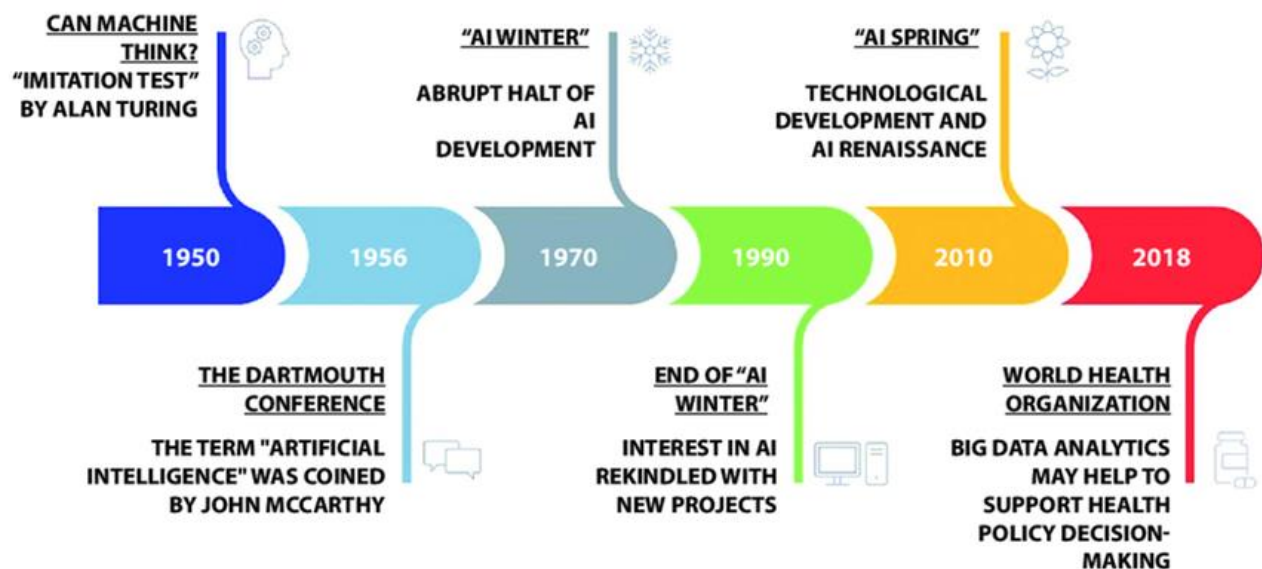
- **Traditional Rule-Based Systems:** These machines follow set rules without learning from data.
- **Simple Automation Tools:** Basic tools like timers or calculators do specific tasks but do not think or learn.
- **Mechanical Devices:** Machines like pulleys or gears work based on physics but do not learn or think.

- **Fixed-Function Hardware:** Devices like microwave ovens perform tasks without learning or thinking.
- **Non-Interactive Systems:** Machines that do not change based on new information, like a basic electric fan.
- **Basic Sensors:** Sensors collect data but do not analyze or understand it.

Artificial Intelligence machines are different. They learn from data and can make decisions on their own. For example, a smart washing machine can adjust its settings based on what it is washing. AI goes beyond just following rules; it can learn, adapt, and make decisions based on data and context.

2. Evolution of AI

The history of AI can be traced back to ancient times, with philosophical discussions about the nature of intelligence and the possibility of creating artificial beings. However, the modern era of AI began in the mid-20th century with significant developments and milestones:



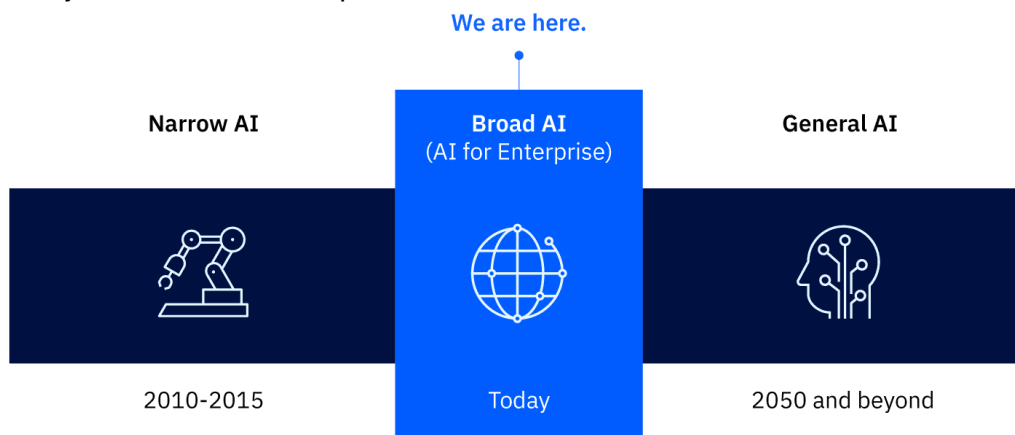
Source: https://www.researchgate.net/figure/Timeline-diagram-showing-the-history-of-artificial-intelligence_fig1_364826401

Time Period	Key Events and Developments
1950	1950 was a landmark year for the question of machine intelligence because of Alan Turing's famous paper "Computing Machinery and Intelligence." In this paper, Turing proposed a thought experiment called the "imitation game" (later known as the Turing test).
1956	The Dartmouth Conference was organized by McCarthy that marked the birthplace of AI as a field. The term "Artificial Intelligence" was coined by John McCarthy. McCarthy, along with Turing, Minsky, and Simon, laid the foundation for AI.
1960-1970	Significant progress in AI research that led to the development of expert systems, early neural networks, exploration of symbolic reasoning, and problem-solving techniques.

Time Period	Key Events and Developments
1980-1990	Mixed optimism and skepticism about AI with breakthroughs in machine learning, and neural networks led to "AI winter".
21st Century	Resurgence of interest and progress in AI with advancements in computing power, data availability, and algorithmic innovation. Also, there were breakthroughs in machine learning, deep learning, and reinforcement learning. That led to transformative applications of AI in healthcare, finance, transportation, and entertainment.

3. Types of AI

Computer scientists have identified three levels of AI based on predicted growth in its ability to analyze data and make predictions.



1. Narrow AI:

- Focuses on single tasks like predicting purchases or planning schedules.
- Rapidly growing in consumer applications, such as voice-based shopping and virtual assistants like Siri.
- Capable of handling specific tasks effectively, but lacks broader understanding.

2. Broad AI:

- Acts as a midpoint between Narrow and General AI.
- More versatile than Narrow AI, capable of handling a wider range of related tasks.
- Often used in businesses to integrate AI into specific processes, requiring domain-specific knowledge and data.

3. General AI:

- Refers to machines that can perform any intellectual task a human can.
- Currently, AI lacks abstract thinking, strategizing, and creativity like humans.
- Artificial Superintelligence (ASI) may emerge, potentially leading to self-aware machines, but this is far from current capabilities.

4. Domains of AI

Artificial Intelligence (AI) encompasses various fields, each focusing on different aspects of replicating human intelligence and performing tasks traditionally requiring human intellect. These fields are classified based on the type of data input they handle:

- a) **Data Science:** Data Science deals with numerical, alphabetical, and alphanumeric data inputs. It involves the collection, analysis, and interpretation of large volumes of data to extract insights and patterns using statistical methods, machine learning algorithms, and data visualization techniques.
- b) **Natural Language Processing (NLP):** NLP focuses on processing text and speech inputs to enable computers to understand, interpret, and generate human language. It involves tasks such as language translation, sentiment analysis, text summarization, and speech recognition, facilitating communication between humans and machines through natural language interfaces.
- c) **Computer Vision:** Computer Vision deals with visual data inputs, primarily images and videos. It enables computers to interpret and understand visual information, and perform tasks such as object detection, image classification, facial recognition, and scene understanding, enabling applications such as autonomous vehicles, medical imaging, and augmented reality.

Activity:

Divide the students into groups and provide them with a list of real-world applications without specifying which domain each application belongs to. Ask each group to categorize the applications into the three domains: Data Science, Natural Language Processing (NLP), and Computer Vision.

1. *Gesture recognition for human-computer interaction*
2. *Chatbots for customer service*
3. *Spam email detection*
4. *Autonomous drones for surveillance*
5. *Google Translate*
6. *Fraud detection in financial transactions*
7. *Augmented reality applications (e.g., Snapchat filters)*
8. *Sports analytics for performance optimization*
9. *Object detection in autonomous vehicles*
10. *Recommendation systems for e-commerce platforms*
11. *Customer segmentation for targeted marketing*
12. *Text summarization for news articles*
13. *Automated subtitles for videos*
14. *Medical image diagnosis*
15. *Stock prediction*

Data Science	Natural Language Processing	Computer Vision

a. Data Science

Data might be facts, statistics, opinions, or any kind of content that is recorded in some format. This could include voices, photos, names, and even dance moves! It surrounds us and shapes our experiences, decisions, and interactions. For example:

- Your search recommendations, Google Maps history are based on your previous data.
- Amazon's personalized recommendations are influenced by your shopping habits.
- Social media activity, cloud storage, textbooks, and more are all forms of data.

It is often referred to as the "**new oil**" of the 21st century. Did you know? 90% of the world's data has been created in just the last 2 years, compared to the previous 6 million years of human existence.

Type of Data

- Structured Data
- Unstructured Data
- Semi-structured Data

Structured data is like a neatly arranged table, with rows and columns that make it easy to understand and work with. It includes information such as names, dates, addresses, and stock prices. Because of its organized nature, it is straightforward to analyze and manipulate, making it a preferred format for many data-related tasks.

On the other hand, **unstructured data** lacks any specific organization, making it more challenging to analyze compared to structured data. Examples of unstructured data include images, text documents, customer comments, and song lyrics. Since unstructured data does not follow a predefined format, extracting meaningful insights from it requires specialized tools and techniques.

Semi-structured data falls somewhere between structured and unstructured data. While not as organized as structured data, it is easier to handle than unstructured data. Semi-structured data uses metadata to identify certain characteristics and organize data into fields, allowing for some level of organization and analysis. An example of semi-structured data is a social media video with hashtags used for categorization, blending structured elements like hashtags with unstructured content like the video itself.

Unstructured data	Semi-structured data	Structured data																								
The university has 5600 students. John's ID is number 1, he is 18 years old and already holds a B.Sc. degree. David's ID is number 2, he is 31 years old and holds a Ph.D. degree. Robert's ID is number 3, he is 51 years old and also holds the same degree as David, a Ph.D. degree.	<pre><University> <Student ID="1"> <Name>John</Name> <Age>18</Age> <Degree>B.Sc.</Degree> </Student> <Student ID="2"> <Name>David</Name> <Age>31</Age> <Degree>Ph.D. </Degree> </Student> </University></pre>	<table><tr><th>ID</th><th>Name</th><th>Age</th><th>Degree</th></tr><tr><td>1</td><td>John</td><td>18</td><td>B.Sc.</td></tr><tr><td>2</td><td>David</td><td>31</td><td>Ph.D.</td></tr><tr><td>3</td><td>Robert</td><td>51</td><td>Ph.D.</td></tr><tr><td>4</td><td>Rick</td><td>26</td><td>M.Sc.</td></tr><tr><td>5</td><td>Michael</td><td>19</td><td>B.Sc.</td></tr></table>	ID	Name	Age	Degree	1	John	18	B.Sc.	2	David	31	Ph.D.	3	Robert	51	Ph.D.	4	Rick	26	M.Sc.	5	Michael	19	B.Sc.
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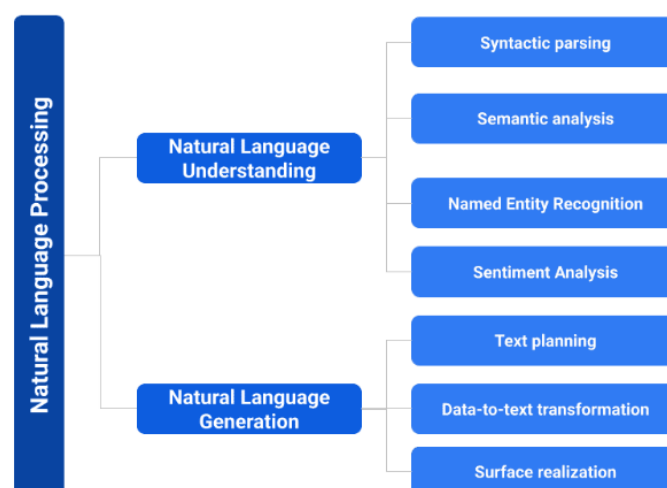
Source: https://www.researchgate.net/figure/Unstructured-semi-structured-and-structured-data_fig4_236860222

b. Natural Language Processing:

It refers to the field of computer science and AI that focuses on teaching machines to understand and process languages in both written and spoken form, just like humans do.

The goal of an NLP-Trained model is to be capable of “understanding” the contents of documents, including the slangs, sarcasm, inner meaning, and contextual definitions of the language in which the text was written.

Differences Between NLP, NLU, and NLG?



Source: <https://www.baeldung.com/cs/natural-language-processing-understanding-generation>

Natural Language Processing (NLP): This is the broad umbrella term encompassing everything related to how computers interact with human language. Think of it as the "what" - what computers can do with human language. *It is like the whole library - filled with different tools and techniques for working with language data.*

Natural Language Understanding (NLU): This is a subfield of NLP that focuses on understanding the meaning of human language. It analyzes text and speech, extracting information, intent, and sentiment. NLU helps computers understand the language and what it means. *Imagine finding a specific book in the library.*

Natural Language Generation (NLG): This is another subfield of NLP, but instead of understanding, it focuses on generating human language. It takes structured data as input and turns it into coherent and readable text or speech. *Think of this as writing a new book based on the information gathered in the library.*

c. Computer Vision:

Computer Vision is like giving computers the ability to see and understand the world through digital images and videos, much like how humans use their eyes to perceive their surroundings. In this domain, computers analyze visual information from images and videos to recognize objects, understand scenes, and make decisions based on what they "see."

When we take a digital image, it is essentially a grid of tiny colored dots called **pixels**. Each pixel represents a tiny portion of the image and contains information about its color and intensity.

Resolution is expressed as the total number of pixels along the width and height of the image. For example, an image with a resolution of 1920x1080 pixels has 1920 pixels horizontally and 1080 pixels vertically. Higher resolution images have more pixels, providing more detail.

Now, here's where AI comes in. To make sense of these images, computers convert them into numbers. They break down the image into a series of numbers that represent the color and intensity of each pixel. This numerical representation allows AI algorithms to process the image mathematically and extract meaningful information from it. For instance, AI algorithms might learn to recognize patterns in these numbers that correspond to specific objects, like cars or faces. By analyzing large amounts of labeled image data, AI systems can "learn" to identify objects accurately.

Cognitive Computing (Perception, Learning, Reasoning)

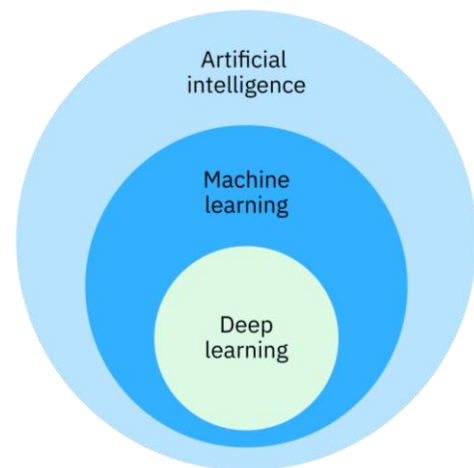
Cognitive Computing is a branch of Artificial Intelligence (AI) that aims to mimic the way the human brain works in processing information and making decisions. It involves building systems that can understand, reason, learn, and interact with humans in a natural and intuitive way.

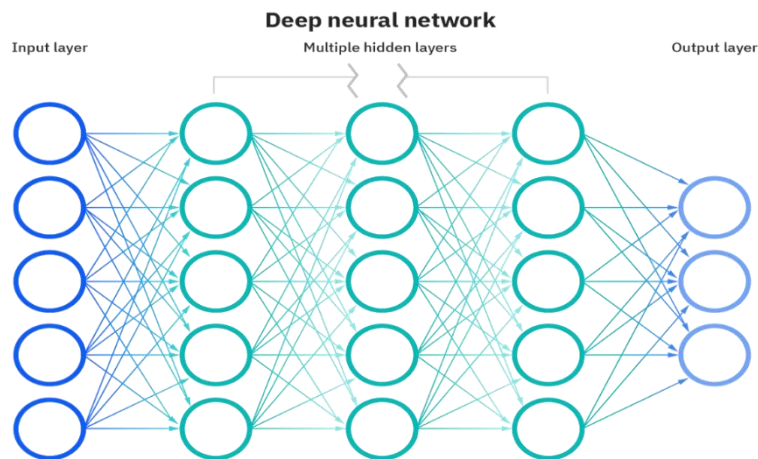
1.This is a platform based on Artificial Intelligence and Signal processing.	2. The platform (Cognitive computing) uses Machine Learning, Reasoning, Natural Language Processing (NLP) and Computer Vision to compute results.
3.Cognitive computing improves human decision making	4.Cognitive computing tries to mimic the human brain
Examples of cognitive computing software: IBM Watson, Deep mind, Microsoft Cognitive service etc.	

In summary, Cognitive Computing integrates Data Science, Natural Language Processing, and Computer Vision to create intelligent systems that can understand and interact with humans in a human-like manner. By combining these technologies, Cognitive Computing enables machines to process and interpret diverse types of data, communicate effectively in natural language, and perceive and understand visual information, thereby extending the capabilities of traditional AI systems.

5. AI Terminologies

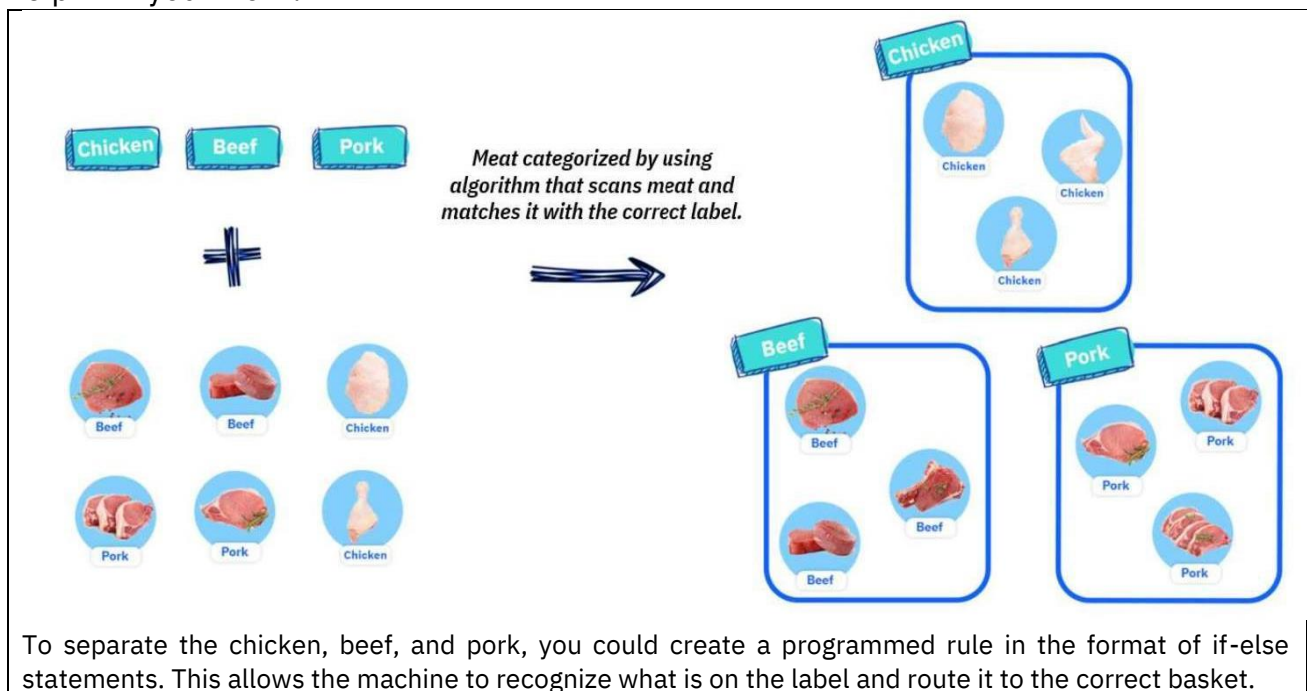
- **Artificial intelligence** machines don't think. They calculate. They represent some of the newest, most sophisticated calculating machines in human history. It is a computer system that can perform tasks that ordinarily require human intelligence or human interference.
- Some can perform what is called **machine learning** as they acquire new data. Machine learning is a subset of artificial intelligence (AI) that focuses on developing algorithms and models that enable computers to learn from data and make predictions or decisions without being explicitly programmed.
- Others, using calculations arranged in ways inspired by neurons in the human brain, can even perform **deep learning** with multiple levels of calculations. Deep learning is an AI function that imitates the working of the human brain in processing data and creating patterns for use in decision making.
 - The structure of Deep Learning is inspired by the structure of the neurons and neuron connections in the human brain.
 - Neural networks, also known as Artificial Neural Networks (ANNs), are a subset of Machine Learning and the core heart and concept of Machine Learning.
 - They comprise of node layers, containing an input layer, one or multiple hidden layers, and an output layer.
 - If the output of any node is above a specified threshold, that node is activated, sending data to the next layer of the network.
 - Otherwise, no data is passed along to the next layer of the network.
 - If the number of Layers including the Input and Output Layer is more than three, then it is called a Deep Neural Network.

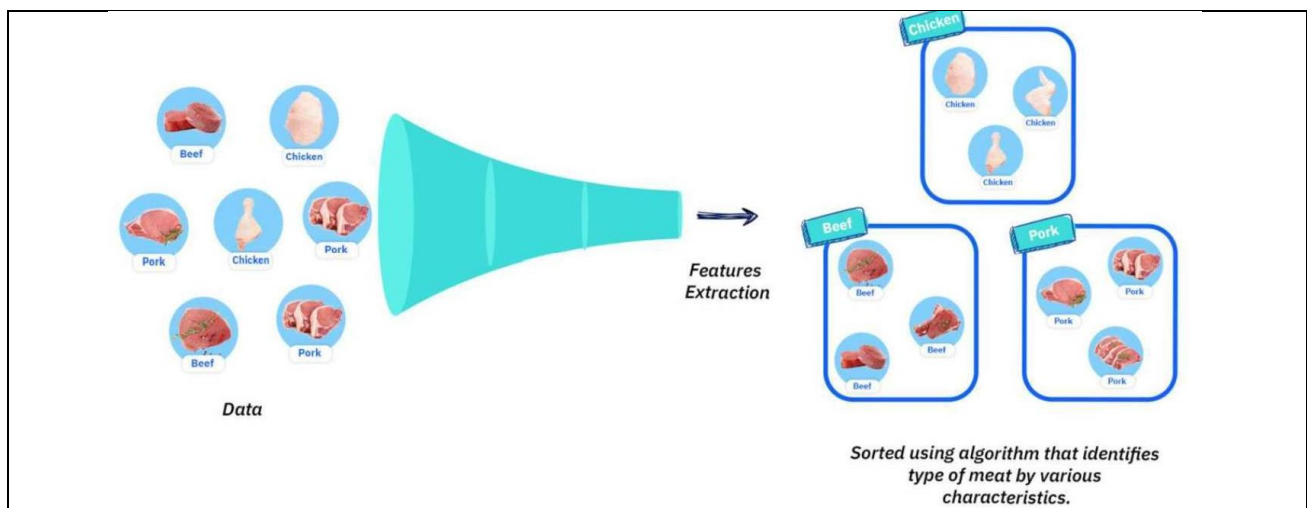




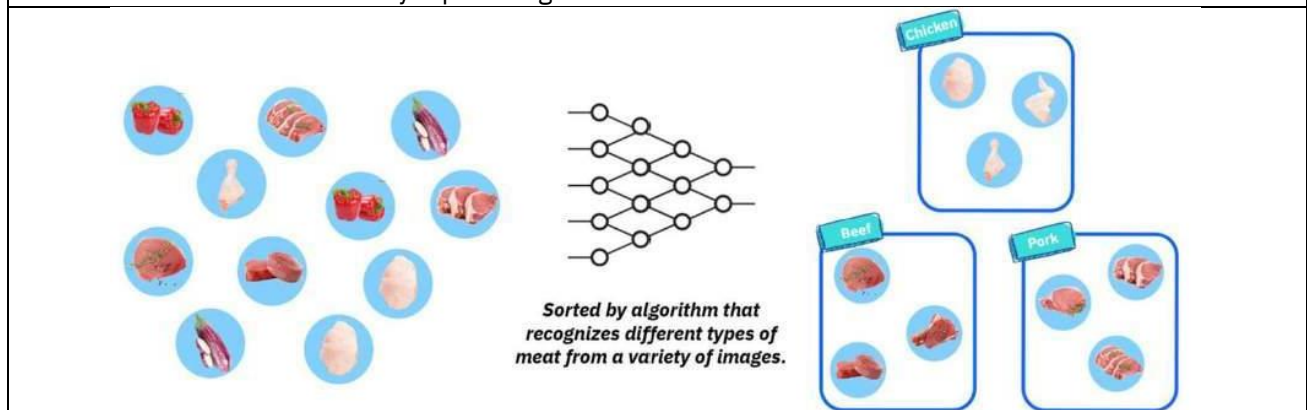
MACHINE LEARNING	DEEP LEARNING
1. Works on small dataset for accuracy	1. Works on Large dataset
2. Dependent on Low-end machine	2. Heavily dependent on high-end machine
3. Divides the tasks into sub-tasks, solves them individually and finally combine the results	3. Solves problem end to end
4. Takes less time to train	4. Takes longer time to train
5. Testing time may increase	5. Less time to test the data

Example: Imagine you are given the job to sort items in the meat department at a grocery store. You realize that there are dozens of products and very less time to sort them manually. How will you use artificial intelligence, machine learning, and deep learning to help with your work?



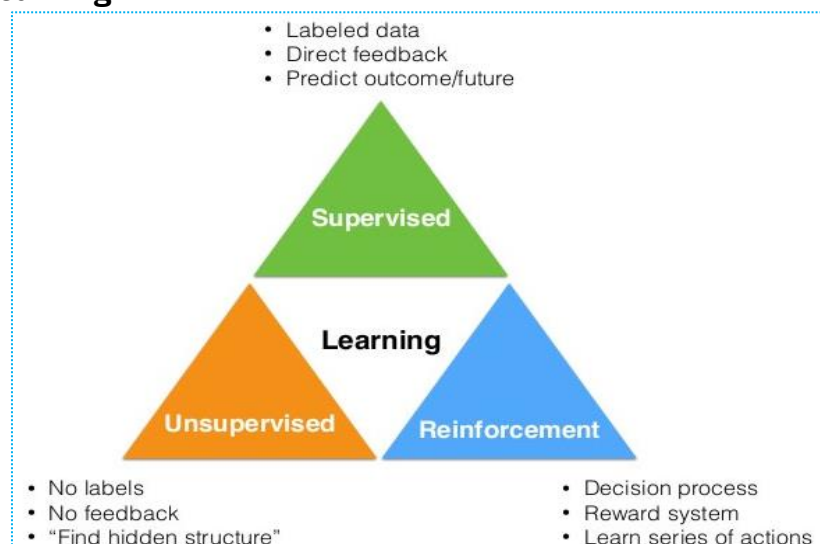


To improve the performance of the machine, you expose it to more data to ensure that the machine is trained on numerous characteristics of each type of meat, such as size, shape, and color. The more data you provide for the algorithm, the better the model gets. By providing more data and adjusting parameters, the machine minimizes errors by repetitive guess work.

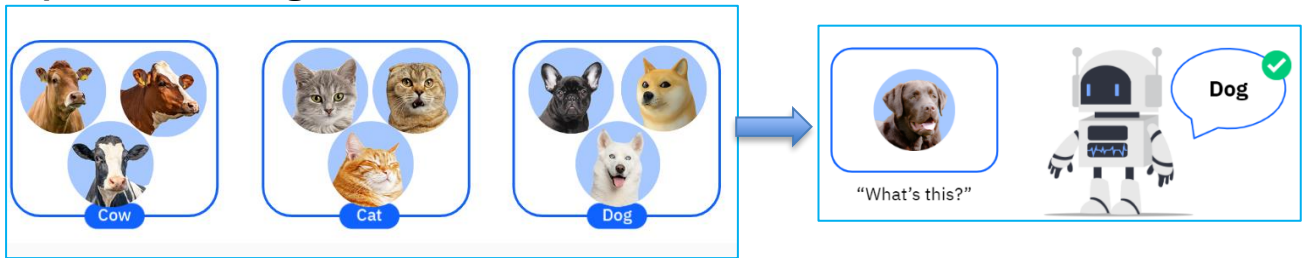


Deep learning models eliminate the need for feature extractions. Decide the algorithms based on deep learning to sort meat by removing the need to define what each product looks like. Feature extraction is built into the process without human input. Once you have provided the deep learning model with dozens of meat pictures, it processes the images through different layers of neural networks. The layers can then learn an implicit representation of the raw data on their own.

Types of Machine Learning

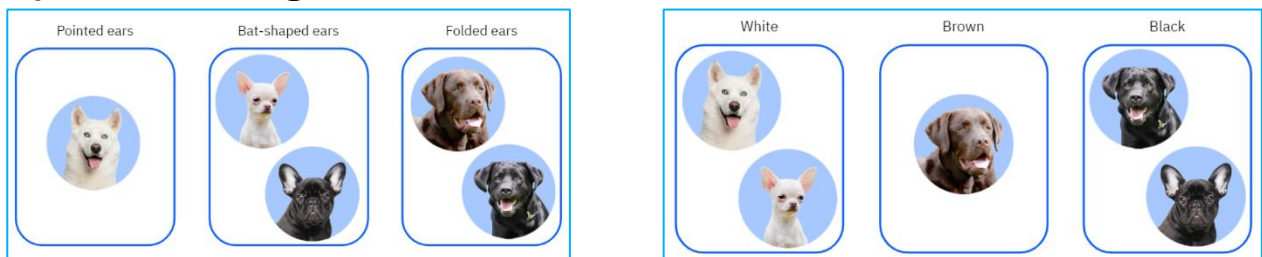


Supervised learning



- Supervised learning is a type of machine learning where the model learns from labelled data, which means that the input data is accompanied by the correct output.
- In supervised learning, the algorithm learns to map input data to output labels based on example input-output pairs provided during the training phase.
- The goal of supervised learning is to learn a mapping function from input variables to output variables so that the model can make predictions on unseen data.
- Examples of supervised learning algorithms include linear regression, logistic regression, decision trees, support vector machines (SVM), and neural networks.

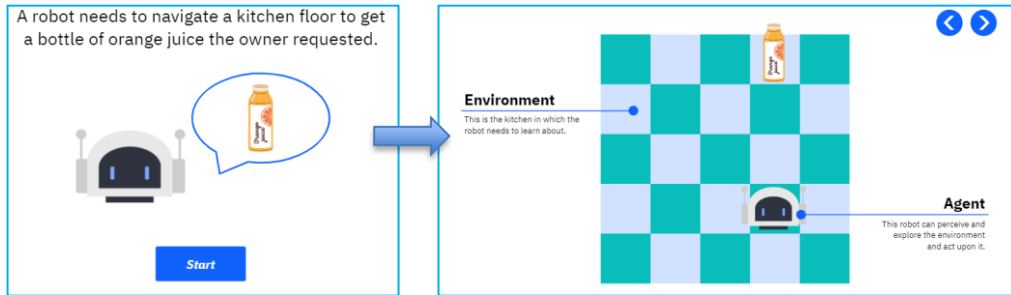
Unsupervised Learning:



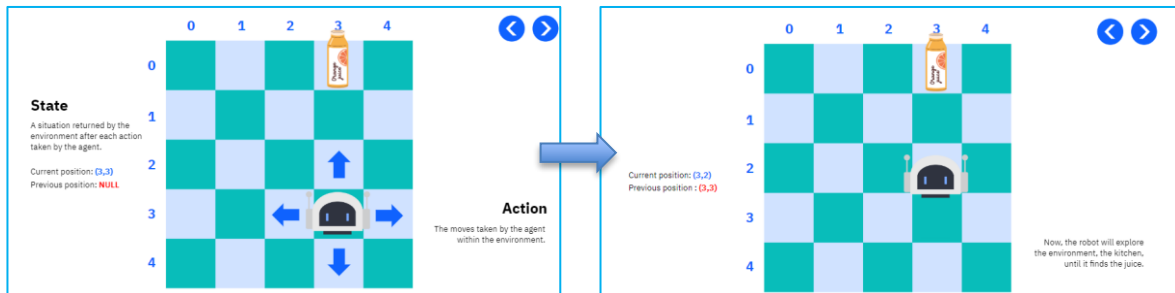
- Unsupervised learning is a type of machine learning where the model learns from unlabelled data, which means that the input data is not accompanied by the correct output.
- In unsupervised learning, the algorithm tries to find hidden patterns or structure in the input data without explicit guidance.
- The goal of unsupervised learning is to explore and discover inherent structures or relationships within the data, such as clusters, associations, or anomalies.
- Examples of unsupervised learning algorithms include k-means clustering, hierarchical clustering, principal component analysis (PCA), and autoencoders.

Reinforcement Learning:

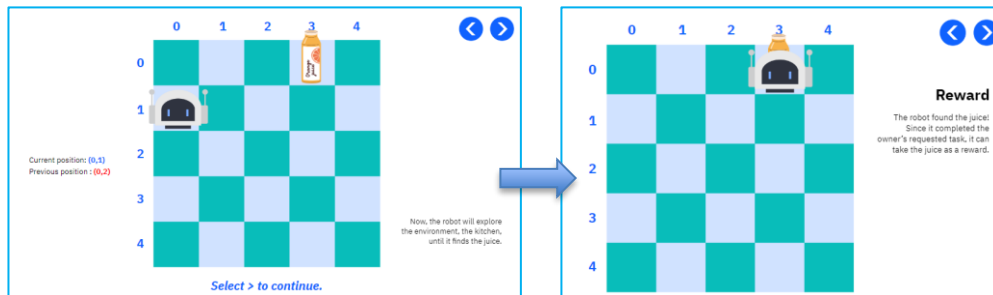
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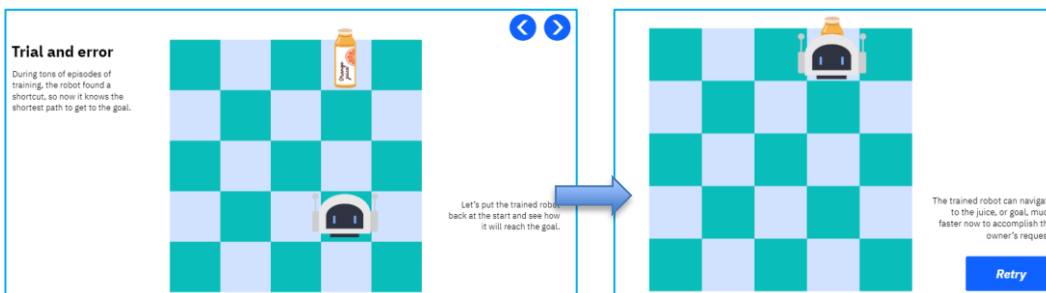
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3.



4.



- Reinforcement learning is a type of machine learning where an agent learns to make decisions by *interacting with an environment* to maximize cumulative rewards.
- In reinforcement learning, the agent learns through *trial and error* by taking actions and receiving feedback from the environment in the form of rewards or penalties.
- The goal of reinforcement learning is to learn a policy or strategy that guides the agent to take actions that lead to the highest cumulative reward over time.
- Reinforcement learning is commonly used in scenarios where the agent must make a sequence of decisions over time, such as playing games, controlling robots, or managing financial portfolios.
- Examples of reinforcement learning algorithms include Q-learning, deep Q-networks (DQN), policy gradients, and actor-critic methods.

6. Benefits and limitations of AI

BENEFITS:

1. **Increased efficiency and productivity:** AI automates tasks, analyzes data faster, and optimizes processes, leading to increased efficiency and productivity across various sectors.
2. **Improved decision-making:** AI analyzes vast amounts of data and identifies patterns that humans might miss, assisting in data-driven decision-making and potentially leading to better outcomes.
3. **Enhanced innovation and creativity:** AI tools can generate new ideas, explore possibilities, and automate repetitive tasks, freeing up human resources for more creative pursuits and innovation.
4. **Progress in science and healthcare:** AI aids in drug discovery, medical diagnosis, and personalized medicine, contributing to advancements in healthcare and scientific research.

LIMITATIONS:

1. **Job displacement:** Automation through AI raises concerns about job displacement and the need for workforce retraining and upskilling.
 2. **Ethical considerations:** Concerns exist around bias in AI algorithms, potential misuse for surveillance or manipulation, and the need for ethical guidelines and regulations.
 3. **Lack of explainability:** Some AI models, particularly complex ones, lack transparency in their decision-making, making it difficult to understand how they arrive at their outputs.
 4. **Data privacy and security:** Large-scale data collection and use for AI development raise concerns about data privacy and security vulnerabilities.
- **Earn a credential on IBM Skills Build on the topic Artificial Intelligence Fundamentals using the link:**
<https://students.yourlearning.ibm.com/activity/PLAN-CC702B39D429>
 - *Semantris*, is an NLP-Based game by Google based on Word association powered by semantic search.
<https://experiments.withgoogle.com/semantris>
 - This is a game built with machine learning. We **draw**, and a neural network tries to guess what you're **drawing**.
<https://quickdraw.withgoogle.com/>
 - *The experiment based on the computer vision domain of AI. It identifies that you draw and suggests the related images. To play the game, visit the following link on any computing device with speakers.*
<https://www.autodraw.com/>

THE GLOBAL DEMAND

Artificial Intelligence (AI) was once confined to the realms of science fiction, but today, it permeates our daily lives in ways we often take for granted. From personalized recommendations on streaming platforms to the algorithms powering autonomous vehicles, AI has become an indispensable part of the modern society. As the field continues to evolve and expand, so do the opportunities it presents for career growth and development.

Amidst the concerns about automation and job displacement, it is essential to recognize the significant demand for AI professionals across various industries. While it is true that AI technologies may replace some traditional roles, they also create a multitude of new and exciting career paths. *Rather than viewing AI as a threat, individuals should embrace it as an opportunity for advancement and innovation.*

Image Source: <https://media.licdn.com>



The global demand for AI talent is skyrocketing, driven by the rapid pace of technological advancements and the increasing integration of AI solutions into diverse sectors. From healthcare and finance to transportation and retail, organizations are harnessing the power of AI to streamline

operations, optimize processes, and deliver enhanced services to consumers. One of the most significant advantages of pursuing a career in AI is the sheer breadth of opportunities it offers. Whether you are passionate about machine learning, natural language processing, robotics, or data analytics, there is a niche within the AI field suited to your interests and skills. Moreover, as AI technologies continue to mature, new specialties and job roles are emerging, creating avenues for specialization and expertise.

SOME COMMON JOB ROLES IN AI:

In today's market, there is a wide range of job roles within the field of Artificial Intelligence (AI) that are in high demand. Some common job roles include:

- **Machine Learning Engineer:** Machine learning engineers bridge software engineering and data science, utilizing big data tools and programming frameworks to develop scalable data science models capable of handling vast volumes of real-time data. Strong mathematical skills, experience in machine learning and deep learning, and proficiency in programming languages like Java, Python, and Scala are essential for success in this role.
- **Data Scientist:** Data scientists leverage machine learning and predictive analytics to extract insights from large datasets, to take proper business decisions. Proficiency in big data platforms like Hadoop, Pig, and Spark, fluency in programming languages such as SQL, Python, and Scala, and a solid understanding of descriptive and inferential statistics are the key requirements for this role.

- **Business Intelligence Developer:** Business intelligence (BI) developers design, model, and maintain complex data sets to analyse business and market trends, enhance organizational profitability and efficiency. Strong technical and analytical skills, along with expertise in data warehouse design and BI technologies, are essential for success in this role.
- **Robotics Engineer:** They design and maintain AI-powered robots, develop mechanical devices capable of performing tasks with human commands. Proficiency in programming, along with expertise in disciplines like robotic engineering, mechanical engineering, and electrical engineering, is crucial for success in this field.
- **Software Engineer:** AI software engineers build and maintain software products for AI applications, staying updated on the latest artificial intelligence technologies. Proficiency in software engineering, programming languages, and statistical analysis is essential, typically requiring a bachelor's degree in computer science, engineering, or related fields.



- **Natural Language Processing (NLP) Engineer:** NLP engineers specialize in human language processing, working on voice assistants, speech recognition, and document processing. A specialized degree in computational linguistics or a combination of computer science, mathematics, and statistics is typically required for this role.
- **Computer Vision Engineer:** Computer vision engineers specialize in developing algorithms and systems that enable computers to analyse and interpret visual information from images or videos. Their expertise lies in creating software solutions that can understand and process visual data, requiring proficiency in image processing techniques and programming languages such as Python and C++.
- **AI Ethicist:** AI ethicists address ethical considerations and implications related to the development and deployment of AI technologies, ensuring that they are used responsibly and ethically. They provide guidance on ethical frameworks, policies, and practices to promote fairness, transparency, and accountability in AI systems, often requiring a background in ethics, philosophy, or law, combined with expertise in AI technology.
- **AI Consultant:** AI consultants offer expert guidance and advice to organizations on how to leverage AI technologies to solve business challenges and drive innovation. They assess business needs, identify opportunities for AI integration, and develop strategic AI initiatives, requiring a deep understanding of AI technologies, business processes, and industry trends, along with strong communication & analytical skills.

Activity 1:

Divide the class into small groups and distribute the list of AI job roles to each group. Using the roles written in the chart, the teams will identify ten companies currently hiring employees for those specific AI positions.

ESSENTIAL SKILLS AND TOOLS FOR PROSPECTIVE AI CAREERS

A successful career in artificial intelligence requires a diverse set of skills that encompass both technical expertise and soft skills. According to industry leaders, here are some of the top skills that AI professionals need:

Technical Skills:

- Expertise in neural networks, machine learning, and deep learning is essential for developing advanced AI applications.
- Knowledge of big data technologies and techniques for handling and analysing large datasets is crucial in AI applications.
- Understanding of frameworks and libraries like TensorFlow, SciPy, and NumPy is vital for building and deploying AI solutions.
- Familiarity with programming languages such as Python, R, Java, and C++ is necessary for developing AI models and algorithms.
- Proficiency in linear algebra, probability, statistics, and signal processing is essential for understanding the mathematical principles underlying AI algorithms.



Soft Skills:

- Effective communication skills are crucial for conveying complex technical concepts to non-technical stakeholders and collaborating with multidisciplinary teams.
- Strong teamwork and collaboration abilities are essential for working effectively in cross-functional teams to develop AI products and solutions.
- Problem-solving, decision-making, and analytical thinking skills are critical for identifying and addressing challenges in AI projects.
- Time management and organizational skills are essential for managing multiple projects and meeting deadlines.
- Business intelligence and critical thinking skills are valuable for understanding business requirements and translating them into AI solutions that deliver tangible value.



Your Professional Toolkit

In addition to acquiring the necessary skills, it is essential for AI professionals to familiarize themselves with popular AI tools, platforms, and programming languages. Here are some essential tools and their purposes:

- **Python:** A versatile programming language with pre-made libraries for advanced computing and scientific computation.
- **R:** A programming language for data collection, organization, and analysis, particularly useful for machine learning and statistical functions.
- **Java:** Widely used in AI for implementing intelligence programming, neural networks, and machine learning solutions.
- **C++:** Known for its flexibility and object-oriented functions, used for procedural programming and hardware manipulation in AI.
- **TensorFlow:** An open-source machine learning platform with tools and libraries for developing sophisticated AI applications.
- **SciPy and NumPy:** Python libraries for scientific computing and mathematical operations, ideal for manipulating and visualizing data.

By acquiring the right combination of technical skills and tools, aspiring AI professionals can position themselves for success in this dynamic and rapidly growing field. Whether you are interested in developing AI algorithms or implementing AI solutions in real-world applications, building a strong foundation of skills and expertise is the key to unlocking exciting career opportunities in artificial intelligence.

Activity 2:

In continuation with the previous activity, list the technical skills and soft skills listed by any two companies for the specific AI position.

Technical Skills	Soft Skills	Link to the website

OPPORTUNITIES IN AI ACROSS VARIOUS INDUSTRIES

Artificial intelligence professionals design and develop AI systems that use machine learning and neural networks to predict trends, provide better customer experiences and recommendations, and offer solutions to difficult problems. While some AI professionals work towards the goal of General AI—systems interconnected and able to be nearly as creative as human beings—others focus on narrower applications. This following table gives you a variety of opportunities to choose from depending upon your choice of subject.

Industry	Employment Opportunities	Some Existing/Expected Job Roles	Relevant Subjects in School
Automobile	Design, manufacturing, and sale of motor vehicles.	1. Autonomous Vehicle Engineer: Develops AI algorithms for self-driving cars. 2. Simulation Engineer: Creates virtual environments for testing autonomous vehicle technologies. 3. Robotics Engineer: Designs AI-powered robots for automotive tasks.	Mathematics, Physics, Computer Science/Artificial Intelligence
Agriculture	Monitoring crop health, optimizing irrigation, and maximizing yields.	1. Precision Agriculture Specialist: Uses AI-powered drones and sensors for monitoring crops. 2. Crop Yield Prediction Analyst: Forecasts crop yields using AI models. 3. Livestock Monitoring Specialist: Tracks the health and productivity of farm animals.	Biology, Mathematics, Computer Science/Artificial Intelligence
Retail	Optimizing inventory, sales forecasting, and enhancing customer experience.	1. Inventory Management Specialist: Optimizes inventory levels using AI algorithms. 2. Sales Forecasting Analyst: Forecasts sales using AI models. 3. Customer Experience Designer: Enhances customer experience using AI-driven insights.	Business Studies, Mathematics, Computer Science/Artificial Intelligence
Media	Creating visual effects, content generation, and audience analysis.	1. Visual Effects Artist: Uses AI tools for creating visual effects. 2. Content Creator: Generates content using AI-generated insights. 3. Audience Analyst: Analyzes audience behavior using AI algorithms.	Fine Arts, Media Studies, Computer Science/Artificial Intelligence
Information Technology	Developing AI algorithms, systems, and infrastructure for various applications.	1. Machine Learning Engineer: Develops AI algorithms and systems. 2. AI Software Developer: Builds AI-powered applications. 3. AI Infrastructure Specialist: Maintains and optimizes AI infrastructure.	Computer Science/Artificial Intelligence, Mathematics, Physics

Healthcare	Medical imaging analysis, personalized healthcare, and drug discovery.	1. Medical Imaging Analyst: Analyzes medical images using AI algorithms. 2. Virtual Nurse Assistant: Provides personalized healthcare recommendations. 3. Drug Discovery Researcher: Identifies potential drug candidates using AI.	Biology, Chemistry, Computer Science/Artificial Intelligence
Finance	Market analysis, fraud detection, risk management, and investment recommendations.	1. Quantitative Analyst: Analyzes market trends using AI algorithms. 2. Fraud Detection Analyst: Identifies fraudulent activities using AI models. 3. Financial Advisor: Offers personalized investment recommendations using AI-driven analytics.	Economics, Mathematics, Computer Science/Artificial Intelligence
Government & Military	Surveillance, predictive analytics, citizen services, and military technologies.	1. National Security Analyst: Uses AI-powered surveillance systems. 2. Defense Contractor: Develops AI-enabled military technologies. 3. Government AI Specialist: Implements AI for citizen services and regulatory compliance.	Political Science, Computer Science/Artificial Intelligence, Mathematics
Tourism	Personalized travel recommendations, customer service, and itinerary planning.	1. Travel Recommendation Engine Developer: Provides personalized travel recommendations. 2. Chatbot for Customer Service: Assists travelers with booking and inquiries. 3. Smart Travel Itinerary Planner: Optimizes travel routes and schedules.	Geography, Business Studies, Computer Science/Artificial Intelligence
Beauty & Wellness	Skincare analysis, virtual styling, and wellness guidance.	1. AI-powered Skincare Assistant: Provides personalized skincare recommendations. 2. Virtual Hair Stylist: Simulates different hairstyles using AI. 3. Wellness Chatbot: Offers guidance on nutrition and fitness.	Chemistry, Biology, Computer Science/Artificial Intelligence
Banking	Loan approval automation, fraud detection, and personalized financial advice.	1. Loan Approval Specialist: Automates loan approval process using AI. 2. Fraud Detection Analyst: Identifies fraudulent transactions using AI algorithms. 3. Financial Advisor: Offers personalized financial advice using AI-driven analytics.	Economics, Mathematics, Computer Science/Artificial Intelligence
Geospatial	Spatial data analysis, remote sensing, and mapping technologies.	1. Geographic Information Systems (GIS) Specialist: Analyzes spatial data using AI. 2. Remote Sensing Analyst: Interprets satellite imagery using AI algorithms.	Geography, Geology, Computer Science/Artificial Intelligence

		3. Mapping Technician: Uses AI-enabled drones for mapping.	
Textile	Fabric design, quality control, and inventory management.	1. AI-powered Fabric Design Specialist: Creates innovative textile patterns using AI. 2. Textile Quality Control Inspector: Ensures product quality using AI-enabled systems. 3. Smart Inventory Management Specialist: Optimizes inventory levels using AI algorithms.	Chemistry, Art & Design, Computer Science/Artificial Intelligence
Design	Design optimization, user experience enhancement, and content creation.	1. Generative Design Assistant: Optimizes design solutions using AI algorithms. 2. AI-powered UX Designer: Enhances user experience using AI-driven insights. 3. AI-powered Content Creator: Generates content using AI tools.	Art & Design, Computer Science/Artificial Intelligence, Mathematics
Sales & Marketing	Campaign automation, customer segmentation, and sales forecasting.	1. Marketing Campaign Automation Specialist: Automates marketing campaigns using AI. 2. Customer Segmentation Analyst: Segments customers based on behavior using AI. 3. Sales Forecasting Analyst: Forecasts sales using AI models.	Business Studies, Mathematics, Computer Science/Artificial Intelligence
Fashion	Personalized fashion recommendations, trend analysis, and virtual try-on.	1. AI-powered Fashion Stylist: Recommends personalized clothing combinations using AI. 2. Trend Analyst: Analyzes fashion trends using AI algorithms. 3. Virtual Clothing Try-on Specialist: Allows virtual try-on of clothing using AI and AR.	Fashion Design, Mathematics, Computer Science/Artificial Intelligence

ADDITIONAL LEARNING RESOURCES:

Here are some resources you can explore, bookmark, and keep in mind if you would like to explore more about AI and stay in touch with the latest developments in the field. This is a curated listing. There are many organizations and websites to explore, depending on your interests.

News and blogs to stay current

- [Analytics Insight](#) offers insights, latest news, and a magazine featuring opinions and views of top industry leaders and executives who share their journeys, experiences, success stories, and knowledge to grow profitable businesses.
- [Towards Data Science](#) is an online publication in which independent authors who follow their rules and guidelines can publish their work, share their knowledge and expertise, and engage a wide audience on [Medium](#).

- [KDnuggets](#) is a leading site on data science, machine learning, AI, and analytics. It contains excellent tutorial materials, courses, webinars, online events.
- [Data Science Central](#) is a leading online resource for data practitioners. From statistics and analytics to machine learning and AI, Data Science Central provides a community experience that includes a rich editorial platform, social interaction, forum-based support, and the latest information on technology, tools, trends, and careers.
- [Datanami](#) is a news portal dedicated to providing insight, analysis, and up-to-the-minute information about emerging trends and solutions in big data.

Free learning opportunities to build skills

Note: You will need to sign up for a free account for the following online learning opportunities.

- You can take advantage of [IBM SkillsBuild](#) to power your future in tech with job skills, courses, digital credentials, and more.
- [Kaggle](#) offers free online micro courses to help you gain the skills you need to do independent data science projects. Kaggle also allows you to grow your data science and machine learning skills by competing in Kaggle competitions. Find help in the [documentation](#) or learn about [Community Competitions](#).
- Udemy offers a variety of free video-based courses on artificial intelligence, including a short, practical hands-on course on artificial intelligence, called [Kickstart Artificial Intelligence](#). Udemy also offers a course called [Artificial Intelligence: Preparing Your Career for AI](#), which covers what you should be doing now to prepare for the coming of AI.
- [freeCodeCamp.org](#) offers a rundown of [All the Math You Need to Know in Artificial Intelligence](#). Jason Dsouza gives you an overview of the core math principles you need to focus on to work in AI.
- [DataCamp](#) offers a free, two-hour [Machine Learning for Everyone](#) course which introduces machine learning without coding involved.
- [W3Schools](#) is the world's largest web developer site that offers a variety of free online tutorials with hands-on practice. The site includes tutorials on some popular data science programming languages, such as [Python](#), [R](#), and [SQL](#).
- [Codecademy](#) offers free coding classes on 12 different programming languages including Python, Java, and C++.

Additional information regarding colleges offering professional course in AI.

- ✓ IIT Madras – Four year Bachelor of Science Degree in Data Science and Applications
<https://study.iitm.ac.in/ds/>
- ✓ AICTE – All India Council for Technical Education's unique website – “Digital Skilling”. Explore this site for a wide variety of course and internships. <https://1crore.aicte-india.org/>
- ✓ Most of the top colleges in India now offer B.Tech courses in AI and ML, Data Science, Robotics and Computer Science with specializations. Students can visit college websites to know more about these courses. Also, some colleges are offering BSc in AI and ML.

REFERENCES : [IBM Skills Build](#)



Introduction to Python

Python is a general-purpose, high level programming language. It was created by Guido van Rossum, and released in 1991. Python got its name from a BBC comedy series – “Monty Python’s Flying Circus”



Features of Python

- High Level language
- Interpreted Language
- Free and Open Source
- Platform Independent (Cross-Platform) – runs virtually in every platform if a compatible python interpreter is installed.
- Easy to use and learn – simple syntax similar to human language.
- Variety of Python Editors – Python IDLE, PyCharm, Anaconda, Spyder
- Python can process all characters of ASCII and UNICODE.
- Widely used in many different domains and industries.



Python Editors

There are various editors and Integrated Development Environments (IDEs) that you can use to work with Python. Some popular options are PyCharm, Spyder, Jupyter Notebook, IDLE etc. Let us look how we can work with Jupyter Notebook.

Jupyter Notebook is an open-source web application that allows you to create and share documents containing live code, equations, visualizations, and narrative text. It's widely used in data science and research. It can be installed using Anaconda or with pip. For more details of installation use the link

<https://docs.jupyter.org/en/latest/install/notebook-classic.html>

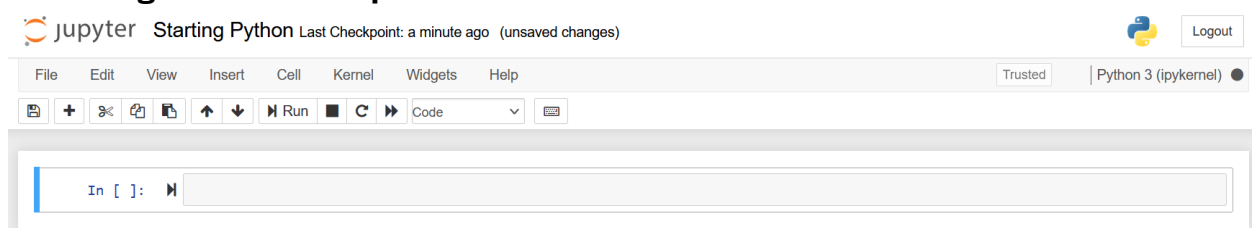
Those who are familiar with Python, open the command prompt in administrative mode and type

```
pip install notebook
```

To run the notebook, Open the command prompt and type

```
jupyter notebook
```

Following window will open



You can type the code in the cell provided. Then click  to see the output just below it.

```
In [1]: ▶ print ( " Hai all this is how to display a text")
```

Hai all this is how to display a text

```
In [4]: ▶ #To Add two numbers
num1=10
num2=20
print (num1 + num2)
```

30

Output – We can work on multiple programs in same window



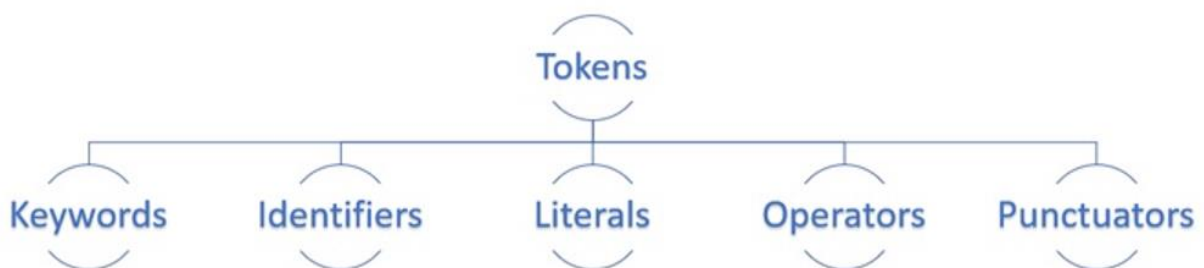
Point to be Noted

Another way to install and start using Jupyter Notebook is through Anaconda. It is the most widely used Python distribution for data science and comes pre-loaded with all the most popular libraries and tools. Jupyter notebook can easily be accessed using the Anaconda Prompt with the help of a local host.



Getting Started with Python Programs

Python program consists of Tokens. It is the smallest unit of a program that the interpreter or compiler recognizes. Tokens consist of keywords, identifiers, literals, operators, and punctuators. They serve as the building blocks of Python code, forming the syntactic structure that the interpreter parses and executes. During lexical analysis, the Python interpreter breaks down the source code into tokens, facilitating subsequent parsing and interpretation processes.



<https://www.studytrigger.com/wp-content/uploads/2022/08/Tokens-in-Python.jpg>

Keywords

Reserved words used for special purpose. List of keywords are given below.

False	None	True	for	in	or	while
and	class	elif	from	is	pass	with
as	continue	else	global	lambda	raise	yield
assert	def	except	if	nonlocal	return	async
break	del	finally	import	not	try	await

Identifier

An identifier is a name used to identify a variable, function, class, module or other object. Generally, keywords (list given above) are not used as variables. Identifiers cannot start with digit and also it can't contain any special characters except underscore.

Literals:

Literals are the raw data values that are explicitly specified in a program. Different types of Literals in Python are String Literal, Numeric Literal (Numbers), Boolean Literal (True & False), Special Literal (None) and Literal Collections.

Operators:

Operators are symbols or keywords that perform operations on operands to produce a result. Python supports a wide range of operators:

- | | |
|---|--|
| <ul style="list-style-type: none">• Arithmetic operators (+, -, *, /, %)• Relational operators (==, !=, <, >, <=, >=)• Assignment operators (=, +=, -=) | <ul style="list-style-type: none">• Logical operators (and, or, not)• Bitwise operators (&, , ^, <<, >>)• Identity operators (is, is not)• Membership operators (in, not in) |
|---|--|

Punctuators:

Common punctuators in Python include

: () [] { } , ; . ` ' ' ' " " / \ & @ ! ? | ~ etc.

Example

In [2]:	<pre>#Finding Square root of a number with the function sqrt() of math Library import math num = 625 root = math.sqrt(num) print("Square root= ", root)</pre>
output	Square root= 25.0

Tokens in the above program are given below

- Keyword - import
- Identifier - num , root (Here it can be said as variables also)
- Literal - 625
- Operator - =
- Punctuator - " " , () .

Point to be Noted

In the above program

-  `print ()` is used to display the output on the screen
-  `#` symbol is used to write comments which are used to increase readability and will not be executed
-  `import` statement is used to load the functions from the library (math)
-  Variables – Named labels whose value can be used and processed during the execution of the program.

Sample Program-1

Display the string “National Animal-Tiger” on the screen

```
In [1]: ▶ print("National animal - Tiger")
```

output

National animal - Tiger

Sample Program-2

Write a program to calculate the area of a rectangle given the length and breadth are 50 and 20 respectively.

```
In [1]: ▶ length = 12  
        breadth = 7  
        area = length * breadth  
        print("Area of Rectangle=", area)
```

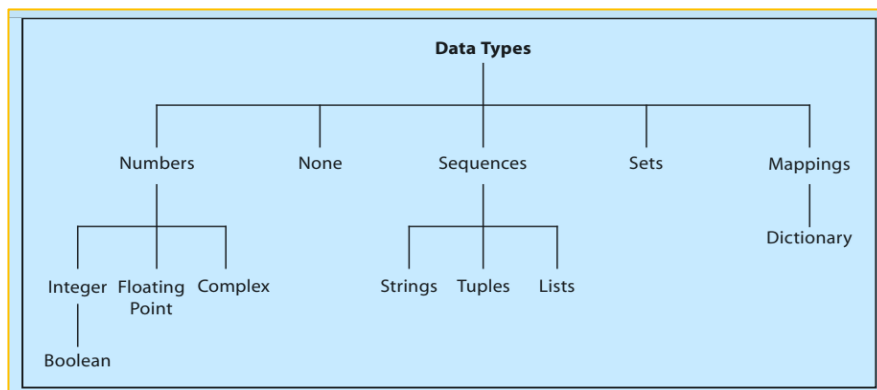
output

Area of Rectangle= 84

Data Types:

Data types are the classification or categorization of data items. It represents the kind of value that tells what operations can be performed on a particular data. Python supports Dynamic Typing. A variable pointing to a value of certain data type can be made to point to a value/object of another data type. This is called Dynamic Typing.

The following are the standard or built-in data types in Python:



Data Type Description

Integer	Stores whole number	a=10
Boolean	Boolean is used to represent the truth values of the expressions. It has two values True & False	Result = True
Floating point	Stores numbers with fractional part	x=5.5
Complex	Stores a number having real and imaginary part	num=a+bj
String	Immutable sequences (After creation values cannot be changed in-place) Stores text enclosed in single or double quotes	name= "Ria")
List	Mutable sequences (After creation values can be changed in-place) Stores list of comma separated values of any data type between square []	lst=[25, 15.6, "car", "XY"]
Tuple	Immutable sequence (After creation values cannot be changed in-place) Stores list of comma separated values of any data type between parentheses ()	tup=(11, 12.3, "abc")
Set	Set is an unordered collection of values, of any type, with no duplicate entry.	s = { 25, 3, 3.5}
Dictionary	Unordered set of comma-separated key:value pairs within braces { }	dict= { 1 : "One", 2: "Two", 3: "Three"}



Accepting values from the user

The input() function retrieves text from the user by prompting them with a string argument. For instance:

```
name = input("What is your name?")
```

Return type of input function is string. So, to receive values of other types we have to use conversion functions together with input function.

Sample Program-3

Write a program to read name and marks of a student and display the total mark.

```
In [3]: name=input("Enter Student's Name")
m1=float(input("Enter the Mark of English"))
m2=float(input("Enter the Mark of Artificial Intelligence"))
m3=float(input("Enter the Mark of Maths"))
Total=m1+m2+m3
print("Name : ", name)
print("Total Marks : ", Total)
```

output

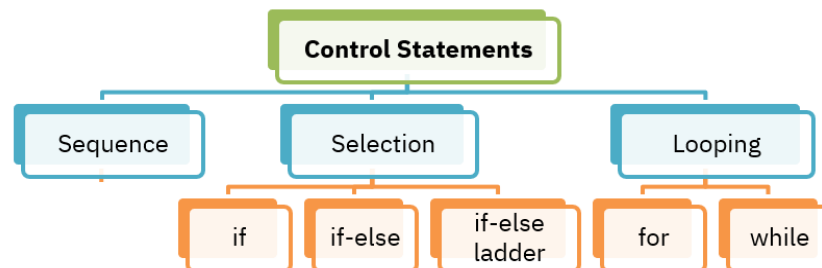
```
Enter Student's Name  M J Anakha
Enter the Mark of English  99
Enter the Mark of Artificial Intelligence  100
Enter the Mark of Maths  96
Name :    M J Anakha
Total Marks :  295.0
```

In the above example float() is used to convert the datatype into floating point. The explicit conversion of an operand to a specific type is called **type casting**.



Control flow statements in Python

Till now, the programs you've created have followed a basic, step-by-step progression, where each statement executes in sequence, every time. However, there are many practical programs where we have to selectively execute specific sections of the code or iterate over parts of the program. This capability is achieved through selective statements and looping statements.



Selection Statement

The if/ if..else statement evaluates test expression and the statements written below will execute if the condition is true otherwise the statements below else will get executed. Indentation is used to separate the blocks.

Syntax:

```
if <test expression>:  
    Statements
```

```
if <test expression>:  
    Body of if  
else:  
    Body of else
```

```
if <test expression>:  
    Body of if  
elif <test expression>:  
    Body of elif  
else:  
    Body of else
```

Let's check out different examples to see the working of if and if-else statements

Sample Program-4

Asmita with her family went to a restaurant. Determine the choice of food according to the options she chooses from the main menu.

Case 1: All Members are vegetarians. They prefer to have veg food. No other options.

(menu-veg)

Program & Output

```
choice=input("Enter the choice of food")  
if choice == "veg":  
    print("Welcome to Vegetarian Food House")
```

```
Enter the choice of foodveg  
Welcome to Vegetarian Food House
```


Case 2: Family Members may choose non-vegetarian foods also if veg foods are not available. (menu-veg/Nonveg)

```
choice=input("Enter the choice of food")
if choice == "veg":
    print("Welcome to Vegetarian Food House")
else:
    print("Welcome to Non-vegetarian Foods")
```

```
Enter the choice of food Nonveg
Welcome to Non-vegetarian Foods
```

Case 3: Family members can choose from variety of options

```
print("Menu: Veg | Nonveg | Mixed")
choice=input("Enter the choice of food")
if choice == "veg":
    print("Welcome to Vegetarian Food House")
elif choice == "Nonveg":
    print("Welcome to Non-vegetarian Foods")
else:
    print("Welcome to your Favourite Choice of Foods")
```

```
Menu: Veg | Nonveg | Mixed
Enter the choice of food Mixed
Welcome to your Favourite Choice of Foods
```

Sample Program-5

Write a program to get the length of the sides of a triangle and determine whether it is equilateral triangle or isosceles triangle or scalene triangle,

```
# Sample program to classify a triangle based on its sides

# Get user input for the lengths of the sides
side1 = float(input("Enter the length of side 1: "))
side2 = float(input("Enter the length of side 2: "))
side3 = float(input("Enter the length of side 3: "))

# Check the type of triangle based on the lengths of its sides
if side1 == side2 == side3:
    print("It is an equilateral triangle.")
elif side1 == side2 or side1 == side3 or side2 == side3:
    print("It is an isosceles triangle.")
else:
    print("It is a scalene triangle.")
```

```
Enter the length of side 1: 7
Enter the length of side 2: 7
Enter the length of side 3: 11
It is an isosceles triangle.
```

Looping Statements

Looping statements in programming languages allow you to execute a block of code repeatedly. In Python, there are mainly two types of looping statements: for loop and while loop.

For loop

For loop iterates through a portion of a program based on a sequence, which is an ordered collection of items.

The **“for”** keyword is used to start the loop. The loop variable takes on each value in the specified sequence (e.g., list, string, range). The colon (:) at the end of the for statement indicates the start of the loop body. The statements within the loop body are executed for each iteration. Indentation is used to define the scope of the loop body. All statements indented under the for statement are considered part of the loop. It is advisable to utilize a for loop when the exact number of iterations is known in advance.

Syntax

```
for <control-variable> in <sequence/items in range>:  
    <statements inside body of the loop>
```

Example -1

```
for i in range(5):  
    print("Python")
```

```
Python  
Python  
Python  
Python  
Python
```

Example-2

```
for i in range(5):  
    print(i)
```

```
0  
1  
2  
3  
4
```

In the above program

- ✚ range (5) returns the values 0,1,2,3,4
- ✚ For each iteration of the loop variable i receives these values.
- ✚ First iteration of the loop i=0 (one time print(“Python”) executes, similarly with i=1,2,3,4 also print statement works.
- ✚ Whatever is given inside the loop executes repeatedly. In the first example, 5 times Python was printed, but in example-2, as i is to be printed it displayed 0 1 2 3 4

The **for** loop iterates over each item in the sequence until it reaches the end of the sequence or until the loop is terminated using a **break** statement. It's a powerful construct for iterating over collections of data and performing operations on each item.

Sample Program-6

Write a program to display even numbers and their squares between 100 and 110.

```
for num in range(100,110,2):
    square=num*num
    print(num, "squared is", square)
```

```
100 squared is 10000
102 squared is 10404
104 squared is 10816
106 squared is 11236
108 squared is 11664
```

Sample Program-7

Write a program to read a list, display each element and its type. (use type() to display the data type.)

```
lst = [ 25, "fruit", 17.7, ('a', 'b'), 100]
for word in lst:
    print (word, type(word))
```

```
25 <class 'int'>
fruit <class 'str'>
17.7 <class 'float'>
('a', 'b') <class 'tuple'>
100 <class 'int'>
```



Point to be Noted

In the above program

- ✚ the control variable word gets each element of the list. Hence in print statement each element and its type is displayed
- ✚ Same program can be written using the following code also
for i in range (len (lst)):
 print (lst[i] , type (lst[i])
Here we take i as index number, lst[0]= 25 & lst[-1] = 100

Sample Program-8

Write a program to read a string. Split the string into list of words and display each word.

```
Str="Iam studying in Jyothis Central School"
wordlist=Str.split()
print ("Words in list format", wordlist)
for word in wordlist:
    print(word)
```

```
Words in list format ['Iam', 'studying', 'in', 'Jyothis', 'Central', 'School']
Iam
studying
in
Jyothis
Central
School
```

Sample Program-9

Write a simple program to display the values stored in dictionary

```
dict = {'S1':"Bio-Math", 'S2':"Math-Comp", 'S3': "Bio-Psy", 'S4': "Math-AI"}
for key in dict:
    print(dict[key])
```

```
Bio-Math
Math-Comp
Bio-Psy
Math-AI
```



UNDERSTANDING CSV file (Comma Separated Values)

CSV files are delimited files that store tabular data (data stored in rows and columns). It looks similar to spread sheets, but internally it is stored in a different format. In csv file, values are separated by comma. Data Sets used in AI programming are easily saved in csv format. Each line in a csv file is a data record. Each record consists of more than one fields(columns). The csv module of Python provides functionality to read and write tabular data in CSV format.

Let us see an example of opening, reading and writing formats for a file student.csv with file object file. student.csv contains the columns rollno, name and mark.

importing library	<code>import csv</code>
Opening in reading mode	<code>file= open("student.csv", "r")</code>
Opening in writing mode	<code>file= open("student.csv", "w")</code>
closing a file	<code>file.close()</code>
writing rows	<code>wr=csv.writer(file)</code> <code>wr.writerow([12, "Kalesh", 480])</code>
Reading rows	<code>details = csv.reader(file)</code> <code>for rec in details:</code> <code> print(rec)</code>

Sample Program-10

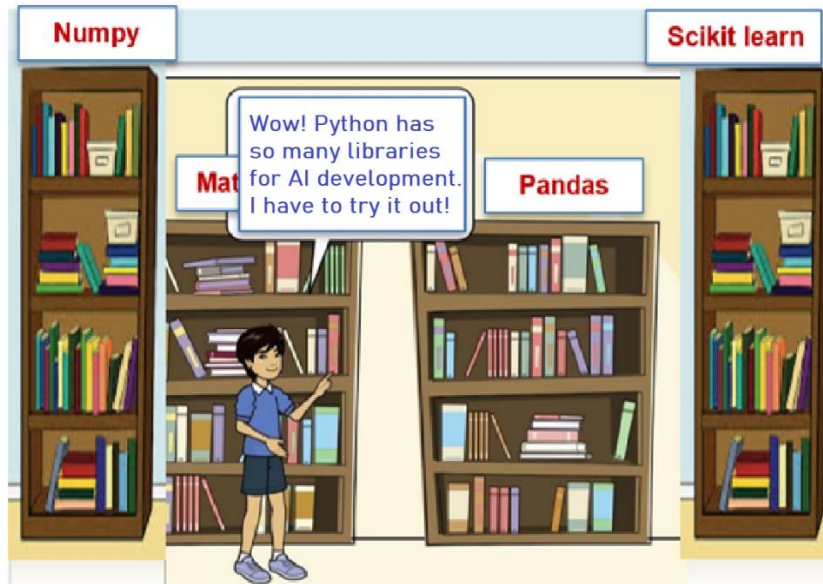
Write a Program to open a csv file students.csv and display its details

```
import csv
file = open("D:\JPB\Python\students.csv", "r")
details=csv.reader(file)
for rec in details:
    print(rec)
```

```
['RollNo', 'Name', 'class', 'TrName']
['11', 'Akshith', 'II', 'Sruthy ']
['12', 'Ashmitha', 'VII', 'Ruby']
['13', 'M J Anakha', 'X', 'Jayasankar']
```

INTRODUCING LIBRARIES

A **library** in Python typically refers to a collection of reusable modules or functions that provide specific functionality. Libraries are designed to be used in various projects to simplify development by providing pre-written code for common tasks. Concept of libraries are very easy to understand.



In Python, functions are organized within libraries similar to how library books are arranged by subjects such as physics, computer science, and economics. For example, the "math" library contains numerous functions like `sqrt()`, `pow()`, `abs()`, and `sin()`, which facilitate mathematical operations and calculations. To utilize a library in a program, it must be imported. For example, if we wish to use the `sqrt()` function in our program, we include the statement "**import math**". This allows us to access and utilize the functionalities provided by the math library.

Python offers a vast array of libraries for various purposes, making it a versatile language for different domains such as web development, data analysis, machine learning, scientific computing, and more. Now, let us explore some libraries that are incredibly valuable in the realm of Artificial Intelligence.

NUMPY

NumPy, which stands for Numerical Python, is a powerful library in Python used for numerical computing. It is a general-purpose array-processing package. NumPy provides the **ndarray** (N-dimensional array) data structure, which represents arrays of any dimension. These arrays are homogeneous (all elements are of the same data type) and can contain elements of various numerical types (integers, floats, etc.)

Where and why do we use the NumPy library in Artificial Intelligence?

Suppose you have a dataset containing exam scores of students in various subjects, and you want to perform some basic analysis on this data. You can utilize NumPy arrays to store exam scores

for different subjects efficiently. With NumPy's array operations, you can perform various calculations such as calculating average scores for each subject, finding total scores for each student, calculating the overall average score across all subjects, identifying the highest and lowest scores. NumPy's array operations streamline these computations, making them both efficient and convenient. This makes NumPy an indispensable tool for data manipulation and analysis in data science applications.

NumPy can be installed using Python's package manager, pip.

```
pip install numpy
```

Creating a Numpy Array - Arrays in NumPy can be created by multiple ways. Some of the ways are programmed here:

➡ Using List of Tuples

```
import numpy as np
ar = np.array( [(99, 88, 77), (44, 55, 66)])
print ("Numpy Array:\n", ar)
```

```
Numpy Array:
[[99 88 77]
 [44 55 66]]
```

➡ Using values from the user (using empty()) -- *The empty() function in Python is used*

```
import numpy as np
n=int(input("Enter the size of an array"))
ar=np.empty(n)
for i in range(n):
    ar[i]=int(input("Enter a number"))
print("Array\n", ar)
```

```
Enter the size of an array4
Enter a number34
Enter a number67
Enter a number85
Enter a number92
Array
[34. 67. 85. 92.]
```

to return a new array of a given size)

PANDAS

The name "Pandas" has a reference to both "Panel Data", and "Python Data Analysis". Pandas is a powerful and versatile library that simplifies tasks of data manipulation in Python. Pandas is built on top of the NumPy library which means that a lot of structures of NumPy are used or replicated in Pandas and Pandas is particularly well-suited for working with tabular data, such as spreadsheets or SQL tables. Its versatility and ease of use make it an essential tool for data analysts, scientists, and engineers working with structured data in Python.

Where and why do we use the Pandas library in Artificial Intelligence?

Suppose you have a dataset containing information about various marketing campaigns conducted by the company, such as campaign type, budget, duration, reach, engagement metrics, and sales performance. We use Pandas to load the dataset, display summary statistics, and perform group-wise analysis to understand the performance of different marketing campaigns. We then visualize the sales performance and average engagement metrics for each campaign type using Matplotlib, a popular plotting library in Python.

Pandas provides powerful data manipulation and aggregation functionalities, making it easy to perform complex analysis and generate insightful visualizations. This capability is invaluable in AI and data-driven decision-making processes, allowing businesses to gain actionable insights from their data.

Pandas can be installed using:

```
pip install pandas
```

Pandas generally provide two data structures for manipulating data, they are: Series and DataFrame.

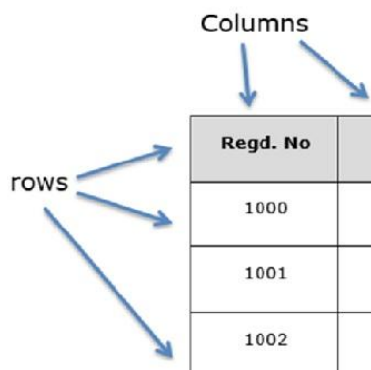
Series

A Series is a one-dimensional array containing a sequence of values of any data type (int, float, list, string, etc.) which by default have numeric data labels starting from zero. The data label associated with a particular value is called its **index**. We can also assign values of other data types as index. We can imagine a Pandas Series as a column in a spreadsheet as given here.

In data science, we often encounter datasets with two-dimensional structures. This is where Pandas DataFrames come into play.

A Data Frame is used when we need to work on multiple columns at a time, i.e., we need to process the tabular data.

Index	Data
0	Mark
1	Justin
2	John
3	Vicky



Regd. No	Name	Marks%
1000	Steve	86.29
1001	Mathew	91.63
1002	Jose	72.90
1003	Patty	69.23
1004	Vin	88.30

For example, the result of a class, items in a restaurant's menu, reservation chart of a train, etc.

A DataFrame is a two-dimensional labeled data structure like a table of MySQL. It contains rows and columns, and therefore has both a row and column index. Each column can have a different type of value such as numeric, string, boolean, etc., as in tables of a database.

Creation of DataFrame

There are several methods to create a DataFrame in Pandas, but here we will discuss two common approaches:

➔ Using NumPy ndarrays-

```
import numpy as np
import pandas as pd
array1 = np.array([10,20,30])
array2 = np.array([100,200,300])
array3 = np.array([-10,-20,-30])
dFrame = pd.DataFrame( [array1, array2, array3] , columns = ['col1', 'col2', 'col3'])
print(dFrame)
```

	col1	col2	col3
0	10	20	30
1	100	200	300
2	-10	-20	-30

➔ Using List of Dictionaries

```
listDict = [{'Dance':10, 'Music':20}, {'Dance':15,'Music':10,'Painting':20}, {'Painting': 12}]
a= pd.DataFrame(listDict, index=['X', 'XI', 'XII'])
print(a)
```

	Dance	Music	Painting
X	10.0	20.0	NaN
XI	15.0	10.0	20.0
XII	NaN	NaN	12.0

Point to be Noted

- ➔ Dictionary keys become column labels by default in a DataFrame, and the lists become the rows.
- ➔ NaN (Not a Number) is inserted if a corresponding value for a column is missing.
- ➔ Pandas uses isnull() function to identify NaN values in a DataFrame.

Dealing with Rows and Columns

Based on the DataFrame **'Result'** provided below, we can observe various operations related to rows and columns. Each operation statement is accompanied by its corresponding output from the Result DataFrame

DataFrame: **Result**

Result						
	Rajat	Amrita	Meenakshi	Rose	Karthika	
Maths	90	92	89	81	94	
Science	91	81	91	71	95	
Hindi	97	96	88	67	99	



Adding a New Column to a DataFrame:

We can add a new column 'Fathima', by mentioning column name as given below

```
Result['Fathima']=[89,78,76]
print(Result)
```

	Rajat	Amrita	Meenakshi	Rose	Karthika	Fathima
Maths	90	92	89	81	94	89
Science	91	81	91	71	95	78
Hindi	97	96	88	67	99	76



Adding a New Row to a DataFrame:

We can add a new row to a DataFrame using the DataFrame.loc[] method. Let us add marks for English subject in Result →

```
Result.loc['English'] = [90, 92, 89, 80, 90, 88]
print(Result)
```

	Rajat	Amrita	Meenakshi	Rose	Karthika	Fathima
Maths	90	92	89	81	94	89
Science	91	81	91	71	95	78
Hindi	97	96	88	67	99	76
English	90	92	89	80	90	88



Deleting Rows and Columns from a DataFrame:

We need to specify the names of the labels to be dropped and the axis from which they need to be dropped. To delete a row, the parameter axis is assigned the value 0 and for deleting a column, the parameter axis is assigned the value 1.

Deleting a row "Hindi"

```
Result = Result.drop('Hindi', axis=0)
print(Result)
```

	Rajat	Amrita	Meenakshi	Rose	Karthika	Fathima
Maths	90	92	89	81	94	89
Science	92	84	90	72	96	88
English	90	92	89	80	90	88

➡ **Delete the columns having labels 'Rajat', 'Meenakshi' and 'Karthika':**

```
Result = Result.drop(['Rajat', 'Meenakshi', 'Karthika'], axis=1)
print(Result)
```

	Amrita	Rose	Fathima
Maths	92	81	89
Science	84	72	88
English	92	80	88



Point to be Noted

During Data Analysis, DataFrame.drop() method is used to remove the rows and columns.



Accessing DataFrame Elements

Data elements in a DataFrame can be accessed using different ways. Two common ways of accessing are using loc and iloc. DataFrame.loc[] uses label names for accessing and DataFrame.iloc[] uses the index position for accessing the elements of a DataFrame. Let us check an example

```
Result.loc['Science']
```

```
Result.iloc[1]
```

```
Rajat      91
Amrita     81
Meenakshi  91
Rose       71
Karthika   95
Fathima    78
Name: Science, dtype: int64
```



Understanding Missing Values

Missing Data or Not Available data can occur when no information is provided for one or more items or for a whole unit. During Data Analysis, it is common for an object to have some missing attributes. If data is not collected properly it results in missing data.

In DataFrame it is stored as NaN (Not a Number). For example, while collecting data, some people may not fill all the fields while taking the survey. Sometimes, some attributes are not relevant to all.

Pandas provide a function isnull() to check whether any value is missing or not in the DataFrame. This function checks all attributes and returns True in case that attribute has missing values, otherwise returns False. Now, we can explore different operations related to missing values based on the DataFrame 'listDict' provided below.

	Dance	Music	Painting
X	10.0	20.0	NaN
XI	15.0	10.0	20.0
XII	NaN	NaN	12.0

StudCCA.isnull()



	Dance	Music	Painting
X	True	True	False
XI	True	True	True
XII	False	False	True

Point to be Noted

Finding any missing value in a column → `StudCCA['Music'].isnull().any() → True`
Finding total number of NaN → `StudCCA.isnull().sum() → 3`
Deleting entire row with NaN values → `StudCCA.dropna()`
Replacing NaN values (here by 1) → `StudCCA.fillna(1)`

Attributes of DataFrames

Attributes are the properties of a DataFrame that can be used to fetch data or any information related to a particular DataFrame.

The syntax of writing an attribute is:

DataFrame_name . attribute

Let us understand the attributes of DataFrames with the help of DataFrame Teacher
DataFrame:Teacher

	IX	X	XI	XII
Physics	Jayasankar	Shanthini	Sruthy	Anand Raj
Maths	Snitha	Haripriya	Praseetha	Sobhana Beegum
Artificial Intelligence	Noufiya	Rejila	Lekshmi	Jyoti

→ Displaying Row Indexes - Teacher.index

```
In [2]: Teacher.index
```

```
Out[2]: Index(['Physics', 'Maths', 'Artificial Intelligence'], dtype='object')
```

→ Displaying column Indexes - Teacher.columns

```
In [3]: Teacher.columns
```

```
Out[3]: Index(['IX', 'X', 'XI', 'XII'], dtype='object')
```

→ Displaying datatype of each - Teacher.dtypes

```
In [4]: Teacher.dtypes
```

```
Out[4]: IX      object
        X      object
        XI     object
        XII     object
        dtype: object
```

→ Displaying data in Numpy Array form - Teacher.values

```
In [5]: Teacher.values
```

```
Out[5]: array([[ 'Jayasankar', 'Shanthini', 'Sruthy', 'Anand Raj'],
               [ 'Snitha', 'Haripriya', 'Praseetha', 'Sobhana Beegum'],
               [ 'Noufiya', 'Rejila', 'Lekshmi', 'Jyoti']], dtype=object)
```



Displaying total number of rows and columns (row, column) - Teacher.shape

```
In [6]: Teacher.shape
```

```
Out[6]: (3, 4)
```



Displaying first n rows (here n = 2) - Teacher.head(2)

```
In [7]: Teacher.head(2)
```

```
Out[7]:
```

	IX	X	XI	XII
Physics	Jayasankar	Shanthini	Sruthy	Anand Raj
Maths	Snitha	Haripriya	Praseetha	Sobhana Beegum



Displaying last n rows (here n = 2) - Teacher.tail(2)

```
In [8]: Teacher.tail(2)
```

```
Out[8]:
```

	IX	X	XI	XII
Maths	Snitha	Haripriya	Praseetha	Sobhana Beegum
Artificial Intelligence	Noufiya	Rejila	Lekshmi	Jyoti

Importing and Exporting Data between CSV Files and DataFrames

We can create a DataFrame by importing data from CSV files. Similarly, we can also store or export data in a DataFrame as a .csv file.

Importing a CSV file to a DataFrame

Using the read_csv() function, you can import tabular data from CSV files into pandas dataframe by specifying a parameter value for the file name

Syntax: `pd.read_csv("filename.csv")`

Example: Reading file students.csv

```
import pandas as pd
import csv

df = pd.read_csv("D:\JPB\Python\students.csv", sep=",", header=0)
print(df)
```

	RollNo	Name	class	TrName
0	11	Akshith	II	Sruthy
1	12	Ashmitha	VII	Ruby
2	13	M J Anakha	X	Jayasankar



Point to be Noted

- read_csv() is used to read the csv file with its correct path.
- sep specifies whether the values are separated by comma, semicolon, tab, or any other character. The default value for sep is a space.
- The parameter header marks the start of the data to be fetched. header=0 implies that column names are inferred from the first line of the file. By default, header=0.

Exporting a DataFrame to a CSV file

We can use the `to_csv()` function to save a DataFrame to a text or csv file. For example, to save the DataFrame Teacher into csv file resultout, we should write

Teacher.to_csv(path_or_buf='C:/PANDAS/resultout.csv', sep=',')

When we open this file in any text editor or a spreadsheet, we will find the above data along with the row labels and the column headers, separated by comma.

Scikit-learn (Learn)

Note for Teachers: This topic can be taught after teaching the Machine Learning Unit.

Scikit-learn (Sklearn) is the most useful and robust library for machine learning in Python. It provides a selection of efficient tools for machine learning and statistical modeling via a consistent interface in Python. Sklearn is built on (relies heavily on) NumPy, SciPy and Matplotlib.

Key Features:

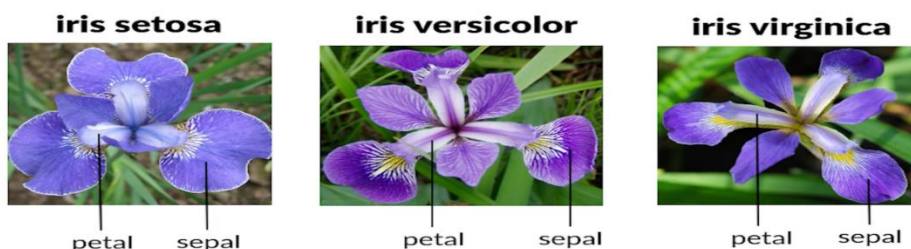
- Offers a wide range of supervised and unsupervised learning algorithms.
- Provides tools for model selection, evaluation, and validation.
- Supports various tasks such as classification, regression, clustering, dimensionality reduction, and more.
- Integrates seamlessly with other Python libraries like NumPy, SciPy, and Pandas.

Install scikit-learn using the statement

```
pip install scikit-learn
```

load_iris (In sklearn.datasets)

The Iris dataset is a classic and widely used dataset in machine learning, particularly for classification tasks. It comprises measurements of various characteristics of iris flowers, such as sepal length, sepal width, petal length, and petal width, along with the corresponding species of iris to which they belong. The dataset typically includes three species: setosa, versicolor, and virginica.



from sklearn.datasets import load_iris	importing iris dataset
iris = load_iris()	calls the “load_iris()” function to load the iris dataset
X = iris.data	X is a variable and assigned as feature vector. The feature vectors contain the input data for the machine learning model
y = iris.target	Y is a variable and assigned as target variable. The target variable contains the output or the variable we want to predict with the model.

Sample output – First 10 rows of X

```
Feature names: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
Target names: ['setosa' 'versicolor' 'virginica']

First 10 rows of X:
[[5.1 3.5 1.4 0.2]
 [4.9 3.  1.4 0.2]
 [4.7 3.2 1.3 0.2]
 [4.6 3.1 1.5 0.2]
 [5.  3.6 1.4 0.2]
 [5.4 3.9 1.7 0.4]
 [4.6 3.4 1.4 0.3]
 [5.  3.4 1.5 0.2]
 [4.4 2.9 1.4 0.2]
 [4.9 3.1 1.5 0.1]]
```

Here, each row represents a sample (i.e., an iris flower), and each column represents a feature (i.e., a measurement of the flower).

For example, the first row [5.1 3.5 1.4 0.2] corresponds to an iris flower with the following measurements:

- Sepal length: 5.1 cm
- Sepal width: 3.5 cm
- Petal length: 1.4 cm
- Petal width: 0.2 cm

train_test_split (In sklearn.model_selection)

Datasets are usually split into training set and testing set. The training set is used to train the model and testing set is used to test the model.

Most common splitting ratio is 80: 20. (Training -80%, Testing-20%)

from sklearn.model_selection import train_test_split	importing train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 1)	
X_train, y_train	the feature vectors and target variables of the training set respectively.
X_test, y_test	the feature vectors and target variables of the testing set respectively.
test_size = 0.2	specifies that 20% of the data will be used for testing, and the remaining 80% will be used for training.
random_state = 1	Ensures reproducibility by fixing the random seed. This means that every time you run the code, the same split will be generated.

KNeighborsClassifier (In sklearn.neighbors)

Scikit-learn has wide range of Machine Learning (ML) algorithms which have a consistent interface for fitting, predicting accuracy, recall etc. Here we are going to use KNN (K nearest neighbors) classifier.

from sklearn.neighbors import KNeighborsClassifier	importing KNeighborsClassifier (type of supervised learning algorithm used for classification tasks.)
knn = KNeighborsClassifier(n_neighbors =3)	we create an instance of the KNeighborsClassifier class . n_neighbors = 3 indicates that the classifier will consider the 3 nearest neighbors when making predictions. This is a hyperparameter that can be tuned to improve the performance of the classifier.
knn.fit(X_train, y_train)	trains the KNeighborsClassifier model using the fit method. it constructs a representation of the training data that allows it to make predictions based on the input features.
y_pred = knn.predict(X_test)	The knn object contains the trained model, make predictions on new, unseen data.

metrics

```
from sklearn import metrics
Accuracy = metrics.accuracy_score(y_test, y_pred))
```

This calculates the accuracy of the model by comparing the predicted target values (y_pred) with the actual target values (y_test). The accuracy_score represents the proportion of correctly predicted instances out of all instances in the testing set.

Scikit-learn offers a variety of modules that simplify the process of building, training, and evaluating machine learning models, making it a popular choice for various tasks in this domain. In our session, we utilized the 'load_iris()' function to load the Iris dataset. Upon loading, we split the dataset into training and test sets using the 'train_test_split' function. Subsequently, we trained our model using the K-Nearest Neighbors Classifier ('KNeighborsClassifier') and evaluated its performance using appropriate metrics. This workflow represents a typical data analysis pipeline in AI project development.

Now, to validate the model's predictive accuracy, we can use some sample data.

```
sample = [[5, 5, 3, 2], [2, 4, 3, 5]]
preds = knn.predict(sample)
for p in preds:
    pred_species.append(iris.target_names[p])
print("Predictions:", pred_species)
```

The provided code snippet demonstrates how to use the trained classifier to make predictions on sample data. After initializing the sample data as [5, 5, 3, 2], the classifier predicts the species of iris flowers based on these measurements. Finally, the predicted species are printed to the console.

Introduction to Capstone Project

A capstone project is a project where students must research a topic, deeply understand the subject matter and integrate all their knowledge gathered about a particular subject to develop a solution to a problem.

The AI Capstone project will give learners an opportunity to implement the AI skills as a final step in the learning path to showcase their expertise in the subject matter.

While doing a Capstone project, students will learn more about problems which they can solve to improve lives and make the world a better place. Also, they learn many important skills including:

- How to work as a team member?
- How to clearly identify an issue? and how it is affecting (the user)?
- How to brainstorm solutions and select the best one?
- How to decide which type of AI may be useful for the proposed solution?
- How to ethically gather and use data to train a computer to help solve the issue?
- How to test the prototype with users and use their feedback to improve the solution?
- How to pitch their solution to people who will be able to help them take action?

Some Examples for Capstone Project are given below

1. Stock Prices Predictor
2. Develop A Sentiment Analyzer
3. Movie Ticket Price Predictor
4. Students Results Predictor
5. Human Activity Recognition using Smartphone Dataset
6. Classifying humans and animals in a photo

The list of Examples is huge as every year students are coming up with new ideas and innovations for new projects.

Asking the right question before starting the AI Capstone Project

Before Starting a Project, one should analyze whether a problem has a pattern associated with it. If there is no pattern, then the problem cannot be solved with AI technology. It is fundamental that this question, **“Is there a Pattern?”** is asked before deciding to embark on an AI development journey.

There are various AI techniques which can be applied to check for patterns and solve the problems. Predictive Analysis helps in identifying the patterns by asking different questions which fall into different approaches. Some of the questions are given below:



- 1) Which category? (Classification)
- 2) How much or how many? (Regression)
- 3) Which group? (Clustering)
- 4) Is this unusual? (Anomaly Detection)
- 5) Which option should be taken? (Recommendation)

Now, once these questions are answered, we can decide that the problem can be solved using AI. Complex problems cannot be dealt with as such. Then we try to simplify the complex problem to analyze it. This is known as Problem Decomposition.

Problem decomposition steps

1. Understand the problem and then restate the problem in your own words.
 - Know what the desired inputs and outputs are?
 - Ask questions for clarification (in class these questions might be to your instructor, but most of the time they will be asking either yourself or your collaborators)
2. Break the problem down into a few large pieces.
 - Write these down, either on paper or as comments in a file.
3. Break complicated pieces down into smaller pieces.
 - Keep breaking down until all of the pieces are small.
4. Code one small piece at a time.
 - Think about how to implement it? and write the code/query.
 - Test it on its own and fix problems, if any.



Activity

1. Many people are facing serious issues with traffic jams during office hours which may be caused due to multiple reasons. Considering this as a major issue to resolve, write down the problem decomposition steps to make it simple to solve.

First Step → Understand the Problem and Restate it.

Public is facing difficulty due to the problem of traffic jams especially during the busy office hours.

Second Step → Break the problem down into a few large pieces.

Split the traffic issue separately for each reason

- a. Bad Roads
- b. Accidents
- c. Office Hours
- d. Rash Driving
- e. Inappropriate Signals
- f. Over Crowded Area

Third Step → Break complicated pieces down into smaller pieces.

From the above sub steps (reasons), if any reason is of complicated type, we can split it into more simpler units. For example, Rash Driving can be further split into:

- a. Over Speed
- b. New Drivers with less experience
- c. Careless Driving

Fourth Step → Take each issue at a time and try to find the solution.

Critical and Creative Thinking

Definition: Critical thinking is the ability to analyze a situation and make a judgment based on facts and data. Creative thinking on the other hand refers to the ability to come up with new ideas or rather solutions.

So, these two skills are very important to raise questions against problems, gather and assess relevant information, brainstorm ideas and formulate well-reasoned conclusions and solutions. Creative thinking and critical thinking are necessary skills for equipping individuals to be the social change makers, leaders and innovators. Complex problems can be easily solved using critical and creative thinking which has a systematic approach to address any problem which has to be solved. This approach is known as Design Thinking.

Introduction to Design Thinking

Definition: Design thinking is a non-linear, iterative process that expert teams use to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test.

Design Thinking framework

Design Thinking is a design methodology that provides a solution-based approach to solving problems. It's extremely useful in tackling complex problems that are ill-defined or unknown.

The five stages of Design Thinking are as follows: Empathize, Define, Ideate, Prototype, and Test.

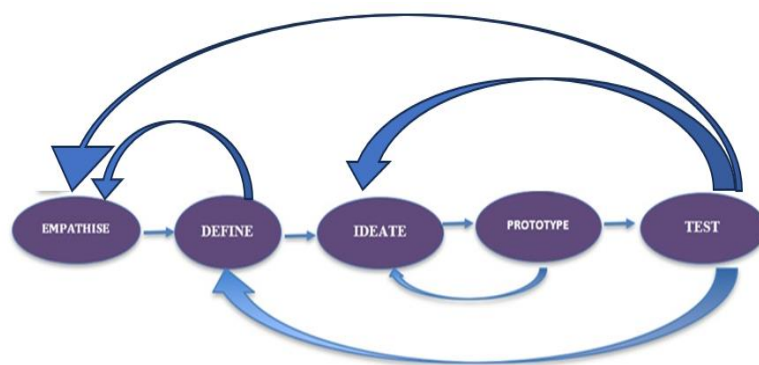


Figure 1: Stages of Design Thinking Framework

i) Empathize

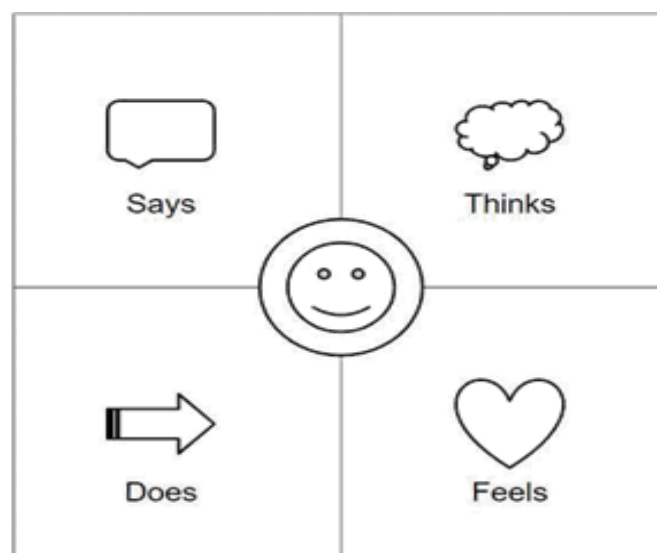
Design thinking begins with empathy. This requires doing away with any preconceived notions and immersing oneself in the context of the problem for better understanding. In simple words, through empathy, one is able to put oneself in other people's shoes and connect with how they might be feeling about their problem, circumstance, or situation.

There is a challenge one needs to solve. How does one approach it? Empathy starts from here. As a designer of the solution to a challenge, one should always understand the problem from the end-user perspective. This is done by observation, interaction or by imagination.

Designers are expected to interact with customers / users very frequently to gather detailed facts about the problems and user's expectations. A detailed analysis of these facts leads to approaching the problem in the best possible way.

Empathy Map

An extremely useful tool for understanding the users' needs and gaining a deeper insight into the problem at hand is the empathy map. It also helps in deepening that understanding, gaining insight into the user's behavior.



To create a “persona” or profile for the user, you can use the empathy map activity to create a realistic general representation of the user or users. Personas can include details about a user's education, lifestyle, interests, values, goals, needs, thoughts, desires, attitudes, and actions.

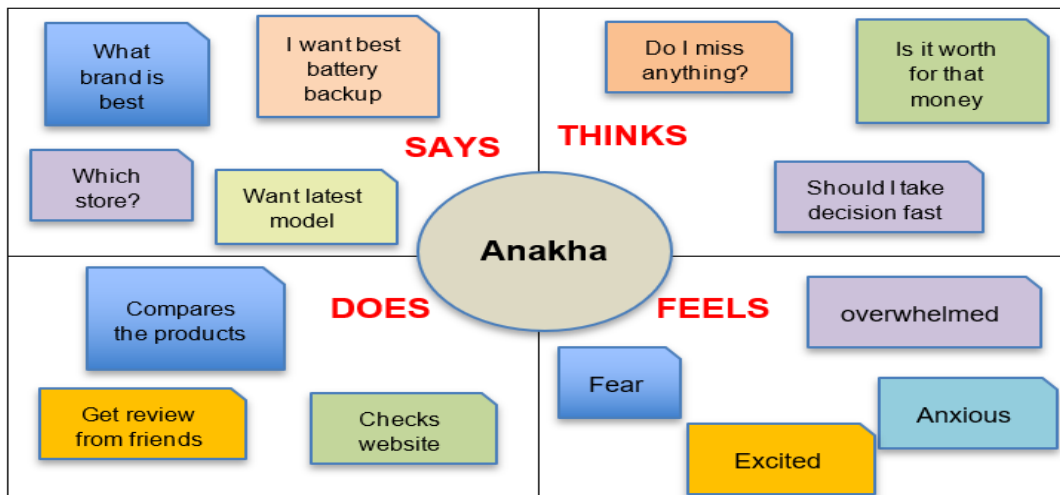
An Empathy Map is divided into 4 quadrants. —Says, Thinks, Does, Feels

- Says – This quadrant contains whatever the user says aloud
- Thinks – This quadrant depicts the thoughts which user have about the problem
- Does – This quadrant depicts the actions of user
- Feels – This quadrant displays the emotional status of the user

Creating Empathy Map

Draw the Empathy map grid as shown above. Write down the respective observations in corresponding quadrants. These observations can be written directly or can paste a sticky note with the observations in the quadrants. Let us understand through an example. Anakha is having a desktop computer. She is planning to buy one new Laptop for her educational purpose. Let us create an Empathy map for Anakha.

Empathy Map for Anakha



Online tool to create Empathy Map

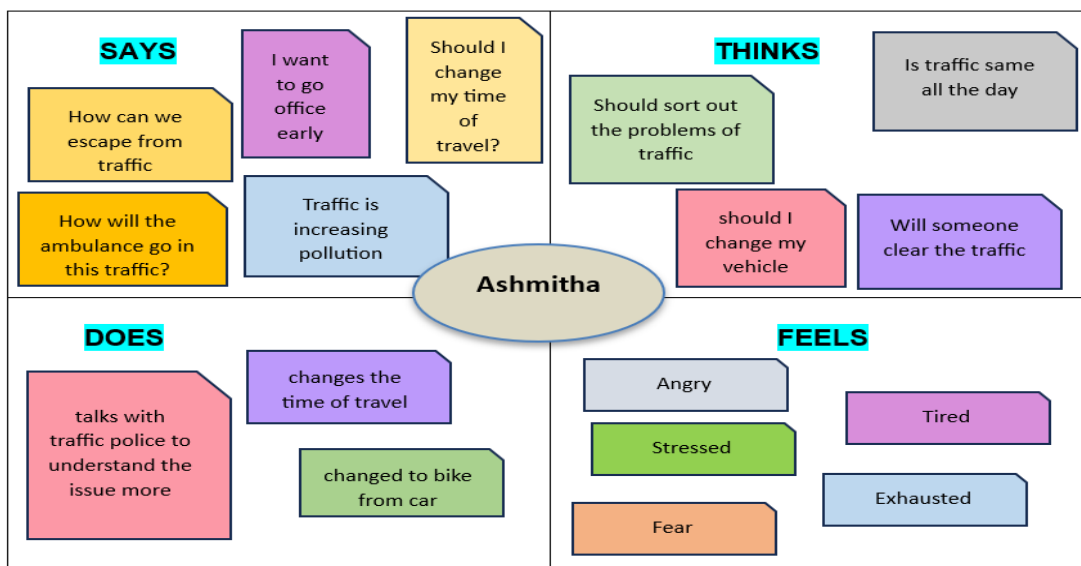
<https://online.visual-paradigm.com/diagrams/features/empathy-map-template/>

Reference: <https://www.ibm.com/design/thinking/page/toolkit/activity/empathy-map>



Activity

1. Ashmitha daily drives to her office and back. The office is hardly 30 minutes' drive from her home. However, due to traffic jams it takes more than 1 hour. Ashmitha is hoping for a solution to this traffic issue. Prepare an Empathy map related to Ashmitha.



ii) Define

In the Define stage, information collected during Empathize is used to draw insights and is instrumental in stating the problem that needs to be solved. It's an opportunity for the design thinker to define the challenge or to write the problem statement in a human-centered manner with a focus on the unmet needs of the users.

In order to extract / gather relevant facts and information from users/customers, it is recommended to use this simple and reliable method of questioning: the **5W1H method**—i.e. asking questions starting with Who? What? When? Where? Why? and How?

For Example: For the traffic issue in a city, we could define the problem in detail after getting the answers of 5W1H questions. Sample questions of the problem is given below

Who ?	What ?	Where ?
- Who are the sufferers? - Who is responsible? - Who can control it? - Who all are involved in it? - Who needs immediate help? - Who questions it? - Who identifies it?	- What can be done? - What should be done? - What is the issue? - What causes the issue? - What steps are necessary? - What to be avoided? - What may be the reason?	- Where is the problem seen? - Where else it can occur? - Where to start the solution? - Where to concentrate more? - Where are the issues severe? - Where can we ignore issue? - Where is solution impossible
When ?	Why ?	How ?
- When does traffic occurs? - When can we start clearing the traffic? - When is it more problematic? - When to think about a quick solution? - When is traffic ignorable?	- Why traffic occurs? - Why it becomes a problem? - Why should we find a solution to it? - Why to clear traffic in that city? - Why can't we ignore traffic?	- How will we find a solution? - How to deal with the traffic? - How to manage the users affected by traffic? - How to solve without creating more issues? - How to identify the traffic?



Activity

1. Considering Ashmitha's traffic issue answer the following questions given below in order to help define the problem.

a. Who is the sufferer in the Problem?

b. What is the Problem?

c. Where does it happen?

d. When can it happen usually?

e. Why does it happen?

f. How can it be solved?

iii) Ideate

By now the problem is obvious and it is time to brainstorm ways and methods to solve it. At this stage, numerous ideas are generated as a part of the problem-solving exercise. In short, ideation is all about idea generation. During brainstorming, one should not be concerned if the generated ideas are possible, feasible, or even viable. The only task of the thinkers is to think of as many ideas as possible for them. It requires "going wide" mentally in terms of concepts and outcomes. There are many brainstorming tools that can be used during this stage. Here is an overview of the most essential ideation techniques employed to generate numerous ideas.

Brainstorm

During a Brainstorming session, students leverage the synergy of the group to generate new innovative ideas by building on others' ideas. Participants should be able to discuss their ideas freely without fear of criticism. A large number of ideas are collected so that different options are available for solving the challenge.

Brain dump

Brain dump is very similar to Brainstorm; however, it's done individually. It allows the concerned person to open the mind and let the thoughts be released and captured onto a piece of paper. The participants write down their ideas onto paper or post-it notes and share their ideas later with the larger group.

Brain writing

Brain writing is also very similar to a Brainstorm session and is known as 'individual brainstorming'. At times only the most confident of team members share their ideas while the introverts keep the ideas to themselves. Brainwriting gives introverted people time to write them down instead of sharing their thoughts out loud with the group. The participants write down their ideas on paper and, after a few minutes, pass on their own piece of paper to another participant who then elaborates on the first person's ideas and so forth. In this way all participants pass their papers on to someone else and the process continues. After about 15 minutes, the papers are collected and posted for instant discussion.

During brainstorming, one should not be concerned if the generated ideas are possible, feasible, or even viable. The only task of the thinkers is to think of as many ideas as possible for them. It requires "going wide" mentally in terms of concepts and outcomes. It is all about creativity and imagination; all types of ideas are encouraged, whether stupid or wise – it hardly matters as long as the solution is imagined.

After everyone shares their ideas, specific measures are applied to evaluate the ideas without being judgmental or critical to narrow the list. It may so happen that the solution comes from the unlikeliest of ideas. So, at this point focus is on quantity over quality of ideas. The most feasible ideas are chosen for further exploration. Storyboarding, or making a visual mock-up of an idea, can also be useful during ideation.



Activity

1. For the traffic problems in a city, brainstorm and generate different ideas as a team.

iv) Prototype

The prototype stage involves creating a model designed to solve consumers' problems which is tested in the next stage of the process. Creating a prototype is not a detailed process. It may include a developing simple drawing, poster, group role-playing, homemade “gadget, or a 3D printed product.” The prototypes must be quick and easy to develop and cheap. Therefore, prototypes are visualized as rudimentary forms of what a final product is expected to look like. Prototyping is intended to answer questions that get you closer to your final solution. Prototypes, though quick and simple to make, bring out useful feedback from users. Prototypes can be made with everyday materials also.



Activity

1. Through a poster or simple drawing depict a prototype for any one solution for the traffic issue.

v) Test

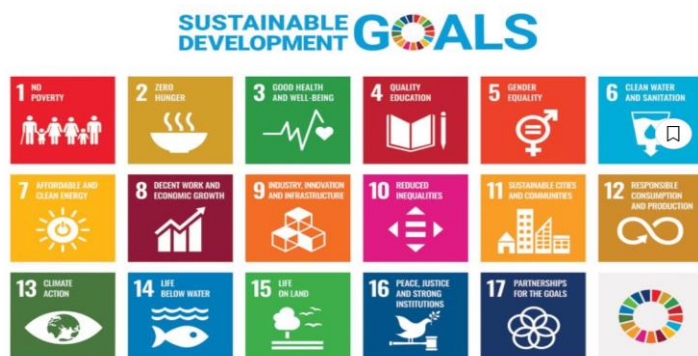
One of the most important parts of the design thinking process is to test the prototypes with the end users. This step is often seen going parallel to prototyping. During testing, the designers receive feedback about the prototype(s), and get another opportunity to interact and empathize with the people they are finding solutions for. Testing focuses on what can be learned about the user and the problem, as well as the potential solution.



Activity

1. Show the prototype made by you to your parents/teachers/relatives who usually get stuck in traffic issues. Collect the feedback from them and understand whether your prototype will work or not. Write down the views your users shared and the conclusion you arrived at.

Sustainable Development Goals



Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs. The global community is witnessing a transition towards a sustainable future. To achieve a sustainable future, 17 goals have been announced by the

United Nations which are termed as the Sustainable Development Goals.

Projects undertaken all over the world play a crucial role in fulfilling this task by aligning their strategies and vision with the sustainable development goals and targets.

Students can select any problem based on SDG to find an AI solution for it.

For example:

SDG 2 was adopted to achieve “Zero Hunger”. That means it is essential to recognize that the successful transformation of our current food systems (shift to a sustainable food system) should tackle the main global challenges in a sustainable, healthy, and inclusive manner. Healthy food habits, food Supply to all areas, consumption of food as per the production, Seasonal Foods etc. are the areas where students can concentrate for identifying problems and try to derive simple AI based solution.

Some other use cases of the SDGs are given below

- Good health and well being
- Gender equality
- Renewable energy
- Combating hunger and poverty
- Protecting resources
- Promoting responsible consumption and production
- Providing access to education
- Sustainable and safe cities and communities



Activity

1. Write down a few problems which you notice from any five SDGs which can be taken up as a Capstone project.

Project Abstract Creation Using Design Thinking Framework

We can now choose a problem (capstone project) and check how it can be detailed using Design Thinking Framework.

Format for Project Abstract Creation

1. Project Name:
2. Team Members Name:
3. Problem Selection –Important issue which you think can be solved using AI by doing survey or interview and which is aligned to any of the SDGs.
4. Users Affected by the Problem:
5. Empathise – create Empathy Map
6. Define the Problem – Write down the 5W1H questions and write the problem statement
7. Ideate – Brainstorm and write down the ideas
8. Prototype- Create prototype by simple drawing, poster, group role-playing, homemade “gadget, or a 3D printed product.”
9. Test (optional at this stage)

Now let us see an example

- 1. Project Name:** {Suitable name related to your problem and your creativity}
- 2. Team Members Name:** { Students can fill by themselves}
- 3. Problem Selection:** After doing a survey amongst the school students who are in Class X, it was observed that most of the students are confused about the subjects that they have to choose in future.

Details about the issue:

While taking admission to plus-two courses, students are confused with subject selection. To provide quality education students should be provided with the subjects in which they have aptitude. A system to analyze their interests, choose subjects from the available options and choose the institutions which provide those subjects are a big problem. After seeing the issue, it was aligned to SDG-4, Quality Education.

4. Users:

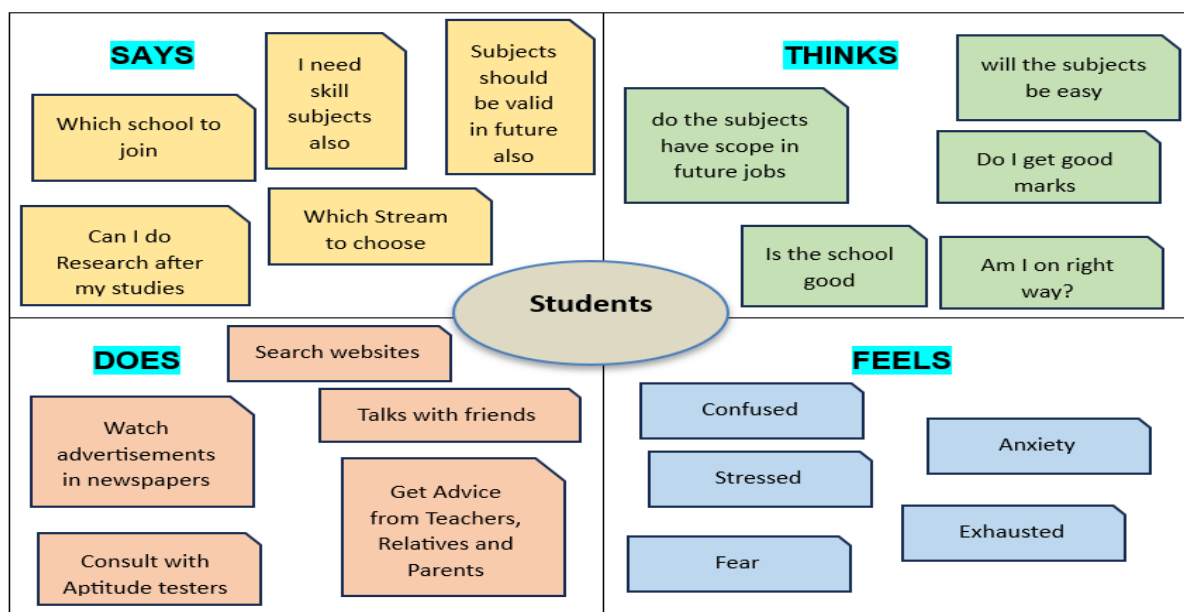
All the students who pass class X face the issue of selecting an appropriate subject for their higher studies as it is their turning point to move on according to their interest.

5. Empathise:

Two categories of students can be met and interviewed.

- First—Those students who are studying in plus-two (to know how difficult it was to choose a subject and a school)
- Second—Those students who are going to take admission (can know about their anxieties and expectations)

After the discussion with students, team members can sit together and finalize the points they understood from the users(students). They can write each in a chit of paper, classify it into the categories SAYS, THINKS, DOES & FEELS. Now they can create an Empathy Map. A sample empathy map is given below.



6. Define:

After Empathetically understanding the students, the problem may be defined. Get the Answers of 5W1H Questions. Write down the Problem Statement.

5W1H Questions & Answers

Who?

1. Who are the users?

Students seeking admission in plus-two course.

2. Who else can be the users?

Parents/well-wishers who seek admission for their wards.

3. What do you know about the users?

Users are students who are interested in taking plus-two admissions. Students have well defined future plans. Now they have to select the appropriate subject as per their plan.

What?

1. What is the Problem?

Not able to select the subject of their choice.

2. What else are the Problems?

They can't choose the school which provides these subjects.

Where?

1. Where is the problem/ In which situation does the problem arises?

All the places where admission takes place & all the place where students wish to take admission the problem persists.

When?

1. When does the problem occur?

Problem occurs during the time of admission.

Why?

1. Why does the problem occur?

Problem occurs due to the lack of knowledge of subjects available as part of the curriculum.

2. Why should it be solved?

Only if students pursue the course, they have aptitude, a well-developed society can be built in future.

How?

1. How can we solve it?

Through a system which will analyze a student's aptitude and suggests the subject to take. After checking the database, the system can suggest the schools giving the subject combinations also.

2. How will the solution improve the situation?

The solution could help students in choosing the appropriate subjects and could give effective education.

PROBLEM STATEMENT

Our students have the problem of not being able to choose the subjects of their interest while taking admission to their plus-two course and it can be solved by developing a system which will analyze students' aptitude and suggest the subject to take.

7. Ideate:

As a team, students can discuss and put forward different ideas to implement the solution.

1. Create an application to input the interest and get suggestions on subjects.
2. Create a chatbot to converse with.
3. Make a robot to discuss with and give advice.
4. Create a mobile application where through different sources, queries can be given and the application will give the result.

8. Prototype:

Select any one idea and prepare the prototype of the solution - Here we select the idea of creating a chatbot to give the solution.

1. WHAT IS DATA LITERACY?

Data can be defined as a representation of facts or instructions about some entity (students, school, sports, business, animals etc.) that can be processed or communicated by human or machines. It is a widely known fact that Artificial Intelligence (AI) is essentially data-driven. AI involves converting large amounts of raw data into actionable information that carry practical value and is usable.

Data literacy means being able to find and use data effectively. This includes skills like collecting data, organizing it, checking its quality, analysing it, understanding the results and using it ethically.

Data may be **structured, semi structured or unstructured**. It should be collected, organized and analysed properly to know whether the input for AI models is valid and appropriate or not. AI data analysis involves using AI techniques and data science to improve the processes of cleaning, inspecting, and modelling both structured and unstructured data. The primary objective is to extract valuable information that can support decision-making and drawing conclusions.

2. DATA COLLECTION

Data collection allows you to capture a record of past events so that we can use data analysis to find recurring patterns. From those patterns, you build predictive models using machine learning algorithms that look for trends and predict future changes.

Data collection means pooling data by scraping, capturing, and loading it from multiple **sources**, including offline and online sources. High volumes of data collection or data creation can be the hardest part of a machine learning project, especially at scale.

How much data you need depends on how many features there are in the dataset. It is recommended to collect as much data as possible for good predictions. You can begin with small batches of data and see the result of the model. The most important thing to consider while data collection is diversity. Diverse data will help your model cover more scenarios. So, when focusing on how much data you need, you should cover all the scenarios in which the model will be used.

The **quantity of data** also depends on the complexity of your model. If it is as simple as license plate detection then you can expect predictions with small batches of data. But if you are working on higher levels of Artificial intelligence like medical AI, you need to consider huge volumes of data.

Before collecting the data, data scientists may understand the problem, its preferable solution and the data requirements. Based on these data requirements, sources of data will be identified and data will be collected. Data is the main ingredient of any Project. throughout the development of the project, data is required. Hence the process of identifying the data requirements, its collection and analysis will be done iteratively.

There are mainly two sources of data collection: Primary and Secondary.

Primary Sources are sources which are created to collect the data for analysis. Some of the examples are given below

Method	Description	Example
Survey	Gathering data from a population through interviews, questionnaires, or online forms. Useful for measuring opinions, behaviors, and demographics.	A researcher uses a questionnaire to understand consumer preferences for a new product.
Interview	Direct communication with individuals or groups to gather information. It can be structured, semi-structured, or unstructured.	An organization conducts an online survey to collect employee feedback about job satisfaction.
Observation	Watching and recording behaviors or events as they occur. Often used in ethnographic research or when direct interaction is not possible.	Observing children's play patterns in a schoolyard to understand social dynamics.
Experiment	Manipulating variables to observe their effects on outcomes. Used to establish cause-and-effect relationships.	Testing the effectiveness of different advertising campaigns on a group of people.
Marketing Campaign (using data)	Utilizing customer data to predict behavior and optimize campaign performance.	A company personalizes email marketing campaigns based on past customer purchases.
Questionnaire	A specific tool used within surveys - a list of questions designed to gather data from respondents. You can collect quantitative (numerical) or qualitative (descriptive) information.	A questionnaire might ask respondents to rate their satisfaction on a scale of 1 to 5 and also provide open-ended feedback.

Secondary data sources are where the data is already stored and ready for use. Data given in Books, Journals, News Papers, Websites, Internal transactional databases etc. can be reused for data analysis. Some methods of collecting secondary data are:

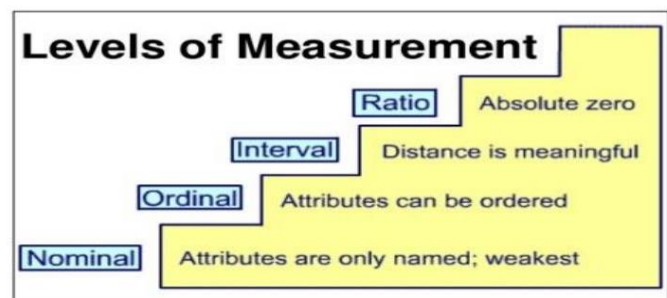
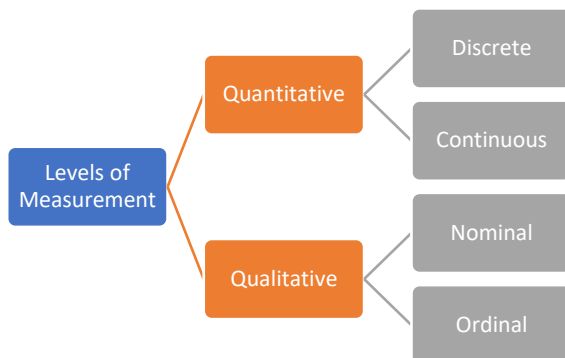
Method	Description	Example
Social Media Data Tracking	Collecting data from social media platforms like user posts, comments, and interactions.	Analyzing social media sentiment to understand audience reception towards a new product launch.
Web Scraping	Using automated tools to extract specific content and data from websites.	Scraping product information and prices from e-commerce websites for price comparison.
Satellite Data Tracking	Gathering information about the Earth's surface and atmosphere using satellites.	Monitoring weather patterns and environmental changes using satellite imagery.
Online Data Platforms	Websites offering pre-compiled datasets for various purposes.	Kaggle, GitHub etc.

3. EXPLORING DATA

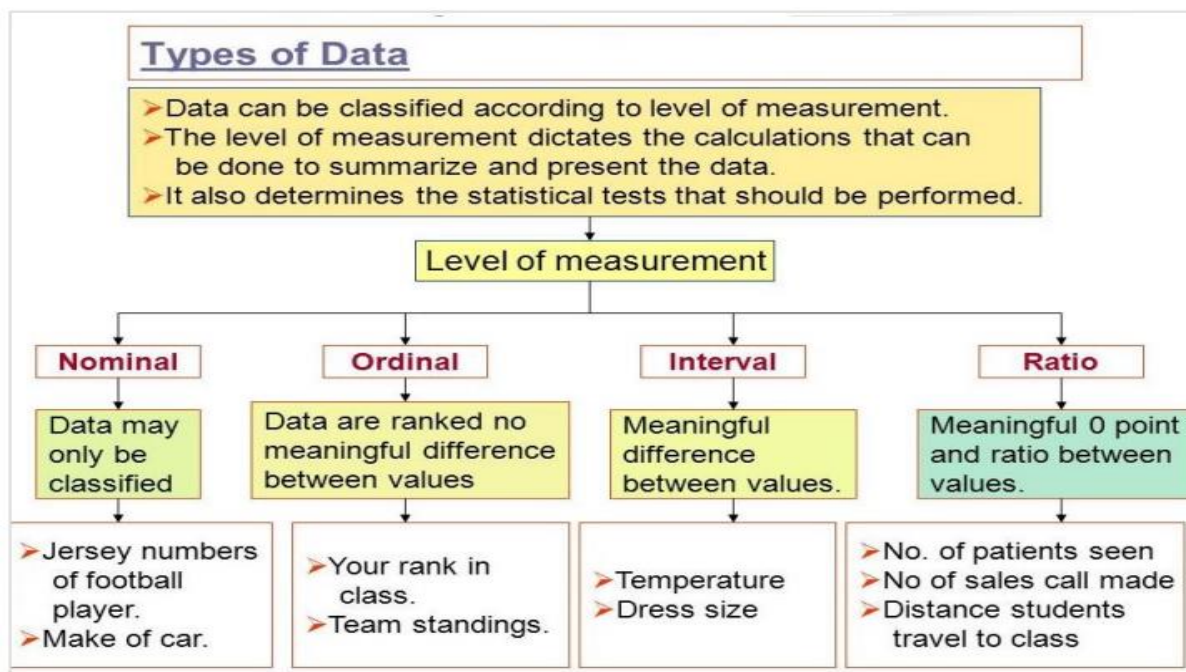
Exploring data is about "getting to know" the data: and its values - whether they are typical, unusual, spread out, or whether they are extremes. More importantly, during the process of exploration one gets an opportunity to identify and correct any problems in the data that would affect the conclusions drawn in any way during analysis.

Levels of Measurement

The way a set of data is measured is called the level of measurement. Not all data can be treated equally. It makes sense to classify datasets based on different criteria. Some are quantitative, and some qualitative. Some datasets are continuous and some are discrete. Qualitative data can be nominal or ordinal. And quantitative data can be split into two groups: interval and ratio.



<https://slideplayer.com/slide/8137745/>



1. Nominal Level

Nominal variables are like categories such as Mercedes, BMW or Audi, or like the four seasons – winter, spring, summer and autumn. They aren't numbers, and cannot be used in calculations and neither in any order or rank. The nominal level of measurement is the simplest or lowest of the four ways to characterize data. Nominal means "in name only".

Colours of eyes, yes or no responses to a survey, gender, smartphone companies, etc all deal with the nominal level of measurement. Even some things with numbers associated with them, such as a number on the back of a cricketer's T-shirt are nominal since they are used as "names"

for individual players on the field and not for any calculation purpose.

2. Ordinal Level

Ordinal data, is made up of groups and categories which follow a strict order. For e.g. if you have been asked to rate a meal at a restaurant and the options are: unpalatable, unappetizing, just okay, tasty, and delicious. Although the restaurant has used words not numbers to rate its food, it is clear that these preferences are ordered

from negative to positive or low to high, thus the data is qualitative, ordinal. However, the difference between the data cannot be measured. Like the nominal scale data, ordinal scale data cannot be used in calculations. A Hotel industry survey where the responses to questions about the hotels are accepted as, "excellent," "good," "satisfactory," and "unsatisfactory." These responses are ordered or ranked from the excellent service to satisfactory response to the least desired or unsatisfactory. But the differences between the two pieces of data as seen in the previous case cannot be measured. Another common example of this is the grading system where letters are used to grade a service or good. You can order things so that A is higher than a B, but without any other information, there is no way of knowing how much better an A is from a B.

Example 1:
Please indicate your marital status.
☐ Married ☐ Single ☐ Separated ☐ Divorced ☐ Widowed

Example 2:
Do you like or dislike chocolate ice cream?
☐ Like ☐ Dislike

Example 3:
Which of the following supermarkets have you shopped at in the last 30 days? Please check all that apply.
☐ Albertson's ☐ Winn-Dixie ☐ Publix ☐ Safeway ☐ Walmart

Example 4:
Please indicate your gender:
☐ Female ☐ Male ☐ Transgender

Example 1:
How likely are you to recommend the Santa Fe Grill to a friend?
Definitely Will Not Recommend 1 2 3 4 5 6 7 Definitely Will Recommend

Example 2:
Using a scale of 0–10, with "10" being Highly Satisfied and "0" being Not Satisfied At All, how satisfied are you with the banking services you currently receive from (read name of primary bank)?
Answer: _____

Example 3:
Please indicate how frequently you use different banking methods. For each of the banking methods listed below, circle the number that best describes the frequency you typically use each method.

Banking Methods	Never Use											Use Very Often
Inside the bank	0	1	2	3	4	5	6	7	8	9	10	
Drive-up window	0	1	2	3	4	5	6	7	8	9	10	
24-hour ATM	0	1	2	3	4	5	6	7	8	9	10	
Debit card	0	1	2	3	4	5	6	7	8	9	10	
Bank by mail	0	1	2	3	4	5	6	7	8	9	10	
Bank by phone	0	1	2	3	4	5	6	7	8	9	10	
Bank by Internet	0	1	2	3	4	5	6	7	8	9	10	

3. Interval Level

Data that is measured using the interval scale is similar to ordinal level data because it has a definite ordering but there is a difference between the two data. The differences between interval scale data can be measured though the data does not have a

starting point i.e. zero value. Temperature scales like Celsius ($^{\circ}\text{C}$) and Fahrenheit ($^{\circ}\text{F}$) are measured by using the interval scale.

In both temperature measurements, 40° is equal to 100° minus 60° . Differences make sense. But 0 degrees does not because, in both scales, 0 is not the absolute lowest temperature. Temperatures like -20°F and -30°C exist and are colder than 0. Interval level data can be used in calculations, but the comparison cannot be done. 80°C is not four times as hot as 20°C (nor is 80°F four times as hot as 20°F). There is no meaning to the ratio of 80 to 20 (or four to one)

Interval level:

- One category is higher than another (Ordered).
- There is a constant unit of measurement.
- Zero is just a point on the scale; or there is no natural zero point.
- Division of two numbers does not make sense.
- Scale or rank are good examples

- **EXAMPLE:** Temperature on the Fahrenheit scale.
 - Zero is just a point on the scale.
- **EXAMPLE:** Shoe size and dress size.
 - There is no natural zero point
- **EXAMPLE:** Years in which Whole Foods Market Inc. stock split.
 - Division of 1992 and 1993 does not make sense.
- **EXAMPLES:** Rank of Indi 500 results, Test scores.

4. Ratio Scale Level

Ratio scale data is like interval scale data, but it has a 0 point and ratios can be calculated. For example, the scores of four multiple choice statistics final exam questions were recorded

Examples

- **Discrete ratio data**
 - Number of children in a household
 - Number of vehicles owned in a specific period (5 years)
 - Number of male students in a classroom
- **Continuous ratio data**
 - Years of working experience
 - Number of hours spent in a waiting room
 - Driving speed (Mph)

as 80, 68, 20 and 92 (out of a maximum of 100 marks). The grades are computer generated. The data can be put in order from lowest to highest: 20, 68, 80, 92 or vice versa. The differences between the data have meaning. The score 92 is more than the score 68 by 24 points. Ratios can be calculated. The smallest score is 0. So, 80 is four times 20. The score of 80 is four times better than the score of 20. So, we can add, subtract, divide and multiply the two ratio level variables. Egg: Weight of a person. It has a real zero point, i.e. zero weight means that the person has no weight. Also, we can add, subtract, multiply and divide weights at the real scale for comparisons

Activity-1

Student Health Survey – Fill in the response and mention appropriate Level of Measurement.

Query	Response	Level of Measurement
Sex (Male/ Female)		
Height (in metres)		
Weight (in kilograms)		
Rate overall health (Excellent; Good; Average; Below Average; Poor)		
Pulse rate (in BPM)		
Body temperature (in Fahrenheit)		
Country of residence		

Activity-2.

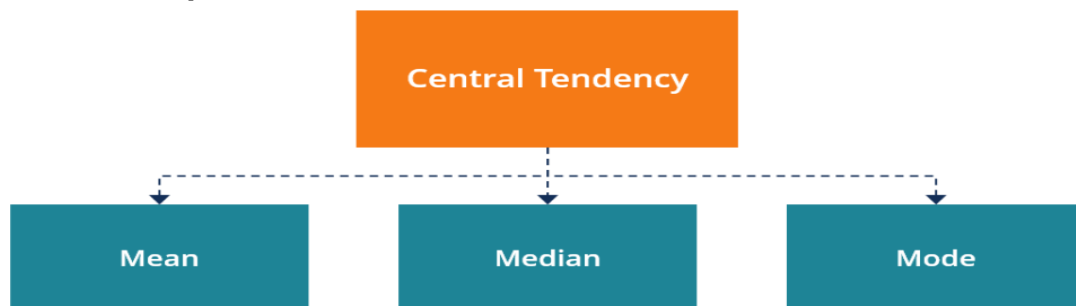
Indicate whether the variable is ordinal or not. Write the variable type, if it is not ordinal.

- ❖ Opinion about a new law (favour or oppose) _____
- ❖ Letter grade in an English class (A, B, C, etc.) _____
- ❖ Student rating of teacher on a scale of 1 – 10. _____

4. STATISTICAL ANALYSIS OF DATA

Measure of Central Tendency

Statistics is the science of data, which is in fact a collection of mathematical techniques that helps to extract information from data. For the AI perspective, statistics transforms observations into information that you can understand and share. Usually, Statistics deals with large dataset and Central tendency is used for the understanding and analysis purpose of data. **“Central tendency” is stated as the summary of a dataset in a single value that represents the entire distribution of data domain (or dataset).**



We can perform Statistical Analysis using Python programming language. For that we have to import the library statistics into the Program. Some important functions which we will use in future programs in this module are

- mean () → returns the mean of the data
- median () → returns the median of the data
- mode () → returns the mode of the data
- variance () → returns the variance of the data
- stdev () → returns the standard deviation of the data

Mean

In statistics, the mean (more technically the arithmetic mean or sample mean) can be estimated from a sample of examples drawn from the domain. It is a quotient obtained by dividing the total of the values of a variable by the total number of their observations or items.

$$M = \sum fx / n$$

where M = Mean

$\sum$ = Sum total of the scores

f = Frequency of the distribution

x = Scores

n = Total number of cases

Example -1

The set S = {5,10,15,20,30}

Mean of set S = $5+10+15+20+30/5 = 80/5 = 16$

Example- 2

Calculate the mean of the following grouped data

Class	Frequency
2 - 4	3
4 - 6	4
6 - 8	2
8 - 10	1

Solution

Class	Frequency (f)	Mid value (x)	f·x
2 -4	3	3	9
4 - 6	4	5	20
6 - 8	2	7	14
8 - 10	1	9	9
	n=10		$\sum f \cdot x = 52$

$$\text{Mean (M)} = \sum fxn / n$$

$$= 52 / 10$$
$$= 5.2$$

Program-1

There are 25 students in a class. Their heights are given below. Write a Python Program to find the mean.

heights → 145, 151, 152, 149, 147, 152, 151, 149, 152, 151, 147, 148, 155, 147, 152, 151, 149, 145, 147, 152, 146, 148, 150, 152, 151

```
import statistics
height = [145,151, 152, 149, 147, 152, 151,149,
          152, 151, 147, 148, 155, 147,152,151,
          149,145, 147, 152,146, 148, 150, 152, 151]
print ("Mean height of students", statistics.mean(height))
```

OUTPUT

```
Mean height of students 149.56
```

Median

The median is another measure of central tendency. It is positional value of the variables which divides the group into two equal parts, one part comprising all values greater than median and other part smaller than median.

Example-3

Following series shows marks in mathematics of students learning AI

17	32	35	15	21	41	32	11	10	20	27	28	30
----	----	----	----	----	----	----	----	----	----	----	----	----

We arrange this data in an ascending or descending order.

10, 11, 15, 17, 20, 21, 27, 28, 30, 32, 32, 35, 40

As 27 is in the middle of this data position wise, therefore Median = 27

Program-2

There are 25 students in a class. Their heights are given below. Write a Python Program to find the median.

heights → 145, 151, 152, 149, 147, 152, 151, 149, 152, 151, 147, 148, 155, 147, 152, 151, 149, 145, 147, 152, 146, 148, 150, 152, 151

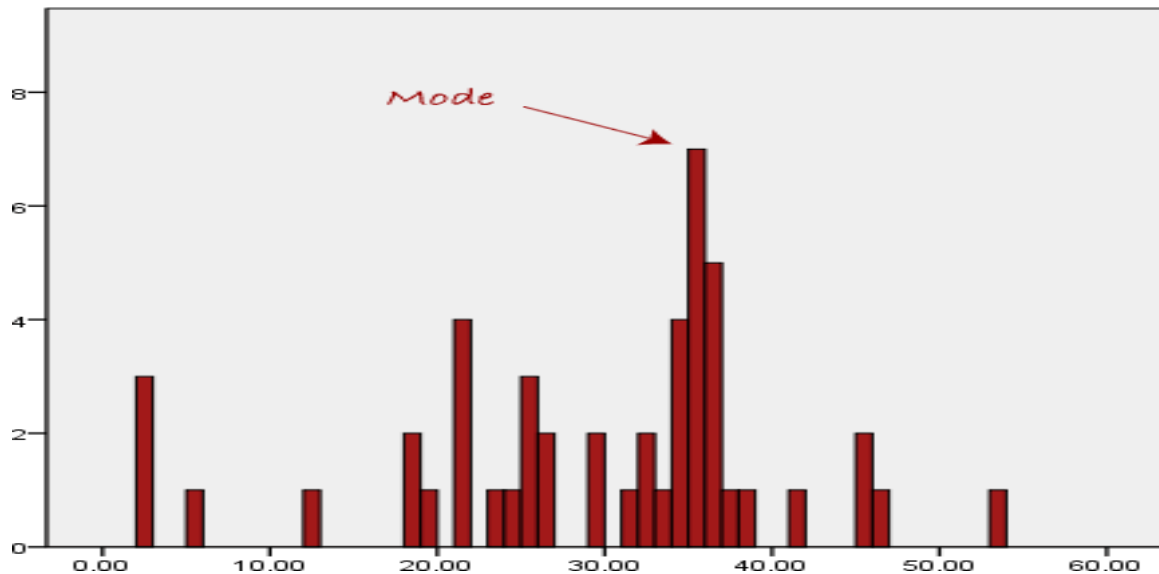
```
import statistics
height = [145,151, 152, 149, 147, 152, 151,149,
          152, 151, 147, 148, 155, 147,152,151,
          149,145, 147, 152,146, 148, 150, 152, 151]
print ("Median of height of students", statistics.median(height))
```

OUTPUT

```
Median of height of students 150
```

Mode

Mode is another important measure of central tendency of statistical series. It is the value which occurs most frequently in the data series. It represents the highest bar in a bar chart or histogram. An example of a mode is presented below:



Example-4 Age of 15 students of a class

Age (years) 22, 24, 17, 18, 17, 19, 18, 21, 20, 21, 20, 23, 22, 22, 22, 21, 24

- We arrange this series in ascending order as
17, 17, 18, 18, 19, 20, 20, 21, 21, 22, 22, 22,
- An inspection of the series shows that 22 occurs most frequently, hence

Mode=22



USING PYTHON

Program – 3

Write a program to find the mode

(heights → 145, 151, 152, 149, 147, 152, 151, 149, 152, 151, 147, 148, 155, 147, 152, 151, 149, 145, 147, 152, 146, 148, 150, 152, 151)

```
import statistics
height = [145, 151, 152, 149, 147, 152, 151, 149,
          152, 151, 147, 148, 155, 147, 152, 151,
          149, 145, 147, 152, 146, 148, 150, 152, 151]
print ("Mode of height of students", statistics.mode(height))
```

OUTPUT

Mode of height of students 152

In summary, when do we use mean, median and mode:

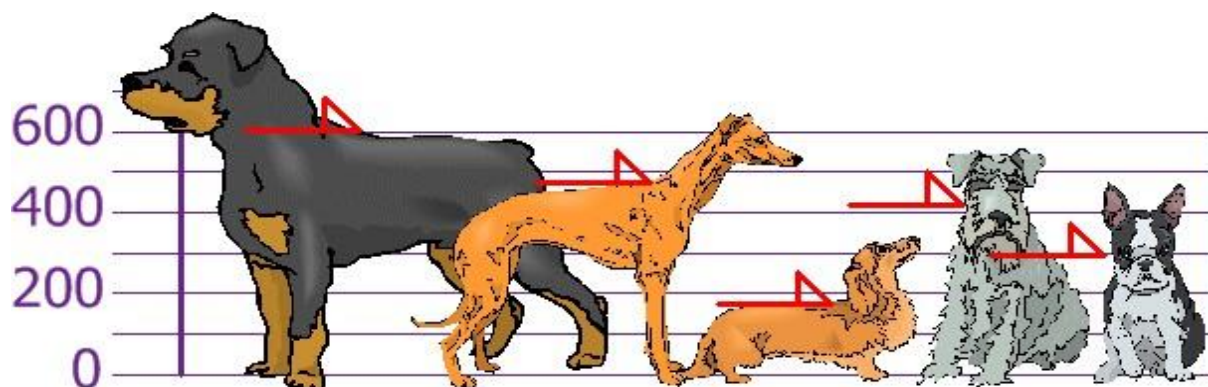
Mean	Median	Mode
The mean is a good measure of the central tendency when a data set contains values that are relatively evenly spread with no exceptionally high or low values.	The median is a good measure of the central value when the data include exceptionally high or low values. The median is the most suitable measure of average for data classified on an ordinal scale.	Mode is used when you need to find the distribution peak and peak may be many. For example, it is important to print more of the most popular books; because printing different books in equal numbers would cause a shortage of some books and an oversupply of others.

Variance and Standard Deviation

Measures of central tendency (mean, median and mode) provide the central value of the data set. Variance and standard deviation are the measures of dispersion (quartiles, percentiles, ranges), they provide information on the spread of the data around the centre.

Let us understand these two using a diagram:

Measure the height (at the shoulder) of 5 dogs (in millimetres)



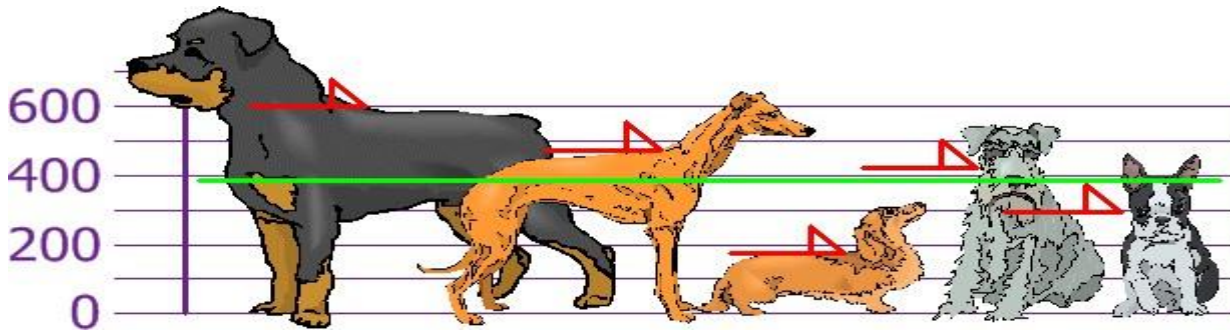
As you can see, their heights are:

600mm,
470mm,
170mm,
430mm and
300mm.

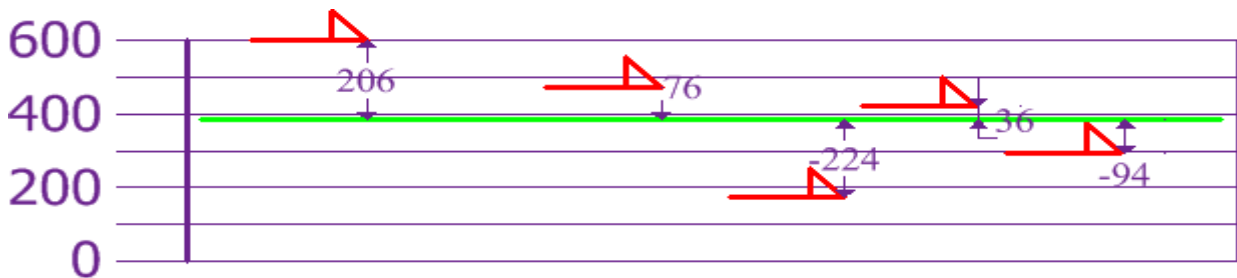
Let us calculate their mean,

$$\text{Mean} = (600 + 470 + 170 + 430 + 300) / 5 = 1970 / 5 = 394 \text{ mm}$$

Now let us plot again after taking mean height (The green Line)



Now, let us find the deviation of dogs' height from the mean height



Calculate the difference (from mean height), square them, and find the average. This average is the value of the variance.

$$\text{Variance} = [(206)^2 + (76)^2 + (-224)^2 + (36)^2 + (-94)^2] / 5 = 108520 / 5 = 21704$$

And standard deviation is the square root of the variance.

$$\text{Standard deviation} = \sqrt{21704} = 147.32$$

FORMULA

Variance	$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$
Standard deviation	$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$

Some important facts about variance and standard deviation

- A small variance indicates that the data points tend to be very close to the mean, and to each other.
- A high variance indicates that the data points are very spread out from the mean, and from one another.
- A low standard deviation indicates that the data points tend to be very close to the mean.
- A high standard deviation indicates that the data points are spread out over a large range of values.

Program -4

Write a program to find the variance and standard deviation. (heights → 145,151, 152, 149, 147, 152, 151, 149, 152, 151, 147, 148, 155, 147, 152, 151, 149, 145, 147, 152, 146, 148, 150, 152, 151)

```
import statistics
height = [145,151, 152, 149, 147, 152, 151,149,
          152, 151, 147, 148, 155, 147,152,151,
          149,145, 147, 152,146, 148, 150, 152, 151]
print ("Variance in the height of students", statistics.variance(height))
print ("Standard Deviation", statistics.stdev(height))
```

OUTPUT

```
Variance in the height of students 6.756666666666667
Standard Deviation 2.5993588953175872
```

5. REPRESENTATION OF DATA

According to Wikipedia, “Statistics is the discipline that concerns the collection, organization, analysis, interpretation and presentation of data. To achieve this task, statisticians summarize a large amount of data in a format that is compact and produces meaningful information. Without displaying values for each observation (from populations), it is possible to represent the data in brief while keeping its meaning intact using certain techniques called 'data representation'. It can also be defined as a technique for presenting large volumes of data in a manner that enables the user to interpret the important data with minimum effort and time.

Data representation techniques are broadly classified in two ways:

Non-Graphical technique:

Tabular form and case form: This is the old format of data representation not suitable for large datasets. Non-graphical techniques are not so suitable when our objective is to make some decisions after analysing a set of data.

Graphical Technique:

The visual display of statistical data in the form of points, lines, dots and other geometrical forms is most common. For a complex and large quantity of data, human brain is more comfortable in dealing if represented through visual format means Graphical or pictorial representation of the data using graph, chart, etc. is known as **Data visualization**.

It would not be possible to discuss the methods of construction of all types of diagrams and maps primarily due to time constraint. We will, therefore, describe the most commonly used graphs and the way they are drawn.

- Line graphs
- Bar diagrams
- Pie diagram
- Scatter Plots
- Histogram



Data Visualization is possible in python using the library **Matplotlib**. It is a comprehensive library that can be used to create a wide variety of plots, including line plots, bar charts, histograms, scatter plots, and more. Matplotlib is also highly customizable, allowing users to control the appearance of their plots in great detail. **pyplot** is a submodule of Matplotlib that provides a MATLAB-like interface to the library. pyplot also provides a number of convenience functions that make it easy to create simple plots.

Installing Matplotlib

```
pip install matplotlib
or
python -m pip install -U matplotlib
```

In the program we have to import the library.

import matplotlib.pyplot

Some of the common functions and its description is given below

Function Name	Description
title ()	Adds title to the chart/graph
xlabel ()	Sets label for X-axis
ylabel ()	Sets label for Y-axis
xlim ()	Sets the value limit for X-axis
ylim()	Sets the value limit for Y-axis
xticks ()	Sets the tick marks in X-axis
yticks()	Sets the tick marks in Y-axis
show ()	Displays the graph in the screen
savefig("address")	Saves the graph in the address specified as argument.
figure (figsize = <i>value in tuple format</i>)	Determines the size of the plot in which the graph is drawn. Values should be supplied in tuple format to the attribute figsize which is passed as argument.

List of Markers and its descriptions:

marker	symbol	description	marker	symbol	description
"."	•	point	"P"	+	plus (filled)
"."	.	pixel	"*"	★	star
"o"	●	circle	"h"	⬡	hexagon1
"v"	▼	triangle_down	"H"	⬢	hexagon2
"^"	▲	triangle_up	"+"	+	plus
"s"	■	square	"x"	×	x
"p"	⬠	pentagon	"D"	◆	diamond

List of Graph Colour Codes:

b	c	k
g	m	w
r	y	Colours

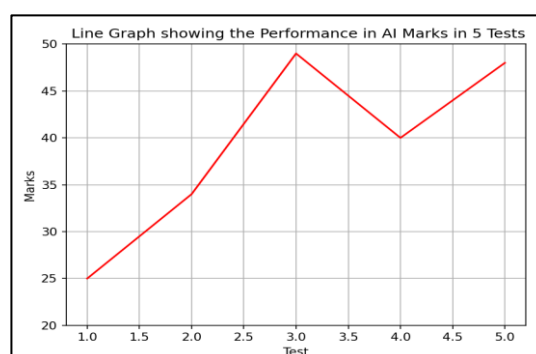
1. Line Graph

A line graph is a powerful tool used to represent **continuous data** along a numbered axis. It allows us to visualize **trends and changes** in data points over time. Line graphs are suitable for data that can take on any value within a specific range. The line can slope upwards, indicating an **increase**, or downwards, signifying a **decrease**, reflecting the changes in the data over time.

Example-5:

Kavya's AI marks for 5 consecutive tests is given below. Draw a line graph to Analyse her performance.

Test-1	Test-2	Test-3	Test-4	Test-5
25	34	49	40	48





USING PYTHON

Activity -3: Construct a simple line graph to represent the rainfall data of Kerala as shown in the table below

Month	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
Rainfall (cm)	7.5	6.3	3.5	1.8	1.2	25.8	19.7	20.3	15.9	22.4	18.6	11.2

Line chart is plotted in python using the function plot (). Colour of the line can be mentioned by giving the colour codes inside the plot function.

Attributes of plot function which are used inside plot () function are:

line width	sets the width of the line
line style	determines the style of line (solid, dashed, dot, dashdot)
marker, markersize, markeredgecolor	determines the marker's shape, size and marker edge colour respectively

Program-5

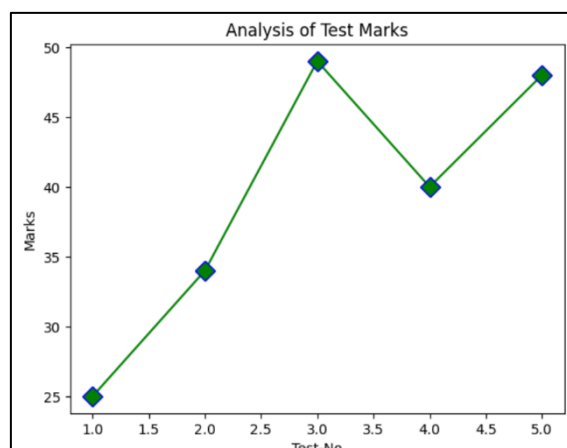
Write a program to draw a line chart, we use plot function (use Example 1)

```
import matplotlib.pyplot as pl
Test=[1,2,3,4,5]
Marks=[25, 34, 49, 40, 48]

pl.title ("Analysis of Test Marks")
pl.xlabel("Test-No")
pl.ylabel("Marks")

pl.plot(Test, Marks,'g', marker='D', markersize=10, markeredgecolor='blue', linestyle='solid')
pl.show()
```

OUTPUT

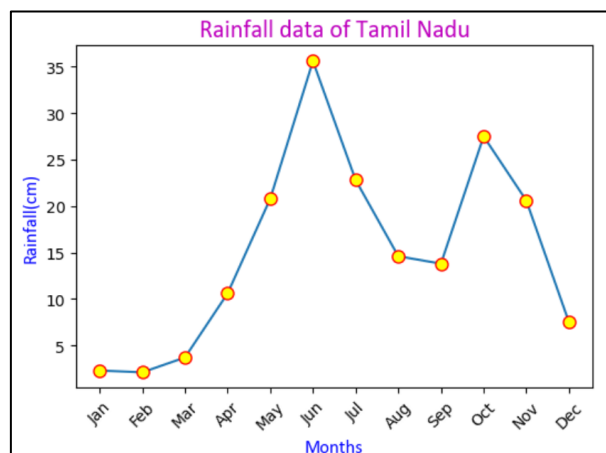


Program -6

Write a program to draw a line chart to visualize the comparative rainfall data for 12 months in Tamil Nadu using the CSV file "rainfall.csv".

```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv("rainfall.csv")
x=df['Months']
y=df['Rainfall(cm)']
plt.figure(figsize=(6,4))
plt.plot(x,y,marker='o', markersize=8, markeredgecolor='red', markerfacecolor='yellow')
plt.xticks(rotation = 45)
plt.xlabel("Months",fontname='Calibri',color='b',fontsize=12)
plt.ylabel("Rainfall(cm)",fontname='Calibri',color='b',fontsize=12)
plt.title("Rainfall data of Tamil Nadu",fontname='Calibri',color='m',fontsize=16)
plt.show()
```

OUTPUT



2. Bar Graph

A bar chart or bar graph is a graph that presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent. It is a good way to show relative sizes, i.e., to show comparison between different categories. The relative sizes of the bars allow for easy comparison between different categories.

Example-6

Create a bar graph to illustrate the distribution of students from various schools who attended a seminar on “Deep Learning”. The total number of students from each school is provided below.

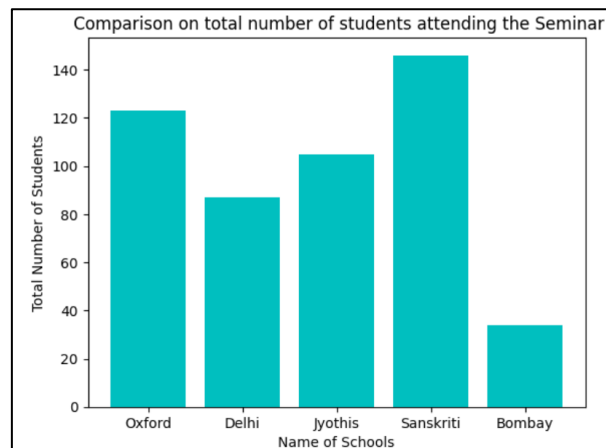
Oxford Public School	Delhi Public School	Jyothis Central School	Sanskriti School	Bombay Public School
123	87	105	146	34

```
import matplotlib.pyplot as plt
a=["Oxford", "Delhi", "Jyothis", "Sanskriti", "Bombay"]
b=[123, 87, 105, 146, 34]

plt.xlabel("Name of Schools")
plt.ylabel("Total Number of Students")

plt.title(" Comparison on total number of students attending the Seminar")
plt.bar(a,b,color='c')

plt.show()
```



Bar chart is plotted in python using the function bar ().

Attributes of bar function which are used inside bar () functions are:

color	determines the color of the bars
edgecolor	determines the colour of the bar edges
width	determines the width of the bars

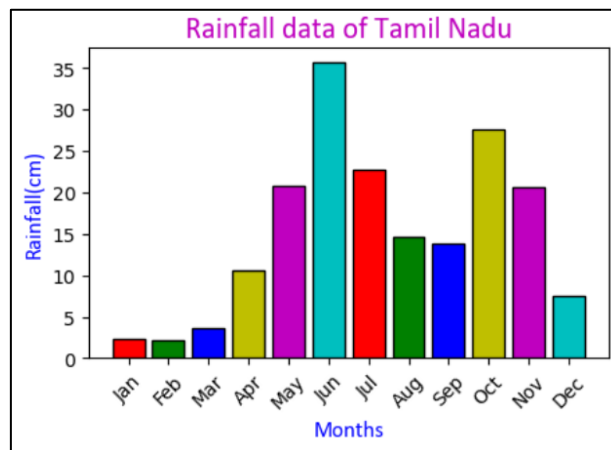
Program – 7



USING PYTHON

Write a program to draw a bar chart to visualize the comparative rainfall data for 12 months in Tamil Nadu using the CSV file "rainfall.csv".

```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv("rainfall.csv")
x=df['Months']
y=df['Rainfall(cm)']
plt.figure(figsize=(6,4))
c=['r','g','b','y','m','c']
plt.bar(x,y,color=c,edgecolor='k')
plt.xticks(rotation = 45)
plt.xlabel("Months",fontname='Calibri',color='b',fontsize=12)
plt.ylabel("Rainfall(cm)",fontname='Calibri',color='b',fontsize=12)
plt.title("Rainfall data of Tamil Nadu",fontname='Calibri',color='m',fontsize=16)
plt.show()
```



3. Histogram

Histograms are graphical representations of data distribution, with vertical rectangles depicting the frequencies of different value ranges. They are drawn on a natural scale, making it easy to interpret the central tendency, such as the mode, of the data. Despite their simplicity and ease of understanding, histograms have a limitation: they can only represent one data distribution per axis.

Example -7

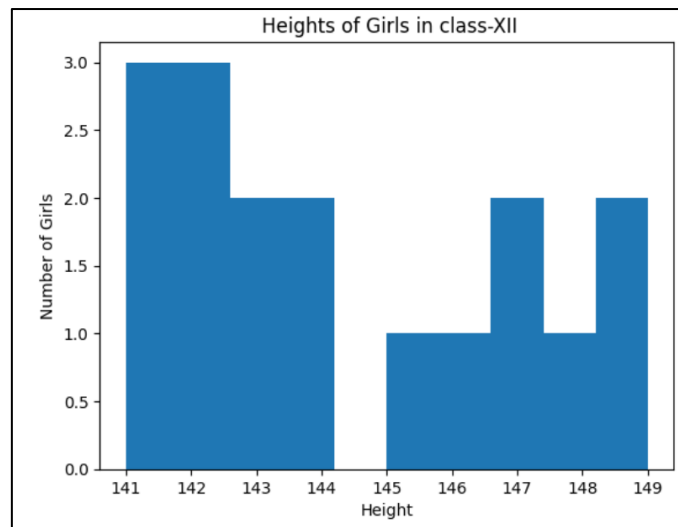
Given a dataset containing the heights of girls in class XII, construct a histogram to visualize the distribution of heights.

141,145,142,147,144,148,141,142,149,144,143,149,146,141, 147, 142, 143

Solution:

To draw a histogram from this, we first need to organize the data into intervals. These intervals are also called logical ranges or **bins**. After computing the number of girls in each interval, draw the graph. Histogram is plotted in python using the function `hist()`.

```
import matplotlib.pyplot as pl
a=[141, 145, 142, 147, 144, 148, 141, 142, 149, 144, 143,
   149, 146, 141, 147, 142, 143]
pl.ylabel("Number of Girls")
pl.xlabel("Height")
pl.title(" Heights of Girls in class-XII")
pl.hist(a)
pl.show()
```



4. Scatter Graph

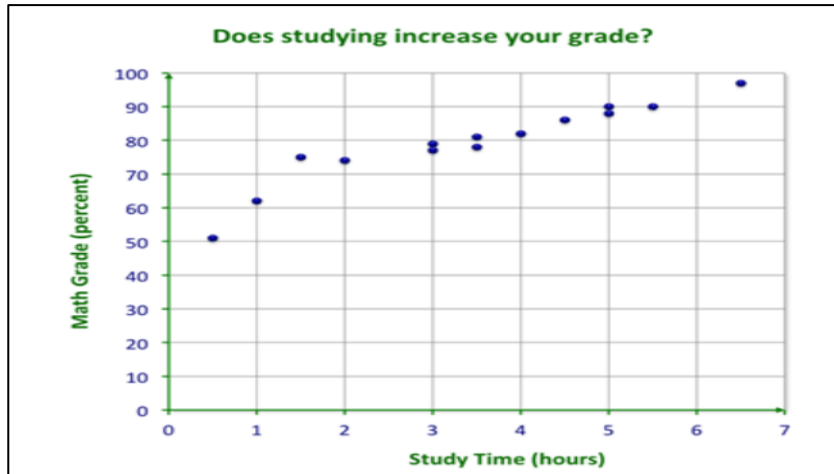
Scatter plots visually represent relationships between two variables by plotting data points along both the x and y axes. They reveal correlations, whether positive or negative, within paired data, showcasing trends and patterns. Essentially, scatter plots illustrate connections between variables through ordered pairs, making them useful for analyzing paired numerical data and situations where the dependent variable varies across different values of the independent variable. Their strength lies in their ability to clearly depict trends, clusters, and relationships within datasets.

Example-8

A student had a hypothesis for a science project. He believed that the more the students studied Math, the better their math scores would be. He took a poll in which he asked students the average number of hours that they studied per week during a given semester. He then found out the overall percentage that they received in their Math classes. His data is shown in the table below:

Study Time (Hours)	4	3.5	5	2	3	6.5	0.5	3.5	4.5	5
Maths Grade (%)	82	81	90	74	77	97	51	78	86	88

To understand this data, he decided to make a scatter plot. The independent variable, or input data, is the study time because the hypothesis is that the Math grade depends on the study time. That means that the Math grade is the dependent variable, or the output data. The input data is plotted on the x-axis and the output data is plotted on the y-axis.



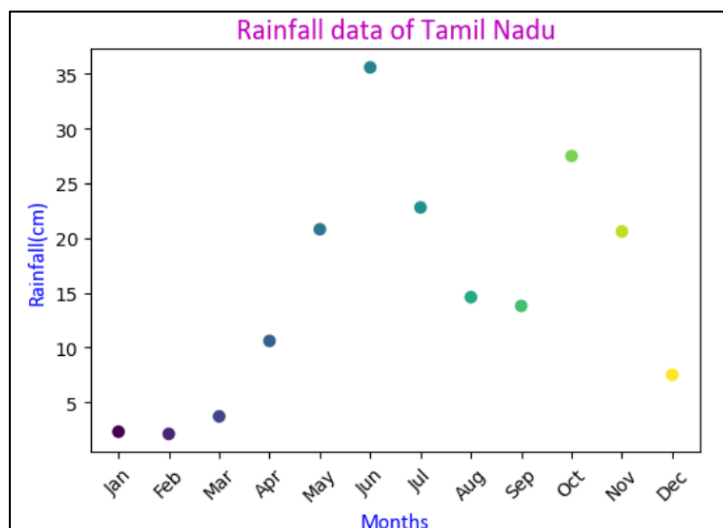
Scatterplot is plotted using the function `scatter ()`

Program-8

Write a program to draw a scatter chart to visualize the comparative rainfall data for 12 months in Tamil Nadu using the CSV file "rainfall.csv".

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv("rainfall.csv")
x=df['Months']
y=df['Rainfall(cm)']
plt.figure(figsize=(5,3))
plt.figure(figsize=(6,4))
colors = np.array([0, 10, 20, 30, 40, 45, 50, 60, 70, 80, 90, 100])
plt.scatter(x,y,c=colors,cmap='viridis')
plt.title("Rainfall data of Tamil Nadu",fontname='Calibri',color='m',fontsize=16)
plt.xticks(rotation = 45)
plt.xlabel("Months",fontname='Calibri',color='b',fontsize=12)
plt.ylabel("Rainfall(cm)",fontname='Calibri',color='b',fontsize=12)
plt.show()
```

OUTPUT



5. Pie Chart

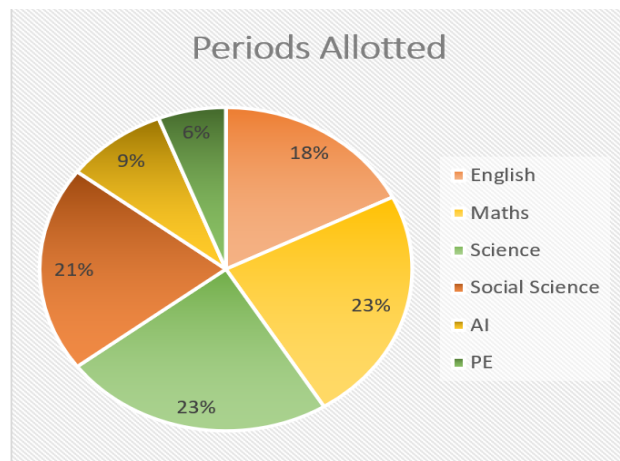
A pie chart is a circular graph divided into segments or sections, each representing a relative proportion or percentage of the total. Each segment resembles a slice of pie, hence the name. Pie charts are commonly used to visualize data from a small table, but it is recommended to limit the number of categories to seven to maintain clarity. However, zero values cannot be depicted in pie charts.

While useful for illustrating compositions or comparing parts of a whole, pie charts can be challenging to interpret and compare with data from other charts. They are not suitable for showing changes over time. Pie charts find applications in various domains such as business, education, and personal finance. In business, they can indicate the success or failure of products or services. In education, they can depict time allocations for different subjects. At home, pie charts can help visualize monthly expenses relative to income.

Example-9

Below given is a Pie chart drawn with the periods allotted for each subject in a week.

Subject	Periods Allotted
English	6
Maths	8
Science	8
Social Science	7
AI	3
PE	2

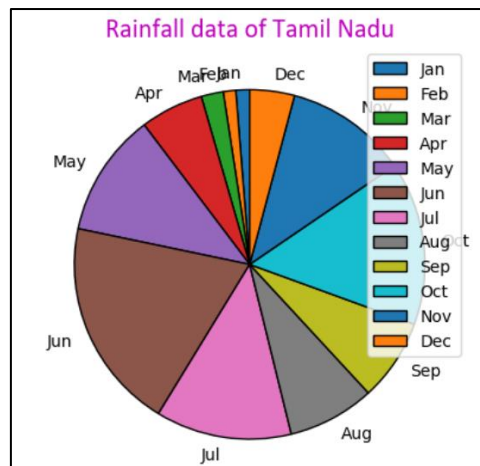


Pie Chart is plotted using the function pie ()

Program-9

Write a program to draw a pie chart to visualize the comparative rainfall data for 12 months in Tamil Nadu using the CSV file "rainfall.csv".

```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv("rainfall.csv")
x=df['Months']
y=df['Rainfall(cm)']
wp = { 'linewidth' : 1, 'edgecolor' : "black" }
plt.pie(y,labels=x, startangle=90,wedgeprops=wp)
plt. legend(loc='upper right')
plt.title("Rainfall data of Tamil Nadu",fontname='Calibri',color='m',fontsize=16)
plt.show()
```

6. INTRODUCTION TO MATRICES

The knowledge of matrices is necessary in all branches of mathematics. Matrix is one of the most powerful tools in Mathematics. In mathematics, matrix (plural matrices) is a rectangular arrangement of numbers. The numbers are arranged in tabular form as rows and columns. Matrices play a huge role in computer vision domain of AI. On the computer, the image is represented as a combination of pixels. This is represented mathematically as matrices!

Let us understand with the help of an example: Consider

Aditi bought 25 pencils 5 erasers

Adit bought 10 pencils 2 erasers

Manu bought 5 pencils 1 eraser

The above information can be arranged in tabular form as follows

	Pencils	Erasers
Aditi	25	5
Adit	10	2
Manu	5	1

And this can be represented as

Row1	$\begin{bmatrix} 25 & 5 \\ 10 & 2 \\ 5 & 1 \end{bmatrix}$
Row2	
Row3	

The entries in the rows represent number of pencils and erasers bought by Aditi, Adit and Manu respectively. Or in another form as

	Col1	Col2	Col3
Row1	$\begin{bmatrix} 25 & 10 & 5 \\ 5 & 2 & 1 \end{bmatrix}$		
Row2			

Here, the entries in the columns represent number of pencils and erasers bought by Aditi, Adit and Manu respectively. We denote matrices by capital letters, for example

$$A = \begin{bmatrix} 5 & 15 \\ -7 & \sqrt{2} \\ 12 & 0 \end{bmatrix}$$

Order of a matrix

A matrix has m rows and n columns. It is called a matrix of *order* $m \times n$ or simply $m \times n$ matrix (read as an m by n matrix). So, the matrix A in the above example is a 3×2 matrix. The number of elements are $m \times n \Rightarrow 3 \times 2 = 6$ elements. Each individual element is represented as a_{ij} where i represents row and j represents column. In general a_{ij} is an element lying in the i^{th} row and j^{th} column. We can also call it as the $(i, j)^{\text{th}}$ element of the matrix.

$$P = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

Operations on Matrices

1. **Addition of matrices** - The sum of two matrices is obtained by adding the corresponding elements of the given matrices. Also, the two matrices have to be of the same order. Example:

$$A = \begin{bmatrix} 3 & 2 \\ 4 & -1 \\ 2 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 6 & 3 \\ 5 & 9 \\ 3 & 2 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 3+6 & 2+3 \\ 4+5 & -1+9 \\ 2+3 & 0+2 \end{bmatrix} = \begin{bmatrix} 9 & 5 \\ 9 & 8 \\ 5 & 2 \end{bmatrix}$$

2. **Difference of matrices** - The difference $A - B$ is defined as a matrix where each element is obtained by subtracting the corresponding elements $(a_{ij} - b_{ij})$. Matrices A and B must be of the same order. Example

$$A = \begin{bmatrix} -2 & 1 \\ 6 & 10 \\ 5 & 3 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 3 \\ 2 & 9 \\ 3 & 1 \end{bmatrix}$$

$$A-B = \begin{bmatrix} -2+1 & 1-3 \\ 6-2 & 10-9 \\ 5-3 & 3-1 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 4 & 1 \\ 2 & 2 \end{bmatrix}$$

3. **Transpose of a matrix** - A matrix obtained by interchanging the rows and columns. Transpose of a matrix A is denoted by A' or A^T . Example

$$A = \begin{bmatrix} 8 & 7 \\ 2 & 5 \\ 4 & 6 \end{bmatrix} \quad A^T = \begin{bmatrix} 8 & 2 & 4 \\ 7 & 5 & 6 \end{bmatrix}$$

Order = 3×2

Order = 2×3

Applications of matrices in AI

Matrices are used throughout the field of machine learning for computing:

- Image Processing – Digital images can be represented using matrices. Each pixel on the image has a numerical value. These values represent the intensity of the pixels. A grayscale or black-and-white image has pixel values ranging from 0 to 255. Smaller values closer to zero represent darker shades, whereas bigger ones closer to 255 represent lighter or white shades. So, in a computer, every image is kept as a matrix of integers called a Channel.
- Recommender systems use matrices to relate between users and the purchased or viewed product(s)
- In Natural Language processing, vectors depict the distribution of a particular word in a document. Vectors are one-dimensional matrices.

7. DATA PREPROCESSING

Data preprocessing is a crucial step in the machine learning process aimed at making datasets more machine learning-friendly. It involves several processes to clean, transform, reduce, integrate, and normalize data:

1. Data Cleaning

1. Missing Data:

Missing data occurs when values are absent from the dataset, which can happen due to various reasons. Strategies for handling missing data include deleting rows or columns with missing values, inputting missing values with estimates, or using algorithms that can handle missing data.

2. Outliers:

Outliers are data points that significantly differ from the rest of the data, often due to errors or rare events. Dealing with outliers involves identifying and removing them, transforming the data, or using robust statistical methods to reduce their impact.

3. Inconsistent Data

Data with typographical errors, different data types etc are corrected and consistency among the data is observed.

4. Duplicate Data

Duplicate data will be identified and removed to ensure data integrity.

2. Data Transformation

Categorical variables are converted to Numerical variable. New features are identified and existing features are modified if needed.

3. Data Reduction

Dimensionality reduction, i.e. reducing the number of features of data set is done. If data set is too large to handle sampling techniques are applied.

4. Data Integration and Normalization

If data is stored in multiple sources or formats, they are merged or aggregated together. Then the data is normalized to ensure that all features have a similar scale and distribution which can improve machine learning models.

5. Feature Selection

The most relevant features that contribute the most to the target variable are selected and irrelevant data are removed.

8. DATA IN MODELLING & EVALUATION

After the data is pre-processed, it is split into two --Training dataset and Testing dataset. The training set is used to train the machine learning models, while the testing set is used to evaluate the performance of the trained models. While modelling, appropriate machine learning algorithms are chosen based on the nature of the problem (e.g., classification, regression, clustering) and the characteristics of the dataset.

Techniques such as train-test split, cross-validation, and error analysis are employed to estimate the model's generalization ability and identify areas for improvement. Train-Test Split trains the model with its training set and evaluates using the test set. Cross Validation ensures that the model's performance is consistent across different subsets of the data. Different types of evaluation techniques are applied on the model depending on the data. For classification problems, metrics like accuracy, precision, recall, F1-score, and ROC curve are commonly used. For regression problems, metrics like mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and R-squared are often used.

In today's world, knowing how to work with data is important. As Artificial Intelligence becomes more and more common, understanding data helps us use information better. It is like having a map to find your way through a big city. Being good with data helps us make smart decisions and use technology wisely.

EXERCISES

A. Multiple-choice questions

1. Which of the following best defines data literacy?
 - A) The ability to read and write data
 - B) The ability to find and use data effectively
 - C) The ability to analyse data using AI
 - D) The ability to collect and store data securely
2. What is the purpose of data preprocessing?
 - A) To make data more complex
 - B) To make data less accessible
 - C) To clean and prepare data for analysis
 - D) To increase the size of the dataset
3. How can missing data be handled in a dataset?
 - A) By ignoring it
 - B) By replacing missing values with estimates
 - C) By deleting rows or columns with missing values
 - D) By converting missing values to zero

Students, you have likely experienced the following scenarios in your daily lives:

- When browsing through online shopping apps, you receive recommendations to purchase socks immediately after looking at shoes.
- Online entertainment platforms suggest movies similar to ones you have previously watched.
- Facial recognition technology identifies you in photos on social media platforms.
- Chatbots engage in conversations with you in a natural manner.

MACHINE LEARNING IN A NUTSHELL

Machine Learning (ML) is a part of artificial intelligence (AI) that focuses on teaching computers to learn from data and make decisions without being explicitly programmed. Unlike traditional programming where developers provide precise instructions, ML algorithms learn from patterns and relationships in data. This allows them to generalize and make decisions on new, unseen data.

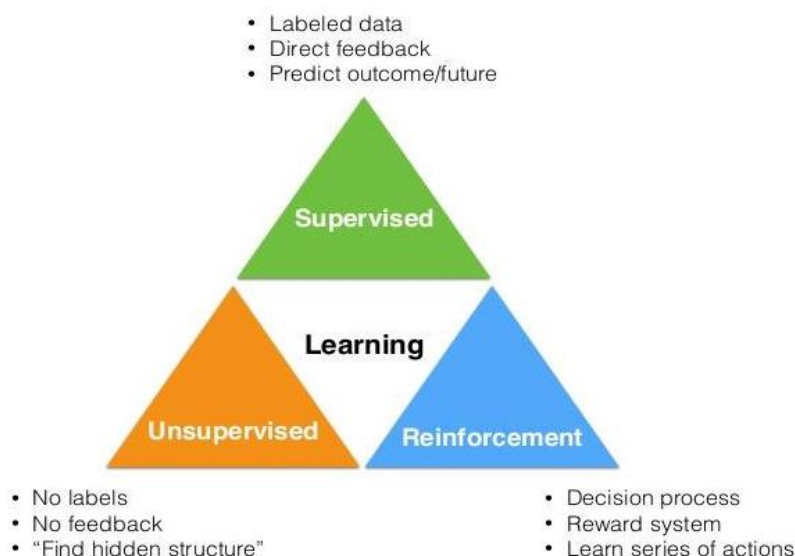
- ML algorithms learn from various **types of data**, including images, text, sensor readings, and historical records. Instead of hardcoding rules, ML models identify patterns and relationships within the data to make predictions or decisions.
- Some common ML **algorithms** include decision trees, neural networks, and support vector machines. Trained models serve as representations of the learned data, such as recognizing handwritten digits using a neural network.
- The **applications** of ML are vast and diverse. It powers recommendation systems like those used by Netflix, speech recognition, medical diagnosis, and autonomous vehicles. ML is also behind chatbots, personalized ads, and fraud detection systems.
- However, ML also presents **challenges**. Overfitting, where models become too specialized on training data, can lead to poor performance on new data. Bias in training data can result in biased predictions, and some models are difficult to interpret, acting as black boxes. Despite these challenges, ML transforms data into knowledge, enabling computers to learn, adapt, and make decisions autonomously.
- Artificial intelligence (AI) and machine learning (ML) have significantly **impacted** various aspects of our lives. From transportation and finance to healthcare and entertainment, AI algorithms are pervasive. They power self-driving cars, fraud detection systems, personalized shopping experiences, and virtual assistants like Siri and Alexa. As technology continues to evolve, the influence of AI and ML is only expected to grow, shaping the future of our society and culture.

Activity 1: [Autodraw](#) - Experience the power of machine learning with Autodraw! Autodraw combines machine learning with the creativity of talented artists, allowing you to draw things quickly and effortlessly.

TYPES OF MACHINE LEARNING

In Chapter 1, we introduced you to the fascinating world of artificial intelligence (AI) and its various learning mechanisms. We discussed three main types of machine learning: Supervised Learning, Unsupervised Learning, and Reinforcement Learning. These terms represent the algorithms that drive AI systems, serving as the building blocks for programming intelligent behavior and decision-making processes. Now, let us delve deeper into how these algorithms shape the landscape of AI applications.

- **Supervised learning** involves the model learning from **labeled data**, where the input data is accompanied by the correct output. The algorithm learns to **map input data to output labels** based on example input-output pairs provided during training. The goal is to learn a mapping function so that the model can make predictions on unseen data. Examples include *linear regression, logistic regression, decision trees, support vector machines, and neural networks*.
- **Unsupervised learning**, on the other hand, deals with **unlabelled data**, where the algorithm tries to **find hidden patterns or structure** without explicit guidance. The goal of this is to explore and discover inherent structures or relationships within the data, such as clusters or associations. Examples include *k-means clustering, hierarchical clustering, principal component analysis, and autoencoders*.
- Finally, **reinforcement learning** involves an agent learning to make decisions by interacting with an environment to maximize cumulative rewards. Through **trial and error**, the agent learns a policy or strategy to take actions that lead to the highest cumulative reward over time. These **rewards** serve as feedback, guiding the agent towards favorable actions, while **penalties** discourage undesirable behavior. This iterative learning approach is particularly effective in scenarios requiring a sequence of decisions, such as game-playing or robot control, where the agent adapts its behavior based on past experiences and outcomes. Examples include *Q-learning, deep Q-networks, policy gradients, and actor-critic methods*.



A. SUPERVISED LEARNING

Supervised learning stands out as one of the foundational types of Machine Learning. It is a powerful approach that allows machines to learn from labeled data, making predictions or decisions based on that learning. Within supervised learning, two primary types of algorithms emerge:

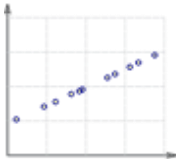
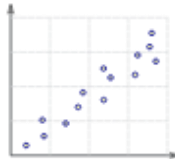
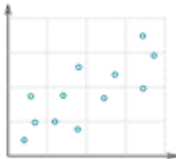
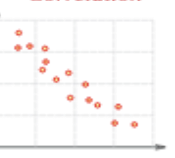
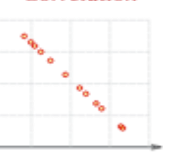
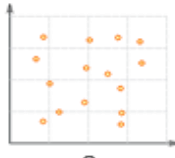
1. Regression – works with continuous data
2. Classification – works with discrete data

1. REGRESSION

Understanding Correlation: The Foundation of Regression Analysis

In data analysis, correlation is a fundamental concept that helps us grasp the relationship between variables, laying the groundwork for predictive modeling and insightful analysis. Correlation is a measure of the strength of a linear relationship between two quantitative variables (e.g. price, sales). If the change in one variable appears to be accompanied by a change in the other variable the two variables are said to be **correlated** and this inter dependence is called **correlation**.

Types of Correlation:

1. Positive Correlation: In a positive correlation, both variables move in the same direction. As one variable increases, the other also tends to increase, and vice versa.	Perfect Positive Correlation  1 High Positive Correlation  0.9 Low Positive Correlation  0.5
2. Negative Correlation: Conversely, in a negative correlation, variables move in opposite directions. An increase in one variable is associated with a decrease in the other, and vice versa.	Low Negative Correlation  -0.5 High Negative Correlation  -0.9 Perfect Negative Correlation  -1
3. Zero Correlation: When there is no apparent relationship between two variables, they are said to have zero correlation. Changes in one variable do not predict changes in the other	No Correlation  0

Correlation can have a value:

- 1 is a perfect positive correlation
- 0 is no correlation (the values don't seem linked at all)
- -1 is a perfect negative correlation

Causation

Causation indicates that one event is the result of the occurrence of the other event. Example: Since there is hot weather, the person will use more sunscreen or eat more ice cream.

Sometimes, these two events may be correlated also. Example: smoking causes an increase in the risk of developing lung cancer or it can correlate with another like-smoking is correlated with alcoholism, but it does not cause alcoholism. Therefore, we can say causation is not always correlation.

PEARSON'S R

Pearson's correlation coefficient (often denoted as Pearson's **r**) is one of the crucial factors to consider when assessing the appropriateness of regression analysis. **Pearson's r measures the strength and direction of the linear relationship between two continuous variables.** In the context of regression analysis, a high degree of correlation between the independent and dependent variables suggests that there may be a meaningful relationship to explore using regression techniques.

The **requirements** when considering the use of Pearson's correlation coefficient are:

1. Scale of measurement should be interval or ratio.
2. Variables should be approximately normally distributed.
3. The association should be linear.
4. There should be no outliers in the data.

Pearson's **r** is calculated using the formula:

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

r can take a range of values from +1 to -1

- A value of 0 indicates that there is no association between the two variables.
- A value greater than 0 indicates a positive association; that is, as the value of one variable increases, so does the value of the other variable.
- A value less than 0 indicates a negative association; that is, as the value of one variable increases, the value of the other variable decreases.

Example 1

In the example below of 6 people with different ages and different weight, let us try calculating the value of the Pearson **r**.

Sr. No	Age (x)	Weight (y)
1	40	78
2	21	70
3	25	60
4	31	55
5	38	80
6	47	66

Solution:

For the Calculation of the Pearson Correlation Coefficient, we will first calculate the following values:

Sr. No	Age (x)	Weight (y)	xy	x ²	y ²
1	40	78	3120	1600	6084
2	21	70	1470	441	4900
3	25	60	1500	625	3600
4	31	55	1705	961	3025
5	38	80	3040	1444	6400
6	47	66	3102	2209	4356
Total (Σ)	202	409	13937	7280	28365

Here the total number of people is 6 so, **n=6**

Now the calculation of the Pearson R is as follows:

E12	$=(6*D10-B10*C10)/SQRT((6*E10-B10^2)*(6*F10-C10^2))$				
	A	D	E	F	G
3	Sr. No	Age (x)	Weight (y)	xy	x ²
4	1	40	78	3120	1600
5	2	21	70	1470	441
6	3	25	60	1500	625
7	4	31	55	1705	961
8	5	38	80	3040	1444
9	6	47	66	3102	2209
10	Total (Σ)	202	409	13937	7280
11					
12	Pearson Correlation Coefficient (r)	0.35			
13					

$$r = (n (\sum xy) - (\sum x)(\sum y)) / (\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]})$$

$$r = (6 * (13937) - (202)(409)) / (\sqrt{[6 * 7280 - (202)^2] * [6 * 28365 - (409)^2]})$$

$$r = (6 * (13937) - (202) * (409)) / (\sqrt{[6 * 7280 - (202)^2] * [6 * 28365 - (409)^2]})$$

$$r = (83622 - 82618) / (\sqrt{[43680 - 40804] * [170190 - 167281]})$$

$$r = 1004 / (\sqrt{[2876] * [2909]})$$

$$r = 1004 / (\sqrt{8366284})$$

$$r = 1004 / 2892.452938$$

$$r = 0.35$$

The value of the Pearson correlation coefficient is **0.35**

It is important to note that, regression analysis may not be suitable in certain situations:

1. **No Correlation:** If there is no correlation between the variables, meaning they change independently of each other, regression analysis will not provide meaningful insights or predictions.
2. **Non-linear Relationships:** While regression can model linear relationships well, it may not capture more complex, non-linear relationships effectively. In such cases, alternative techniques like polynomial regression or non-linear regression may be more appropriate.

3. **Outliers:** Outliers, or extreme data points, can disproportionately influence the regression model and lead to inaccurate predictions. In the presence of outliers, it is essential to assess their impact and consider alternative modeling approaches.
4. **Violation of Assumptions:** Regression analysis relies on certain assumptions, such as the linearity of relationships and the absence of multicollinearity (high correlation between predictor variables). If these assumptions are violated, the results of the regression analysis may be unreliable.

REGRESSION

Regression is a statistical technique used to model the relationship between a dependent variable and one or more independent variables. Its primary objective is to understand and predict the value of the dependent variable based on the values of the independent variables. In simpler terms, regression helps us understand how changes in one or more variables are associated with changes in another variable.

Regression analysis is particularly useful when dealing with **continuous data**, where variables can take on any value within a certain range. For example, variables such as height, temperature, salary, and time are all continuous, meaning they can be measured along a continuous scale. In regression, these continuous variables are used to predict or explain the variability in another continuous variable, known as the dependent variable. By analyzing the relationship between the independent and dependent variables, regression allows us to make predictions and understand how changes in one variable may impact the other. This makes regression a powerful tool for forecasting, prediction, and understanding complex relationships in various fields such as economics, social sciences, and healthcare.

When we make a distribution in which there is an involvement of more than one variable, then such an analysis is called Regression Analysis. It generally focuses on finding or rather predicting the value of the variable that is dependent on the other.

Let there be two variables x and y . If y depends on x , then the result comes in the form of a simple regression. Furthermore, we name the variables x and y as:

y – Regression / Dependent / Explained Variable.
It is the variable we want to predict or understand.

x – Independent / Predictor / Explanator Variable
It is used to predict or explain changes in the dependent variable.

Therefore, if we use a simple linear regression model where y depends on x , then the regression line of y on x is:

$$y = a + bx + e$$

In this equation,

- **a** represents the **intercept** of the regression line with the y -axis.
- **b** represents the **slope** of the regression line, indicating the rate of change in y for a unit change in x .
- **e** represents the **error or residual**, which accounts for the difference between the observed values of y and the values predicted by the regression equation.

FINDING THE LINE

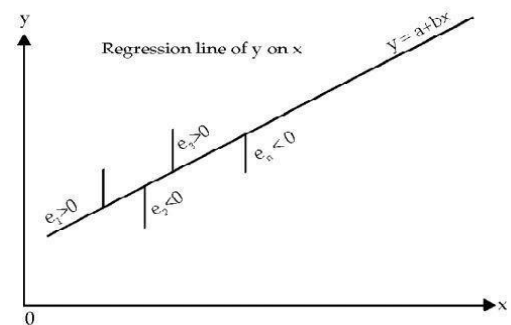
Regression analysis relies on the concept of the regression line or curve, which represents the best-fit relationship between the variables involved. This line or curve is determined by minimizing the differences between the observed values of the dependent variable and the values predicted by the regression model.

The **least squares method** is commonly employed to find this best-fit line or curve. This method minimizes the squared differences between observed and predicted values, ensuring that the regression line captures the overall trend or pattern in the data as accurately as possible. By systematically estimating the parameters of the regression model, such as the intercept and slope, the least squares method provides a precise description of the relationship between the variables.

Through the least squares method, regression analysis yields estimate of the regression coefficients that define the best-fit relationship between the variables. These coefficients allow for making predictions about the dependent variable based on the values of the independent variable(s) with greater accuracy and reliability. As a result, this is widely used in regression analysis.

Properties of the Regression line:

- The line minimizes the sum of squared difference between the observed values (actual y-value) and the predicted value ($\hat{y}$ value)
- The line passes through the mean of independent and dependent features.



Example 1

In the example of 6 people with different ages and different weight, let us draw the line of best fit in Excel.

Sr. No	Age (x)	Weight (y)
1	40	78
2	21	70
3	25	60
4	31	55
5	38	80
6	47	66

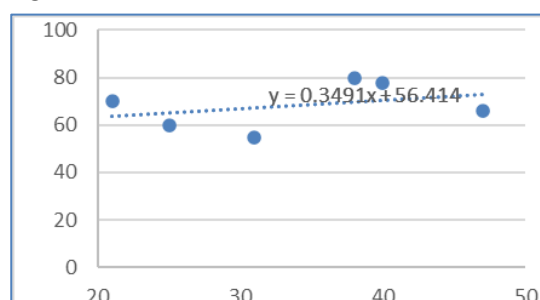
Solution:

Step 1: Select the Age and Weight.

Step 2: Insert a scatter chart and make changes to the following:

Trendline Name: **Linear**, check **Display Equation on Chart**

X axis **minimum : 20**



Step 3: Let us verify the values of slope and intercept using slope() and intercept() function in excel.

Step 4: Click on any cell and type =slope(Now, select the values of Weight, and then type comma. Now, select the values of Age and press enter.

Step 5: Click on any cell and type =intercept(Now, select the values of Weight, and then type comma. Now, select the values of Age and press enter.

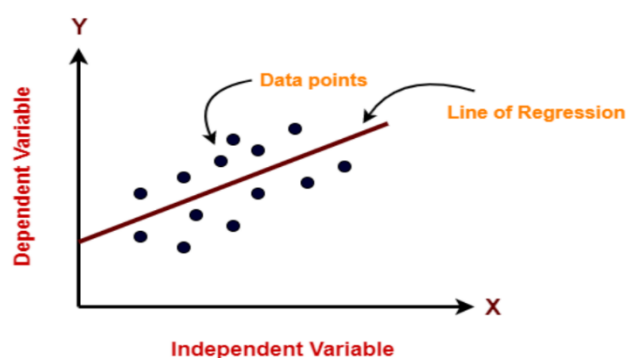
Some of the regression algorithms include Linear Regression, Logistic Regression, Decision Tree Regression, Random Forest Regression. Let us learn about Linear Regression.

Linear Regression

Linear regression is one of the most basic types of regression in machine learning. The linear regression model consists of a predictor variable and a dependent variable related linearly to each other. In case the data involves more than one independent variable, then linear regression is called multiple linear regression models.

Linear regression is further divided into two types:

- a) **Simple Linear Regression:** The dependent variable's value is predicted using a single independent variable in simple linear regression.
- b) **Multiple Linear Regression:** In multiple linear regression, more than one independent variable is used to predict the value of the dependent variable.



Applications of Linear Regression:

- **Market Analysis:** Linear regression helps understand how different factors like pricing, sales quantity, advertising, and social media engagement relate to each other in the market.
- **Sales Forecasting:** It predicts future sales by analyzing past sales data along with factors like marketing spending, seasonal trends, and consumer behavior.
- **Predicting Salary Based on Experience:** Linear regression estimates a person's salary based on their years of experience, education, and job role, aiding in recruitment and compensation planning.
- **Sports Analysis:** Linear regression analyzes player and team performance by considering statistics, game conditions, and opponent strength, assisting coaches and team management in decision-making.
- **Medical Research:** Linear regression examines relationships between factors like age, weight, and health outcomes, helping researchers identify risk factors and evaluate interventions.

Advantages of Linear regression

- Simple technique and easy to implement
- Efficient to train the machine on this model

Disadvantages of Linear regression

1. Sensitivity to outliers, which can significantly impact the analysis.
2. Limited to linear relationships between variables.

<https://www.javatpoint.com/linear-regression-in-machine-learning>

For Advanced Learners – Python program for Linear regression

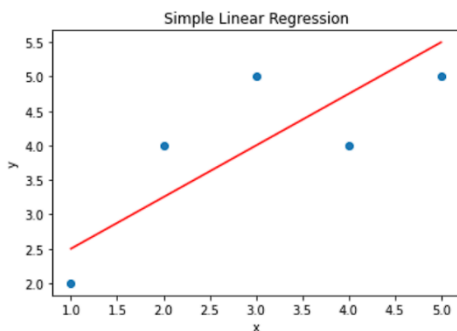
Import **scipy** and draw the line of Linear Regression:

```
import numpy as np
import matplotlib.pyplot as plt
# Sample data
x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 4, 5, 4, 5])
# Calculate mean and standard deviation
x_mean = np.mean(x)
y_mean = np.mean(y)
x_std = np.std(x)
y_std = np.std(y)
# Calculate covariance and slope
covariance = np.sum((x - x_mean) * (y - y_mean)) / (len(x) - 1)
slope = covariance / (x_std**2)
# Calculate y-intercept (b)
intercept = y_mean - slope * x_mean
# Predicted values
y_pred = slope * x + intercept
# Plot data and regression line
plt.scatter(x, y)
plt.plot(x, y_pred, color='red')
# Add labels and title
plt.xlabel('x')
plt.ylabel('y')
plt.title('Simple Linear Regression')
# Show the plot
plt.show()
# Print slope and intercept
print(f"Slope: {slope:.2f}")
print(f"Intercept: {intercept:.2f}")
```

This program:

- Imports numpy for numerical calculations and matplotlib.pyplot for plotting.
- Defines sample data for x and y. You can replace this with your own data.
- Calculates mean, standard deviation, covariance, and slope.
- Calculates y-intercept based on slope and mean.
- Predicts y values for given x using the linear equation.
- Plots the data points and the regression line.
- Prints the estimated slope and intercept values.

The expected output of the above program would be



Slope: 0.75
Intercept: 1.75

REFERENCES

Video links:

- <https://www.youtube.com/watch?v=rHeaoaiBM6Y>
- https://www.youtube.com/watch?v=VWCRDH1_rv0
- https://www.youtube.com/watch?v=YUPagM-OB_M
- <https://www.youtube.com/watch?v=CtsRRUddV2s&t=70s>

2. CLASSIFICATION

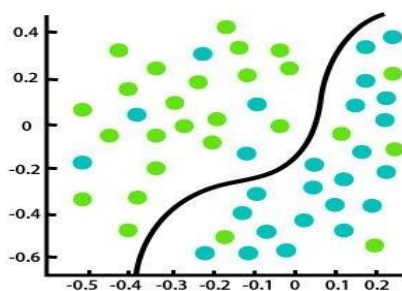
Classification is a fundamental concept in artificial intelligence and machine learning that involves categorizing data into predefined classes or categories. The main objective of classification is to assign labels to data instances based on their features or attributes. In classification, the data is typically labeled with class labels or categories, and the goal is to build a model that can accurately assign these labels to new, unseen data instances. This process is supervised learning, where the model learns from labeled training data to make predictions on unseen data.

For example, let us say, you live in a gated housing society and your society has separate dustbins for different types of waste: paper waste, plastic waste, food waste and so on. What you are basically doing over here is classifying the waste into different categories and then labeling each category. In the picture given below, we are assigning the labels 'paper', 'metal', 'plastic', and so on to different types of waste.

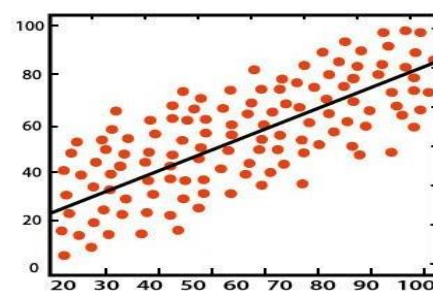


Look at the two graphs below and suggest which graph represents the classification problem.

Graph 1



Graph 2



How Classification Works

In classification tasks within machine learning, the process revolves around categorizing data into distinct groups or classes based on their features. Here is how it typically works:

- **Classes or Categories:** Data is divided into different classes or categories, each representing a specific outcome or group. For example, in a binary classification scenario, there are two classes: positive and negative.
- **Features or Attributes:** Each data instance is described by its features or attributes, which provide information about the instance. These features are crucial for the classification model to differentiate between different classes. For instance, in email classification, features might include words in the email text, sender information, and email subject.
- **Training Data:** The classification model is trained using a dataset known as training data. This dataset consists of labelled examples, where each data instance is associated with a class label. The model learns from this data to understand the relationship between the features and the corresponding class labels.

- **Classification Model:** An algorithm or technique is used to build the classification model. This model learns from the training data to predict the class labels of new, unseen data instances. It aims to generalize from the patterns and relationships in the training data to make accurate predictions.
- **Prediction or Inference:** Once trained, the classification model is used to predict the class labels of new data instances. This process, known as prediction or inference, relies on the learned patterns and relationships from the training data..

Types of classification

The four main types of classification are:

- 1) Binary Classification
- 2) Multi-Class Classification
- 3) Multi-Label Classification
- 4) Imbalanced Classification

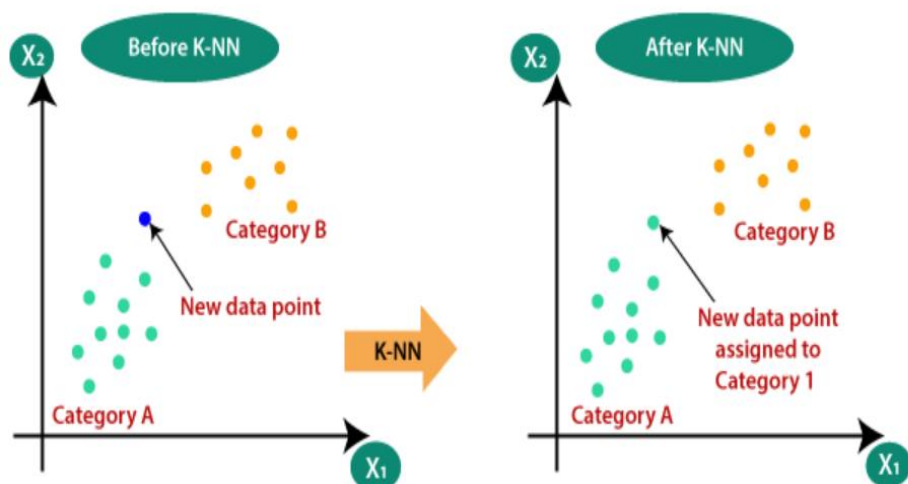
Classification Type	Binary Classification	Multi-Class Classification	Multi-Label Classification	Imbalanced Classification
Description	Classification tasks with two class labels.	Classification tasks with more than two class labels.	Classification tasks where each example may belong to multiple class labels.	Classification tasks with unequally distributed class labels, typically with a majority and minority class.
Example	• Email spam detection - spam or not • Conversion prediction - buy or not • Medical test - Cancer detected or not • Exam results - pass/fail	• Face classification • Plant species classification • Optical character recognition • Image classification into thousands of classes	• Photo classification - objects present in the photo (bicycle, apple, person, etc.)	• Fraud detection • Outlier detection • Medical diagnostic tests

K- Nearest Neighbour algorithm (KNN)

The K-Nearest Neighbors algorithm, commonly known as KNN or k-NN, is a versatile non-parametric supervised learning technique used for both classification and regression tasks. It operates based on the principle of proximity, making predictions or classifications by considering the similarity between data points.

Why KNN Algorithm is Needed:

KNN is particularly useful when dealing with classification problems where the decision boundaries are not clearly defined or when the dataset does not have a well-defined structure. It provides a simple yet effective method for identifying the category or class of a new data point based on its similarity to existing data points.



Steps involved in k-NN

- Select the number K of the neighbors
- Calculate the Euclidean distance of K number of neighbors
- Take the K nearest neighbors as per the calculated Euclidean distance.
- Among these k neighbors, count the number of the data points in each category.
- Assign the new data points to that category for which the number of the neighbor is maximum.
- Our model is ready.

Applications of KNN:

- Image recognition and classification
- Recommendation systems
- Healthcare diagnostics
- Text mining and sentiment analysis
- Anomaly detection

Advantages of KNN:

- Easy to implement and understand.
- No explicit training phase; the model learns directly from the training data.
- Suitable for both classification and regression tasks.
- Robust to outliers and noisy data.

Limitations of KNN:

- Computationally expensive, especially for large datasets.
- Sensitivity to the choice of distance metric and the number of neighbors (K).
- Requires careful preprocessing and feature scaling.
- Not suitable for high-dimensional data due to the curse of dimensionality.

For advanced learners – Python Program for K Nearest Neighbor Algorithm

importing libraries

import numpy as nm

import matplotlib.pyplot as mtp

import pandas as pd

#importing datasets

data_set= pd.read_csv('user_data.csv')

#Extracting Independent and dependent Variable

x= data_set.iloc[:, [2,3]].values

y= data_set.iloc[:, 4].values

Splitting the dataset into training and test set.

from sklearn.model_selection **import** train_test_split

x_train, x_test, y_train, y_test= train_test_split(x, y, test_size= 0.25, random_state=0)

#feature Scaling

from sklearn.preprocessing **import** StandardScaler

st_x= StandardScaler()

x_train= st_x.fit_transform(x_train)

x_test= st_x.transform(x_test)

Reference: <https://www.javatpoint.com/k-nearest-neighbor-algorithm-for-machine-learning>

Note to the teacher: Can introduce Teachable Machine to develop any image classifier problem.

Website: <https://teachablemachine.withgoogle.com/>

REFERENCES

Video Session:

Classification: <https://www.youtube.com/watch?v=hBKI7XvD8R8>

KNN Algorithm: <https://www.youtube.com/watch?v=0p0o5cmgLdE>

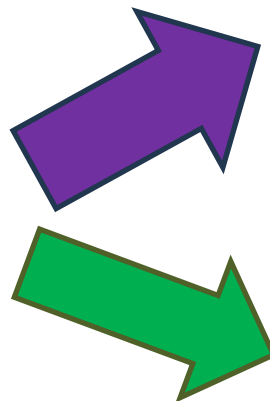
B. UNSUPERVISED LEARNING

3. CLUSTERING

Clustering, or cluster analysis, is a machine learning technique used to group **unlabeled dataset** into clusters or groups based on similarity. Clustering aims to organize data points into groups where points within the same group are more similar to each other than to those in other groups. It involves finding patterns or structures in the data without the need for predefined class labels. It does it by finding some similar patterns in the unlabelled dataset such as shape, size, color, behavior, etc., and divides them as per the presence and absence of those similar patterns. It is an **unsupervised learning** method, hence no supervision is provided to the algorithm, and it deals with the unlabeled dataset.

The clustering technique is commonly used for **statistical data analysis**.

Example: Let us consider the clustering technique using a real-world example. Imagine you are visiting a shopping center where items are grouped together based on their similarities. For instance, in the fruits section, you will find apples, bananas, and grapes neatly arranged together. This organization makes it convenient for shoppers to locate specific items they are looking for.



Based on colour



Based on size

In a similar way, clustering algorithms group similar data points together based on common characteristics or features. This approach helps in organizing and making sense of large datasets in various tasks, such as market segmentation, image recognition, and customer segmentation.

How Clustering Works:

To cluster data effectively, follow these key steps:

- 1) **Prepare the Data:** Select the right features for clustering and make sure the data is ready by scaling or transforming it as needed.
- 2) **Create Similarity Metrics:** Define how similar data points are by comparing their features. This similarity measure is crucial for clustering.
- 3) **Run the Clustering Algorithm:** Apply a clustering algorithm to group the data. Choose one that works well with your dataset size and characteristics.
- 4) **Interpret the Results:** Analyze the clusters to understand what they represent. Since clustering is unsupervised, interpretation is essential for assessing the quality of the clusters.

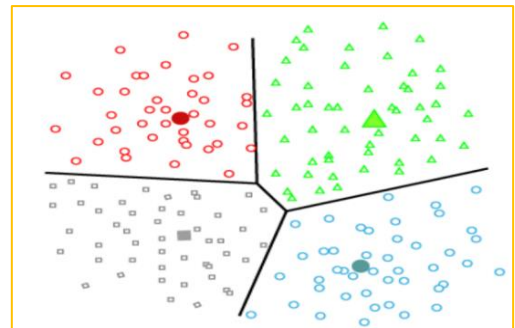
Types of Clustering Methods

Some of the common clustering methods used in Machine learning are:

- 1) Partitioning Clustering
- 2) Density-Based Clustering
- 3) Distribution Model-Based Clustering
- 4) Hierarchical Clustering

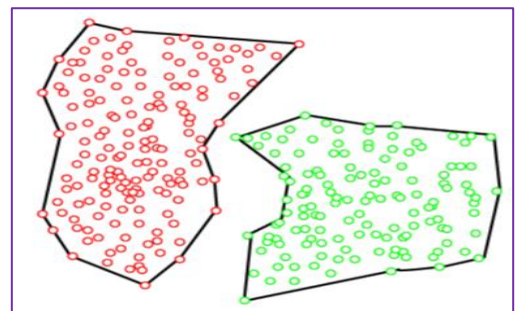
1. Partitioning Clustering

It is a type of clustering that divides the data into non-hierarchical groups. It is also known as the **centroid-based method**. The most common example of partitioning clustering is the K-Means Clustering algorithm. In this type, the dataset is divided into a set of k groups, where k is used to define the number of pre-defined groups. The cluster center is created in such a way that the distance between the data points of one cluster is minimum as compared to another cluster centroid.



2. Density-Based Clustering

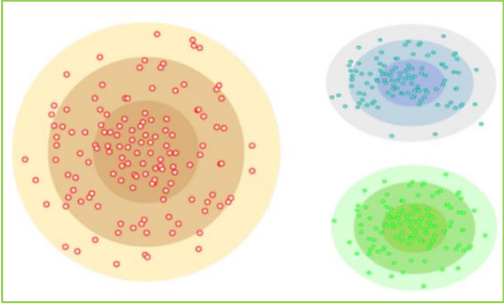
The density-based clustering method connects the highly-dense areas into clusters, and the arbitrarily shaped distributions are formed as long as the dense region can be connected. This algorithm does it by identifying different clusters in the dataset and connects the areas of high densities into clusters. The dense areas in data space are divided from each other by sparser areas. These algorithms can face difficulty in clustering the data points if the dataset has varying densities and high dimensions.



3. Distribution Model-Based Clustering

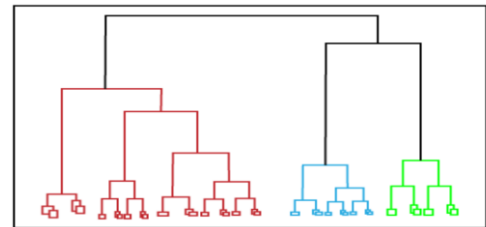
In the distribution model-based clustering method, the data is divided based on the probability of how a dataset belongs to a particular distribution. The grouping is done by assuming some distributions commonly **Gaussian Distribution**.

The example of this type is the **Expectation-Maximization Clustering algorithm** that uses Gaussian Mixture Models (GMM).



4. Hierarchical Clustering

Hierarchical clustering can be used as an alternative for the partitioned clustering as there is no requirement of pre-specifying the number of clusters to be created. In this technique, the dataset is divided into clusters to create a tree-like structure, which is also called a **dendrogram**. The observations or any number of clusters can be selected by cutting the tree at the correct level. The most common example of this method is the **Agglomerative Hierarchical algorithm**.



K- Means clustering

K-Means Clustering is an unsupervised learning algorithm that is used to solve the clustering problems in machine learning or data science. The k-means algorithm is one of the most popular clustering algorithms. It classifies the dataset by dividing the samples into different clusters of equal variances. The number of clusters must be specified in this algorithm.

Steps involved K-Means Clustering:

The working of the K-Means algorithm is explained in the below steps:

- Select the number K to decide the number of clusters.
- Select random K points or centroids. (It can be other from the input dataset).
- Assign each data point to their closest centroid, which will form the predefined K clusters.
- Calculate the variance and place a new centroid of each cluster.
- Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster.
- If any reassignment occurs, then go to step-4 else go to FINISH.
- The model is ready.

Activity: Visual AI: This tool allows you to visualize K-means clustering in real-time. Upload your own data or use provided examples, adjust parameters, and see how clusters change visually using the link [Visualise k-means](#)

Applications of K-Means Clustering:

- Market Segmentation: group customers based on similar purchasing behaviours or demographics for tailored marketing strategies.
- Image Segmentation: partition images into regions of similar colours to aid in tasks like object detection and compression.
- Document Clustering: categorize documents based on content similarity, aiding in organization and information retrieval.
- Anomaly Detection: identify outliers by clustering normal data points and detecting deviations.
- Customer Segmentation: segment customers for targeted marketing and personalized experiences.

Advantages of K-Means Clustering:

- Easy to implement, making it suitable for users of all levels.
- Handles large datasets with low computational resources.
- Works well with numerous features and data points.
- Are easy to understand, aiding in decision-making.
- Applicable across various domains and data types.

Limitations of K-Means Clustering:

- Results can vary based on initial centroid placement.
- Assumes clusters are spherical, which is not always true.
- Number of clusters must be known beforehand.
- Outliers can distort clusters due to their influence on centroids.
- May converge to suboptimal solutions instead of the global optimum.

For advanced Learners – Python Program for K Means Clustering

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.cluster import KMeans

# Generate synthetic data
X, _ = make_blobs(n_samples=300, centers=4, cluster_std=0.60, random_state=0)

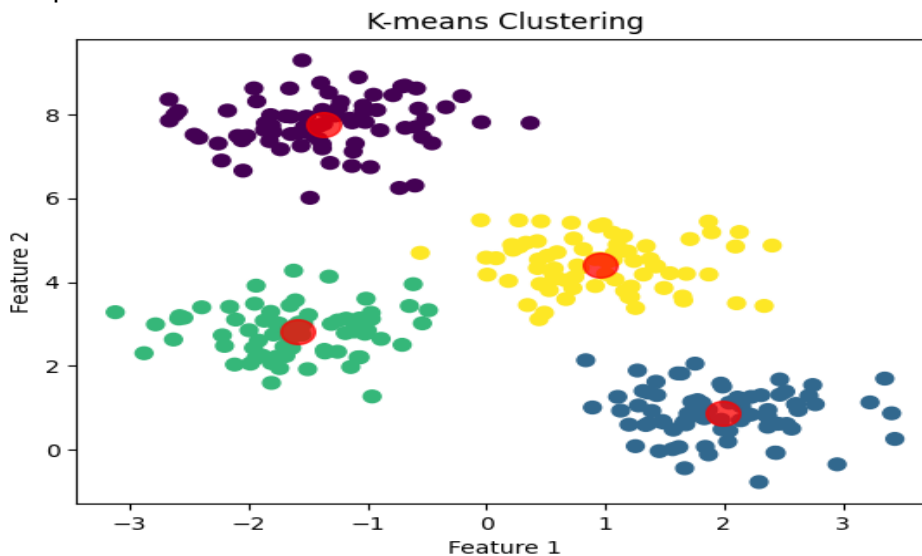
# Apply K-means clustering
kmeans = KMeans(n_clusters=4)
kmeans.fit(X)
y_kmeans = kmeans.predict(X)

# Plot the data points and centroids
plt.scatter(X[:, 0], X[:, 1], c=y_kmeans, s=50, cmap='viridis')
centers = kmeans.cluster_centers_
```



```
plt.scatter(centers[:, 0], centers[:, 1], c='red', s=200, alpha=0.75)
plt.title('K-means Clustering')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.show()
```

Output



This program does the following:

1. Generates synthetic data using `make_blobs` from `sklearn.datasets`.
2. Applies K-means clustering with `n_clusters=4`.
3. Plots the data points colored by their cluster assignments and shows the centroids as red circles.

You can adjust the parameters like the number of clusters, standard deviation, and number of samples in `make_blobs` to observe different clustering scenarios.

REFERENCES

Video Session

Videos: Clustering: <https://www.youtube.com/watch?v=wk2yII1qgU0>

K-means clustering: <https://www.youtube.com/watch?v=4b5d3muPQmA&t=119s>

EXERCISES

A. Multiple Choice Questions

1. Which of the following are the types of correlation?
 - a. Positive correlation
 - b. Negative Correlation
 - c. No correlation
 - d. All of the above
2. Which of the following techniques is an analysis of the relationship between two variables to provide the prediction mechanism?
 - a. Standard error
 - b. Correlation
 - c. Regression
 - d. None of the above

1. Understanding Human Language Complexity

Linguistics is a field of study that focuses on the strategic application of linguistic principles and practices to meet specific objectives or goals. The field of linguistics is often used in marketing, advertising, communications, education, natural language processing, etc. Understanding how language works (including its structure, semantics and pragmatics, as well as sociolinguistics) allows individuals and organizations to tailor their messages, improve communication and influence behavior.

Understanding human language is difficult, even for people who have grown up with it. Human language is incredibly complex, full of strange expressions that seem to contradict each other, metaphors that require cultural knowledge to understand, and grammatical structures that sometimes turn simple ideas into **tongue-twisters**. Machines require systems that research scientists call **Natural Language Processing**, or NLP, to understand human language.

NLP is a branch of Artificial Intelligence (AI) that allows computers to understand, create, and manipulate human speech. NLP has the capability to query the data with natural-language text or voice. It is also known as “language in”. Most consumers have interacted with NLP. For example, NLP is at the core of the technology behind virtual assistants like ODA, Siri, Cortana, Alexa, etc. NLP can be applied to both written text and speech. Some examples of tools that are powered by NLP are: Web search Email spam filtering Auto-translate text or speech Document summarization Sentiment analysis Grammar/spell checking.

Activity 1: Find five interesting facts about “IBM Project Debater” and write in the space provided below.

2. Introduction to Natural Language Processing

Computers are best at working with structured data, in which everything is neatly grouped and labelled. Unfortunately for machines, human language is anything but structured. You have been using language for most of your life. Your brain accomplishes this through some of the most complicated neural circuitry on Earth. But it is very difficult to create machines that can work with human language.

In NLP, machines segment sentences and extract meaning from “tokens” of human language. Human language is unstructured. Although it is loosely held together by rules of grammar, our language expresses information in many confusing ways. Unlike structured information, which can be arranged in tables or matrices with neatly labeled rows and columns, unstructured information is messy and difficult to understand. To see why, consider this famous joke by Groucho Marx.

“One morning I shot an elephant in my pajamas. How he got in my pajamas, I don’t know.”

-Adapted from Groucho Marx, 20th century comedian and movie star

To deal with the “messiness” of unstructured information, computers begin with one sentence at a time. This is called **sentence segmentation**. Computers then break the information into small chunks of information, called **tokens**, that can be individually classified. Once the tokens in text have been sorted into a structure based on what they mean, NLP can work with them.

The following activities show you how Groucho Marx’s joke can be tokenized into useful categories called **entities** and **relationships**.

- An **entity** is a noun representing a person, place, or thing. It is not an adjective, verb, or other article of speech.

Activity 2:

Now keeping this in mind, identify the entities in the sentence “**I shot an elephant in my pajamas.**”

- A **relationship** is a group of two or more entities that have a strong connection to one another.

Activity 3:

Recall Groucho Marx’s quote:

“One morning, I shot an elephant in my pajamas. How he got my pajamas, I don’t know.”

Now keeping this in mind, identify the relationship between the entities in Groucho Marx’s statement:

I + elephant, I + pajamas, in + pajamas, elephant + pajamas, I + shot

Relationship between two entities

Not a relationship between two entities

Once an AI has classified entities and relationships in text or speech, the AI can begin structuring the information as a step toward understanding it. Your brain, by the way, does the same thing, which might have helped you find entities and relationships in the previous activities. For example, consider the following two sentences: “Manu broke the toy. He always breaks toys.” Notice that there is a relationship between the two sentences: the word **he** is related to the word **Manu**. The machine uses NLP to identify this relationship.

- A **concept** is something implied in a sentence but not actually stated. This is trickier because it involves matching ideas rather than the specific words present in the sentence.

Activity 4

Recall, once more, Groucho Marx's statement:

"One morning, I shot an elephant in my pajamas. How he got my pajamas, I don't know."

Based on this statement, select the words that are concepts implied, but not stated.

- ☐ Safari
- ☐ Rifle
- ☐ Photographed
- ☐ Pajamas

Emotion Detection and Sentiment Analysis



Aspect	Emotion Detection	Sentiment Analysis
Definition	Identifies distinct human emotion types.	Measures the strength of an emotion.
Examples	Determining if an expression is anger, happiness, etc.	Assessing if data is positive, negative, or neutral.
Use Cases	Analyzing user ratings, comments in surveys, etc.	Reading social media posts, customer service chats, etc.
AI Training	Can be trained to classify emotions.	Utilizes a sliding scale between positive and negative.
Purpose	Identifying emotional tokens to understand context.	Assessing the overall tone or sentiment of text data.

Classification Problem

Here is an old-fashioned riddle:

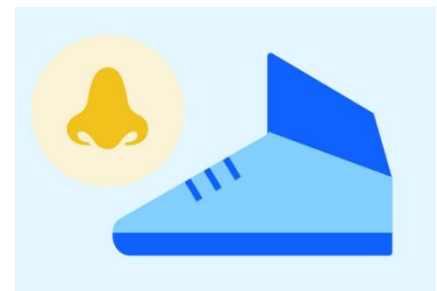
Why does your nose run and your feet smell?

Human language is full of terms that are vague or have double meanings. This is called a **classification problem**.

In this riddle, the phrases "your nose run" and "your feet smell" are used in a humorous way to highlight the ambiguity of language, which poses a classification problem. For instance, "a runny nose" typically refers to having a cold and needing tissues to wipe one's nose, while "a smelly foot" indicates that one's foot has an unpleasant odour. The challenge arises because the words "run" and "smell" have multiple meanings.

Similarly, in everyday language, phrases like "**shipping a box by train**" or "**filling in a form by filling it out**" may seem contradictory or confusing due to the double meanings of the words used. While humans can quickly grasp the intended meaning based on context, AI systems may struggle to accurately classify such phrases without a comprehensive understanding of language nuances and context.

To address this problem,



- An AI system utilizes machine learning techniques such as supervised learning.
- By feeding the system with a large dataset containing examples of language usage and their corresponding classifications, the AI system learns patterns and relationships between words, phrases, and their meanings.
- Over time and with exposure to more data, the AI system improves its classification accuracy by adjusting its internal parameters based on the observed patterns.
- AI systems may not achieve perfect classification accuracy. There will always be some degree of uncertainty or error associated with the system's classifications. To address this, well-designed AI systems not only provide a response but also a confidence value, indicating the system's level of certainty in its classification.

Chatbots

Chatbots are software applications or computer programs designed to simulate conversation with human users, typically through text-based or voice-based interactions. They use artificial intelligence (AI), natural language processing (NLP), and machine learning techniques to understand user queries and provide appropriate responses. Chatbots can be integrated into various platforms such as websites, messaging apps, and voice assistants, and they serve a wide range of purposes including customer support, information retrieval, task automation, and entertainment.

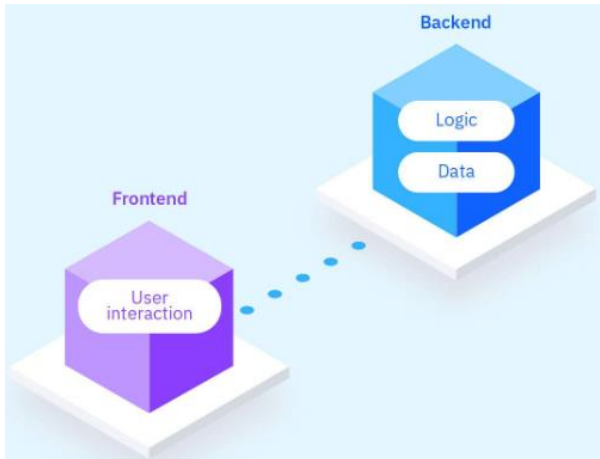
Chatbots, utilized in various fields from retail to healthcare, offer constant availability for online engagement. While not always capable of providing answers, they serve to listen and respond to repetitive inquiries, relieving businesses from the need to allocate human resources for such tasks. *Chatbots operate effectively with small-scale data*, catering to specific needs like movie queries for a cinema chain or broader inquiries for AI systems scanning social media.

Chatbots	Rule-based Chatbots	AI-powered Chatbots
Description	Operate on predefined rules and decision trees. Follow programmed rules to respond to user input.	Utilize natural language processing (NLP) and machine learning algorithms. Also known as chat agents or virtual assistants.
Advantages	- Easy to develop and maintain. - Provide consistent and accurate answers to specific questions.	- 24/7 availability for immediate and consistent support. - Offer personalized interactions based on user preferences and history. - Improve efficiency and cost savings by automating tasks and reducing service costs.
Limitations	- Struggle with understanding complex language. - Unable to adapt to situations beyond programmed rules.	- High development costs and resource requirements. - Prone to biases from training data and lack of transparency in decision-making. - Ethical considerations regarding privacy, manipulation, and responsible use.

Chatbots	Rule-based Chatbots	AI-powered Chatbots
Use Cases	- Customer service tasks like answering common questions and providing order updates. - Guiding users through specific processes.	- Entertainment and Gaming: Engage users with interactive stories and personalized gaming experiences. - Finance and Banking: Answer queries about accounts, transactions, and financial products, and process simple requests.

Structure of a chatbot

A chatbot has a “frontend” and a “backend”.



- The **frontend** of a chatbot serves as the messaging channel through which users interact, providing a user-friendly interface. However, one limitation of the frontend is that it may lack contextual understanding, meaning it might struggle to grasp the full meaning or context of user messages beyond the immediate input.
- The **backend** of a chatbot is where the hard work takes place. The backend operates

application logic and has enough memory to remember earlier parts of a conversation as dialog continues.

Chatbots **understand** a question by breaking it into parts and relating those parts to things in its memory. A chatbot’s goal is to identify **entities** and **intents**, then use what it is found to trigger a **dialog**. Let us understand what intent means.

Intent

An intent is a purpose: the reason why a user is contacting the chatbot. Think of it as something like a **verb**: a kind of action. Users may have various intents when interacting with a chatbot, such as filing a complaint, asking for directions, or speaking to a salesperson. Institutions often have multiple intents that they want their chatbots to address.

For instance, imagine you are tasked with creating a chatbot for a restaurant chain. One potential intent could be to inquire about the restaurant's operating hours. To ensure the chatbot can effectively handle this intent, you might interview a staff member who frequently fields such queries over the phone. Then, your goal would be to compile a comprehensive list of the diverse ways in which customers might inquire about the restaurant's opening hours. The following table provides many examples of possible user inputs that map to this kind of intent.

Intent	Possible user inputs
Open	When do you open?
	What are your hours?
	You open now?
	How late are you open?
	Can I walk in at 7 pm?

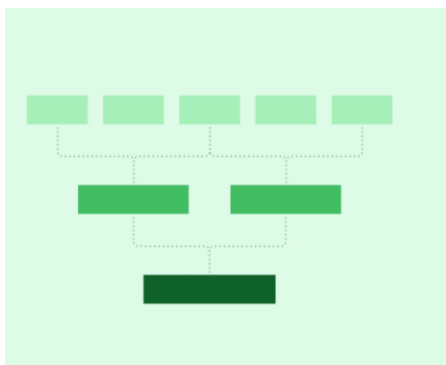
Entity

An entity is a noun: a person, place, or thing. Once you have a list of the intents you want your chatbot to fulfill, you are ready to continue. If a user asks, “What are the hours for the Bangalore office?”, then providing business hours is the **intent** and Bangalore is the **entity**. A chatbot needs a full list of entities to be helpful. The following table lists examples of entities that map to the intent and possible user inputs of the previous restaurant chain example.

Intent	Possible user inputs	Entities
Open	When do you open?	Bangalore
	What are your hours?	Schedule
	You open now?	Time
	How late are you open??	Time
	And so forth; there are many inputs mapping to this intent.	And so forth; there are many entities implied by this intent.

Dialog

A **dialog** is a flowchart—an IF / THEN tree structure that illustrates how a machine will respond to user intents. A dialog is what the machine replies after a human asks a question. Even if a human uses run-on sentences, poor grammar, chat messaging expressions, and so on, artificial intelligence allows the NLP to understand well enough to provide a response.



The dialog represents each possible word or phrase a user might enter, the matched response for the chatbot, and the many possible subsequent replies a user might make next. That is too much for an ordinary flowchart to show (you might need three or four dimensions!), so chatbot software condenses each moment of the conversation into a node. A node contains a statement by the chatbot and a long, expandable list of possible replies.

Creating the flowchart for a chatbot's responses entails a comprehensive process. Each potential user input following the chatbot's initial greeting must be assigned a corresponding reply. For instance, in the case of inquiries about a restaurant's operating hours, all conceivable questions would trigger a single response. This mapping of a vast array of possible questions to a limited number of responses persists throughout the conversation until its conclusion.

Natural Language Processing – Converting Speech to Text & analysing its intent

Natural language processing (NLP) involves a series of five phases that enable machines to analyse, categorize, and understand both spoken and written language. These steps utilize deep neural network-style machine learning techniques to mimic the brain's ability to process data accurately. Businesses leverage tools and algorithms aligned with these NLP stages to extract insights from vast amounts of data and facilitate informed decision-making. Effective comprehension of these NLP stages is essential for optimizing the use of NLP in text and voice applications. Let's explore each step-in detail:

1. Lexical analysis

This step involves understanding and examining the structure of words in a language. It breaks down the text into paragraphs, phrases, and words. Lexical normalization techniques like stemming and lemmatization are commonly used to reduce words to their base forms.

- Stemming reduces words to their root form, such as removing suffixes like "ing", "ly", "es", and "s".
- Lemmatization reduces words to their dictionary form, considering factors like parts of speech (POS) to determine their meaning in context.

2. Syntactical Analysis

Syntactic Analysis is used to check grammar, word layouts, and word relationships.

Example: Mumbai travels to the Anuj.

The line "Mumbai travels to Anuj" makes no sense, hence it is rejected by the Syntactic Analyzer. **Syntactical parsing** is the analysis of words in a sentence for grammar. Dependency Grammar and Part of Speech (POS) tags are significant syntactic elements.

3. Semantic Analysis

Semantic analysis aims to understand the various meanings conveyed by a sentence in a clear and contextually appropriate manner. It extracts relevant insights from the text to comprehend its intended message.

4. Discourse Integration

This involves understanding the context of a statement or word based on preceding sentences or words. It helps interpret references like pronouns and proper nouns by identifying their connections with earlier parts of the conversation.

Example- Arti wants it.

We can observe from the following sentence that the "it" keyword makes no sense. In reality, it applies to anything we don't know. That is all this "it" word depends on the prior sentence, which is not provided. So, if we know what "it" is, we can simply find the reference.

5. Pragmatic Analysis

It denotes the study of meanings in a particular language. Process of extracting insights from a text. It involves verbal repetition, such as "who said what to whom?" It recognizes how individuals communicate with one another, the context in which they are speaking, and many other factors.

Video Session: Five Phases of NLP (Natural Language Processing) (with Tools and Applications) -

<https://www.youtube.com/watch?v=8SSAliFwCy8>

Applications of NLP

Businesses can use natural language processing tools to analyse data, discover insights, automate time-consuming operations, and gain a competitive advantage.

- **Sentiment Analysis:** Natural language processing (NLP) may evaluate consumer comments, social media posts, product reviews, and other text data to determine the sentiment (positive, negative, or neutral) associated with a specific brand, product or service. This data aids firms in understanding client impressions and sentiments.
- **Voice Assistants:** Voice assistants are becoming increasingly popular! Whether it's Siri, Alexa, or Google Assistant, practically everyone uses one to make calls, create reminders, plan meetings, set alarms, browse the internet, and so on. How do they work? They use a complicated blend of voice recognition, natural language interpretation, and natural language processing to understand and respond to what humans say. The long-term purpose of voice assistants is to serve as a bridge between humans and the internet, providing a variety of services through simple speech interaction. However, they are still a long way from achieving that aim, as Siri occasionally fails to grasp what you are saying!
- **Email Filtering:** Email is a part of our daily lives. We find ourselves bombarded with emails about job, study, and a variety of other topics. We receive emails from a variety of sources; some are work-related or from our dream school or institution, while others are spam or promotional in nature. Here, Natural Language Processing comes into play. It classifies incoming emails as "important" or "spam" and assigns them accordingly.
- **Document Analysis:** Document analysis is another use of natural language processing. Companies, institutions, and schools, among other places, are constantly inundated with data that must be properly organized, stored, and searched. All of this may be accomplished with NLP. It not only searches a keyword but also categorizes it according to the instructions, saving us from the tedious and time-consuming task of searching for a single person's information from a large number of files. It is not only restricted to this, but it also assists users in making informed decisions about claims and risk management.
- **Automatic Summarization:** Data has grown in line with technological advancements. This rise of data has broadened the scope of data processing. Still, manual data processing is time-consuming and error-prone. NLP provides a solution for this as well; it can not only summarize the meaning of information but also identify the emotional meaning hiding within it. As a result, the summary process becomes more efficient and precise.

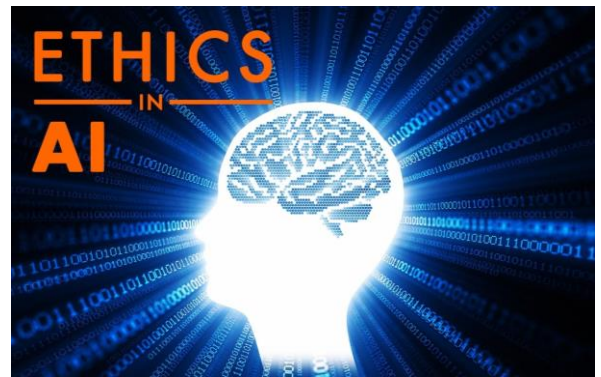
ETHICS IN ARTIFICIAL INTELLIGENCE



In today's rapidly evolving technological landscape, artificial intelligence (AI) has emerged as a transformative force with the potential to revolutionize various aspects of human society. However, with this technological advancement comes a myriad of ethical considerations and challenges. This chapter serves as a foundational exploration into the ethical dimensions of AI, aiming to equip students with the necessary knowledge and

tools to navigate the complex ethical landscape of AI technologies.

Ethics refers to the moral principles that govern human behavior and decision-making. It encompasses concepts such as right and wrong, fairness, justice, and accountability. Ethical considerations guide individuals and organizations in making responsible choices that align with societal values and norms. In the context of AI, ethics play a crucial role in ensuring that AI technologies are developed and deployed in a manner that upholds ethical principles and respects human rights. Ethical considerations are essential in addressing issues such as bias, transparency, accountability, privacy, and the societal impact of AI.



AI ethics refers to the ethical principles and guidelines that govern the design, development, and deployment of AI technologies. AI ethics aims to ensure that AI systems are developed and used in ways that are fair, transparent, accountable, and aligned with human values.

Example 1:

Suppose a CCTV camera was to spot your face in a crowd outside a sports stadium. In the police data center somewhere in the city/ country, an artificial neural network analyzes images from the CCTV footage frame by-frame. A floating cloud in the sky causes a shadow on your face and the neural network (by mistake) finds your face similar to the face of a wanted criminal.

- If the police were to call you aside for questioning and tell you they had reason to detain you, how would you defend yourself?
- Was it your fault that your shadowed face has resemblance by few degrees with a person in the police record?

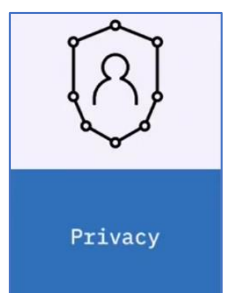
Example 2:

This happened in the USA in 2018. An AI system was being used to allocate care to nearly 200 million patients in the US. It was discovered later that the AI system was offering a lower standard of care to the black patients. Across the board, black people were assigned lower risk scores than white people. This in turn meant that black patients were less likely to be able to access the necessary standard of care.

The problem stemmed from the fact that the AI algorithm was allocating risk values using the predicted cost of healthcare. Because black patients were often less able to pay or were perceived as less able to pay for the higher standard of care, the AI essentially learned that they were not entitled to such a standard of treatment. Though the system was fixed / improved after being discovered, the big question is – whose problem was this? The AI system developers or the US black people data (which was true to an extent)?

THE FIVE PILLARS OF AI ETHICS

- **Explainability** refers to the transparency and interpretability of AI systems, allowing users to understand how algorithms make decisions and predictions. Explainable AI enables stakeholders to comprehend the underlying logic, factors, and considerations driving algorithmic outcomes, fostering trust, accountability, and ethical use of AI technologies. Explainability is essential for ensuring that AI systems are transparent, accountable, and aligned with ethical principles.
- **Fairness** in AI seeks to remove bias and discrimination from algorithms and decision-making models. Machine learning fairness addresses and eliminates algorithmic bias from machine learning models based on sensitive attributes like race and ethnicity, gender, sexual orientation, disability, and socioeconomic class.
- **Robustness** in AI systems indeed refers to their ability to consistently provide accurate and reliable results regardless of the conditions they encounter and for extended periods. It is all about making sure that AI algorithms and systems operate as expected without any unexpected errors or deviations from their intended behavior. This involves ensuring stability in the algorithms, being able to reproduce results, and maintaining consistent performance across different datasets and environments. Achieving reliability in AI systems requires thorough testing, validation, and quality assurance at every stage of development.
- **Transparency** involves openness and disclosure about the design, operation, and implications of AI systems. Transparent AI frameworks provide clear documentation, disclosure, and communication about the data, algorithms, and decision-making processes used in AI applications. Transparency promotes accountability, scrutiny, and informed decision-making, enabling users and stakeholders to assess the ethical implications and societal impacts of AI technologies.
- **Privacy** refers to the right of individuals to control their personal information and to be free from unwanted intrusion into their lives. It encompasses the ability to keep certain aspects of one's life private, such as personal communications, activities, and personal data. Privacy is essential for safeguarding personal autonomy, dignity, and freedom from unwarranted interference.



Activity:

Organize the class into groups. Reflect on the following points based on the video links given

- Video: [AI for Good](#)
- Reflect on the video "[The Ethical Robot](#)" and identify two ethical dilemmas that stood out to you. Document these questions.
- Using "[How to build a moral robot](#)" as a reference point, list the moral and ethical principles you wish to embed in your robot. Consider the video as inspiration, but do not feel constrained by its content. Feel free to expand your thoughts with creativity and innovation.
- Assemble a team of five students and collectively watch the video "[Humans need not apply](#)." It is recommended to view the video multiple times. Following your discussions, compile a group paper summarizing your insights and interpretations from the video.

BIAS, BIAS AWARENESS, SOURCES OF BIAS

Bias, in simple terms, means having a preference or tendency towards something or someone over others, often without considering all the relevant information fairly. It can lead to unfair treatment or decisions based on factors like personal beliefs, past experiences, or stereotypes. In everyday life, bias can affect how we perceive and interact with people, situations, or ideas. In the context of artificial intelligence, bias refers to when AI systems make unfair or inaccurate decisions due to flawed data or built-in assumptions, which can result in unfair outcomes for certain groups of people.

In today's interconnected world, artificial intelligence (AI) technologies play an increasingly prominent role in various aspects of our lives, from healthcare to finance to criminal justice. However, as AI systems become more pervasive, it is essential to recognize and address the presence of bias in these technologies. **Bias awareness** means understanding that AI systems might have unfair preferences because of different things like the information they were taught with, the rules they follow, or the ideas they were built upon. So, being aware of bias in AI is like knowing that sometimes AI might make unfair choices or judgments because of how it was trained or made.

Activity:

Question 1: Why are most images that show up when you do an image search for “vacation” seen as beaches?

Question 2: Why are most images that show up when you do an image search for “nurse” seen as females?

Question 3: Organize students into groups and ask them to find answers for the questions given below after going through the link [Amazon Recruitment Tool](#)

Discussion Questions:

- How do algorithmic hiring systems function, and what criteria are typically used to evaluate job applicants?
- What are the ethical implications of using biased algorithms in hiring processes, particularly regarding fairness, equal opportunity, and diversity?
- How might biased hiring algorithms perpetuate systemic inequalities in employment and hinder efforts to promote inclusivity in the workforce?

Reference: <https://www.ibm.com/blog/shedding-light-on-ai-bias-with-real-world-examples/>

AI bias, also referred to as machine learning bias or algorithm bias, refers to AI systems that produce biased results that reflect and perpetuate human biases within a society, including historical and current social inequality. Bias can be found in the initial training data, the algorithm, or the predictions the algorithm produces. When bias goes unaddressed, it hinders people's ability to participate in the economy and society. It also reduces AI's potential.

The source of bias in AI

Eliminating AI bias requires drilling down into datasets, machine learning algorithms and other elements of AI systems to identify sources of potential bias.

1. Training data bias

AI systems learn to make decisions based on training data, so it is essential to assess datasets for the presence of bias. One method is to review data sampling for over- or underrepresented groups within the training data. For example, training data for a facial recognition algorithm that over-represents white people may create errors when attempting facial recognition for people of color. Similarly, security data that includes information gathered in geographic areas that are predominantly black could create racial bias in AI tools used by police. Bias can also result from how the training data is labeled. For example, AI recruiting tools that use inconsistent labeling or exclude or over-represent certain characteristics could eliminate qualified job applicants from consideration.

2. Algorithmic bias

Using flawed training data can result in algorithms that repeatedly produce errors, unfair outcomes, or even amplify the bias inherent in the flawed data. Algorithmic bias can also be caused by programming errors, such as a developer unfairly weighting factors in algorithm decision-making based on their own conscious or unconscious biases. For example, indicators like income or vocabulary might be used by the algorithm to unintentionally discriminate against people of a certain race or gender.

3. Cognitive bias

When people process information and make judgments, we are inevitably influenced by our experiences and our preferences. As a result, people may build these biases into AI systems through the selection of data or how the data is weighted. For example, cognitive bias could lead to favoring datasets gathered from Americans rather than sampling from a range of populations around the globe.

Examples of AI bias in real life

- **Healthcare**—Underrepresented data of women or minority groups can skew predictive AI algorithms. For example, computer-aided diagnosis (CAD) systems have been found to return lower accuracy results for black patients than white patients.
- **Online advertising**—Biases in search engine ad algorithms can reinforce job role gender bias. Independent research at Carnegie Mellon University revealed that [Google's online advertising system](#) displayed high-paying positions to males more often than to women.
- **Image generation**—[Academic research](#) found bias in the generative AI art generation application Midjourney. When asked to create images of people in specialized professions, it showed both younger and older people, but the older people were always men, reinforcing gender bias of the role of women in the workplace.

Activity: Role Play

Share the following examples of biased AI systems and their potential consequences and ask students to do a role play to present each scenario:

- **Facial Recognition Technology:**

- **Example:** Facial recognition systems have been shown to exhibit bias against certain demographic groups, particularly people with darker skin tones and women.
- **Consequences:** Biased facial recognition algorithms can lead to misidentification and wrongful arrests, disproportionately affecting marginalized communities and eroding trust in law enforcement.

- **Predictive Policing Algorithms:**

- **Example:** Predictive policing algorithms use historical crime data to forecast future criminal activity and allocate law enforcement resources. However, studies have found that these algorithms can perpetuate racial and socioeconomic biases, leading to over-policing of minority neighborhoods.
- **Consequences:** Biased predictive policing algorithms may exacerbate racial profiling and discrimination, fueling tensions between law enforcement agencies and communities of color and undermining public trust in the criminal justice system.

- **Algorithmic Hiring Systems:**

- **Example:** AI-powered hiring systems are used by companies to screen job applications and identify potential candidates. However, research has shown that these systems can perpetuate gender and racial biases, favoring certain demographic groups over others.
- **Consequences:** Biased hiring algorithms may reinforce existing disparities in employment opportunities, leading to discrimination against underrepresented groups and hindering efforts to promote diversity and inclusion in the workforce.

- **Healthcare Algorithms:**

- **Example:** AI algorithms are increasingly used in healthcare for tasks such as diagnosing diseases and predicting patient outcomes. However, studies have identified biases in healthcare algorithms that can result in differential treatment recommendations based on factors such as race or socioeconomic status.
- **Consequences:** Biased healthcare algorithms may lead to disparities in patient care, with certain demographic groups receiving suboptimal or inequitable treatment. This can contribute to worsened health outcomes and perpetuate healthcare inequalities.

- **Credit Scoring Systems:**

- **Example:** AI-powered credit scoring systems are used by financial institutions to assess individuals' creditworthiness and determine their eligibility for loans and other financial products. However, these systems have been found to exhibit biases that disproportionately disadvantage certain demographic groups, such as low-income individuals and people of color.
- **Consequences:** Biased credit scoring algorithms may limit access to financial opportunities for marginalized communities, perpetuating socioeconomic inequalities and hindering economic mobility.

MITIGATING BIAS IN AI SYSTEMS

Mitigating bias in AI systems is essential for several reasons. Firstly, when AI systems have bias, they can make existing problems like unfairness and discrimination even worse. For example, biased algorithms used in hiring processes may unfairly disadvantage certain groups, leading to systemic discrimination. Secondly, biased AI makes people trust technology less. If people don't trust AI to make fair decisions, they might not want to use it, which can cause problems for everyone. Lastly, addressing bias is essential for upholding ethical principles and ensuring that AI technologies are developed and used responsibly.

Strategies for Mitigating Bias

There are several strategies and techniques for mitigating bias in AI systems:

- **Using Diverse Data:** To reduce bias, we should use lots of different kinds of information to teach AI. This way, the AI can learn from many different examples and viewpoints, making it less likely to be biased.
- **Detecting Bias:** We need ways to find and measure bias in AI systems before they are used. This could mean looking at how the AI makes decisions for different groups of people or using special tools to see if the AI is being fair.
- **Fair Algorithms:** We can make AI systems fairer by using special algorithms that are designed to be fair. These algorithms make sure to consider fairness when making decisions, helping to reduce bias.
- **Being Transparent:** It is important for AI systems to be clear and explain how they make decisions. When people understand how AI works, they can see if there is any bias and fix it.
- **Inclusive Teams:** When creating AI, it is helpful to have a team of people from different backgrounds and experiences. This way, they can spot biases that others might miss and make sure the AI is fair for everyone.

Activity:

Allow students to examine various forms of media, such as news articles, advertisements, or social media posts, and identify instances of bias based on factors like race, gender, or socio-economic status. Encourage them to discuss how bias can influence perceptions and stereotypes.

Do you know?

IBM's AI Fairness 360 is an open-source and comprehensive toolkit that offers a wide range of tools and resources for addressing bias in machine learning models. The toolkit includes over 70 fairness metrics which help users detect bias in their models. This suggests that AI Fairness 360 provides robust mechanisms for identifying potential sources of bias. The toolkit offers over 10 algorithms for mitigating bias once it has been detected. These bias mitigating algorithms include optimizing the preprocessing stage, prejudice remover, and regular algorithms, among others. With its diverse set of features, educational resources, and validation mechanisms, it aims to support users in promoting fairness and equity in AI applications.

DEVELOPING AI POLICIES

Developing AI policies is essential for ensuring that AI technologies are used responsibly, safely, and ethically, while also promoting innovation and public trust.

- Rules for AI should start with being good to people and respecting their rights. This means treating everyone fairly, being honest about how AI works, making sure it is safe, and being accountable if something goes wrong.
- We need clear rules and standards for how AI is used. These rules should cover important things like protecting people's information, making sure AI does not have unfair biases, keeping it safe, and making sure people can ask questions about how AI works.
- When making these rules, it is important to talk to lots of different people. This includes government people, business leaders, scientists, community groups, and regular people. Everyone's opinion matters because AI affects everyone.
- Before using AI, we should check to see if there are any problems or risks. This means thinking about what could go wrong and making plans to fix it.

Understanding the components of AI policies involves examining guidelines and principles established by various organizations and regulatory bodies.

1. IBM AI Ethics Board:

Focus: Ethical development and deployment of AI technologies across various industries.

Components:

- Development of ethical principles and guidelines for AI research and development.
- Recommendations for addressing ethical considerations such as fairness, transparency, accountability, and bias mitigation in AI systems.
- Engagement with stakeholders, including researchers, policymakers, and industry partners, to promote dialogue and collaboration on ethical AI practices.
- Support for educational initiatives and resources to raise awareness and understanding of AI ethics among developers, users, and the public.

2. Microsoft's Responsible AI Page:

Focus: Corporate responsibility and ethics in AI

Components:

- Principles for responsible AI development and deployment, including fairness, reliability, privacy, and inclusivity.
- Tools and resources for integrating ethical considerations into AI projects, such as fairness assessments and bias detection algorithms.
- Case studies and best practices for implementing responsible AI practices across various industries and domains.

3. Artificial Intelligence at Google:

Focus: Corporate AI ethics and governance

Components:

- Google's principles for ethical AI development, encompassing areas such as fairness, safety, privacy, and accountability.
- Guidelines for designing AI systems that prioritize human values and societal well-being.

- Commitments to transparency, collaboration, and continuous improvement in AI governance and decision-making.

4. **European Union's Ethics Guidelines for Trustworthy AI—Press Release:**

Focus: Ethical guidelines for AI development and deployment in the EU

Components:

- Principles for trustworthy AI, including respect for human autonomy, prevention of harm, fairness, and accountability.
- Requirements for transparency, explainability, and auditability in AI systems.
- Recommendations for ensuring human oversight and accountability mechanisms in AI applications with high societal impact.

Further reading:

- [IBM's AI Ethics Board](#)
- [Microsoft's Responsible AI page](#)
- [Artificial Intelligence at Google: Our Principles](#)
- [European Union's Ethics Guidelines for Trustworthy AI—Press Release](#)

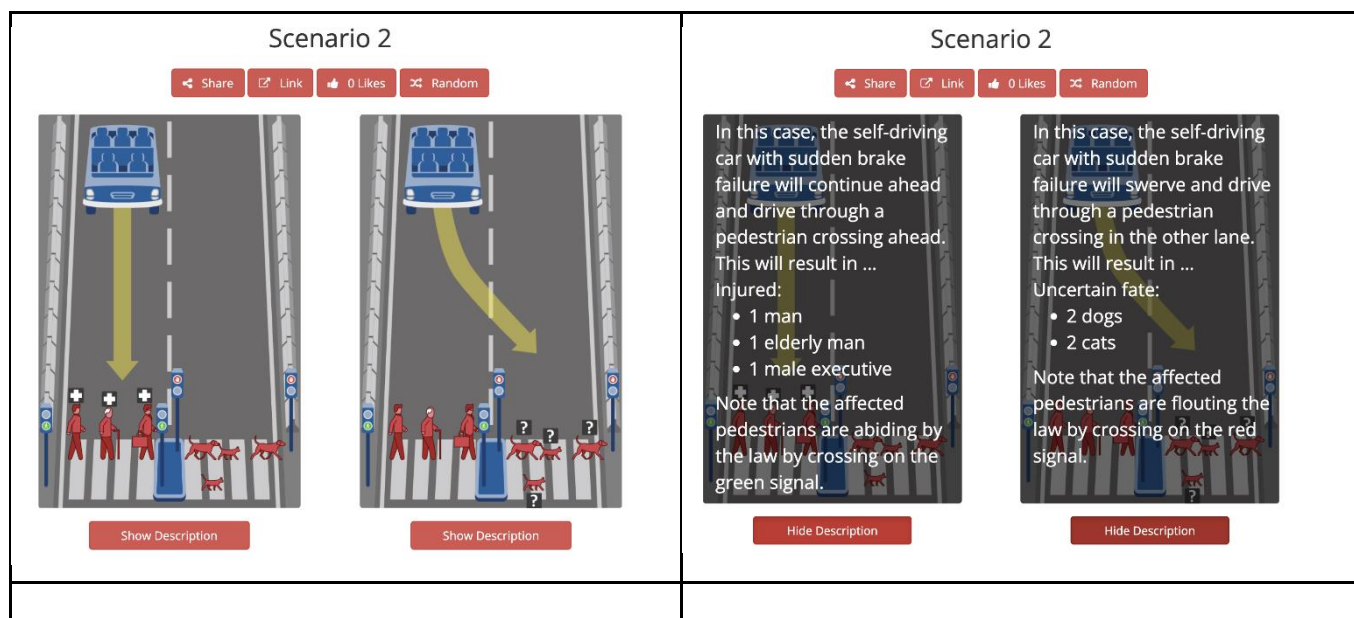
MORAL MACHINE GAME

An **ethical dilemma** is a situation in which a person or group is faced with conflicting moral principles or choices. In other words, it is a scenario where there is no clear "right" or "wrong" decision, and any action taken may have both positive and negative consequences. Ethical dilemmas often involve difficult choices between two or more options, each of which may involve values or principles that are important to the decision-maker.

An **ethical dilemma in the context of artificial intelligence (AI)** arises when there is a conflict between moral principles or values in the design, development, deployment, or use of AI technologies. These dilemmas often emerge due to the complex and multifaceted nature of AI systems, which can have far-reaching implications for individuals, society, and the environment.

The **Moral Machine**, developed by researchers at the Massachusetts Institute of Technology (MIT), is an online platform designed to explore ethical dilemmas in AI through interactive decision-making scenarios. Users are presented with various hypothetical situations where autonomous vehicles must make split-second decisions that could result in harm or even loss of life. The scenarios often involve difficult choices, such as deciding whether to prioritize the safety of passengers or pedestrians, obey traffic laws, or consider factors like age, gender, and social status.

The Moral Machine presents users with a series of thought-provoking scenarios, each with its own set of moral considerations and trade-offs. For example, imagine you are the operator of a self-driving car that encounters a situation where it must choose between swerving to avoid hitting a group of pedestrians, potentially endangering its passengers, or staying the course and risking harm to those on the road. What decision would you make? And more importantly, why?



While the scenarios presented on the Moral Machine are hypothetical, they reflect real-world ethical dilemmas that AI developers, policymakers, and society at large must grapple with. As AI continues to advance and become more pervasive, it is essential that we engage in meaningful dialogue and reflection on the ethical implications of these technologies. The Moral Machine serves as a powerful tool for sparking conversation, raising awareness, and promoting ethical awareness in the age of AI.

Activity:

1. Begin with: [The Ethical Challenges of Autonomous Vehicles](#)

Discussion Points:

- What ethical issues are highlighted by the presenter concerning the application of AI in military contexts, and how do these resonate with your viewpoints?
 - What steps can legislators take to tackle the ethical dilemmas presented by autonomous weaponry, particularly regarding accountability and the preservation of human dignity?
2. Present the Moral Machine game as a tool for delving into ethical decision-making challenges related to AI - accessible at <https://www.moralmachine.net/>

SURVIVAL OF THE BEST FIT GAME

Activity:

Survival of the Best Fit is an educational game about hiring bias in AI. We aim to explain how the misuse of AI can make machines inherit human biases and further inequality.

Students can do this activity in the practical class to understand the concept of bias while people are hired.

Present this tool for deeper understanding of bias during hiring of people in any company or organization- <https://www.survivalofthebestfit.com/>